

ELMFIRE Verification Summary Report

ELMFIRE Verification and Validation Suite

September 2, 2026

Abstract

This document aggregates the standalone reports for every runnable verification case currently discovered under `cases/Verification`. The opening summary records environment and execution configuration and reports each case's scientific decision from its case-local `outputs/metrics.json`; command completion and PDF availability are not substituted for verification evidence.

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26 CASE37_DMS – Case 37: Dead-fuel moisture and extinction sweep	123
27 CASE38_LMS – Case 38: Static live-fuel moisture sweep	127
28 CASE39_DHC – Case 39: Dynamic herbaceous curing sweep	131
29 CASE40_WSD – Case 40: Wind-slope vector and direction sweep	135

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1 Verification summary

1.1 Environment captured for this aggregate

This environment was captured when the aggregate report was generated. It describes the report-build process and must not be interpreted as the execution environment of every case unless the cases were run and aggregated in the same job. ELMFIRE is not executed to obtain this information.

Configuration	Recorded value
Captured at (UTC)	2026-09-02T17:16:45+00:00
Repository revision	6621fd3b869c
Repository state	dirty
Host platform	Linux-7.0.12-linuxkit-aarch64-with-glibc2.35
Python runtime	CPython 3.10.12 (/usr/bin/python3)
Conda environment	not active
ELMFIRE command	elmfire
ELMFIRE resolved path	/elmfire/elmfire/build/linux/bin/elmfire
ELMFIRE version declaration	not declared
ELMFIRE executable	c225b2e00349aa98128285b3d3ff9b2031d0e20944cd9f1c5ee96595d199a920
SHA-256	
MPI launcher	mpirun (Open MPI) 4.1.2
Slurm allocation	not a Slurm build job
Loaded modules	not recorded
Python packages	NumPy 1.26.4; Matplotlib 3.8.4; PyYAML 6.0.3; Rasterio 1.4.4; GDAL 3.4.1; Pandas 2.2.2; GeoPandas 1.1.4; Shapely 2.1.2

1.2 Case-declared execution configuration

These values come only from each case's `case.yaml`. They summarize the declared executable command, namelist, MPI allocation, and namelist contract. A missing declaration is reported explicitly rather than inferred.

Case	ELMFIRE command	Input namelist	MPI ranks	Schema
CASE01_BET	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE02_FGR	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE03_ETC	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE04_STI	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE05_TDWB	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE06_F2D	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE07_IPC	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE08_FBC	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE09_MMD	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE10_WSR	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE11_WSC	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE12_TRS	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE13_PRM	not applicable	not applicable	not declared	1
CASE14_SRS	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE15_CRO	\${ELMFIRE_BIN:-elmfire}	data/inputs/elmfire.data	not declared	1
CASE16_NSP	\${ELMFIRE_BIN:-elmfire}	data/inputs/elmfire.data	not declared	1
CASE17_SCV	\${ELMFIRE_BIN:-elmfire}	data/128/inputs/elmfire.data	not declared	1
CASE18_TCV	\${ELMFIRE_BIN:-elmfire}	data/inputs/elmfire_180.data	not declared	1
CASE19_WTH	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE20_PIG	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE21_WDE	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE22_VSL	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE23_FQD	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE24_CXI	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE25_MQD	\${ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1

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Case	ELMFIRE command	Input namelist	MPI ranks	Schema
CASE26_CNF	#{ELMFIRE_BIN:-elmfire}	variants/1mph/elmfire.data.in	not declared	1
CASE27_FBT	#{ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE28_OVA	#{ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE29_SUP	#{ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE30_MIT	#{ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE31_PFT	#{ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE32_RCV	#{ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE33_SDO	#{ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE34_FMS	#{ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE35_WSS	#{ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE36_SLS	#{ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE37_DMS	#{ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE38_LMS	#{ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE39_DHC	#{ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE40_WSD	#{ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE41_CFP	#{ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1
CASE42_WAF	#{ELMFIRE_BIN:-elmfire}	elmfire.data.in	not declared	1

1.3 Scientific decision summary

This table is generated from each case's case-local `outputs/metrics.json`. A successful run script or report build is not treated as scientific evidence of PASS. Missing, incomplete, blocked, or non-decisional results are reported as NOT EVALUABLE.

Total cases: 42 PASS: 24 FAIL: 4 CHARACTERIZED: 0 NOT EVALUABLE: 14

Case	Category	Title	Decision
CASE01_BET	coupling tests	Biomass Emission-Time Resolution Independence	NOT EVALUABLE
CASE02_FGR	coupling tests	Finite-Duration Firebrand Generation	NOT EVALUABLE
CASE03_ETC	coupling tests	Eulerian Firebrand Transport Convergence	NOT EVALUABLE
CASE04_STI	coupling tests	Steady Transport and Immediate Ignition	NOT EVALUABLE
CASE05_TDW	coupling tests	Firebrand Transport in Time-Dependent Wind	NOT EVALUABLE
CASE06_F2D	coupling tests	Two-Dimensional Firebrand-Driven Spread	NOT EVALUABLE
CASE07_IPC	coupling tests	Ignition Probability and SFT Convergence	NOT EVALUABLE
CASE08_FBC	coupling tests	Deposited Firebrand Consumption	NOT EVALUABLE
CASE09_MMD	coupling tests	Mixed-Mode Ignition-Delay Effects	NOT EVALUABLE
CASE10_WSR	coupling tests	WUI Structural Resolution Sensitivity	NOT EVALUABLE
CASE11_WSC	coupling tests	WUI Surface-Coupling Synchronization	NOT EVALUABLE
CASE12_TRS	coupling tests	WUI Temporal-Resolution Sensitivity	NOT EVALUABLE
CASE13_PRM	coupling tests	WUI Parametric Response Maps	NOT EVALUABLE
CASE14_SRS	coupling tests	Two-Dimensional Spatial-Resolution Sensitivity	NOT EVALUABLE
CASE15_CRO	unit tests	Case 15: Constant rate of spread	PASS
CASE16_NSP	unit tests	Case 16: No-spread limiting behavior	PASS
CASE17_SCV	unit tests	Case 17: Spatial convergence	FAIL
CASE18_TCV	unit tests	Case 18: Temporal convergence	FAIL
CASE19_WTH	coupling tests	Case 19: Transient WUI heat flux	PASS
CASE20_PIG	coupling tests	Case 20: Point ignition and isotropic spread	PASS
CASE21_WDE	coupling tests	Case 21: Wind-driven elliptical spread	PASS
CASE22_VSL	coupling tests	Case 22: Spread across opposing valley slopes	PASS

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Case	Category	Title	Decision
CASE23_FQD	coupling tests	Case 23: Spatially varying fuel quadrants	PASS
CASE24_CXI	coupling tests	Case 24: Combined fuel, slope, and wind interactions	PASS
CASE25_MQD	coupling tests	Case 25: Spatially varying moisture quadrants	PASS
CASE26_CNF	coupling tests	Case 26: Surface, passive-crown, and active-crown fire	PASS
CASE27_FBT	coupling tests	Case 27: Lagrangian firebrand generation and transport	FAIL
CASE28_OVA	coupling tests	Case 28: Overnight spread-rate adjustment	PASS
CASE29_SUP	coupling tests	Case 29: Initial- and extended-attack suppression	FAIL
CASE30_MIT	coupling tests	Case 30: Multiple-ignition topology	PASS
CASE31_PFT	coupling tests	Case 31: Planar-front transport	PASS
CASE32_RCV	coupling tests	Case 32: Raster-orientation covariance	PASS
CASE33_SDO	coupling tests	Case 33: Slope-driven directional spread	PASS
CASE34_FMS	unit tests	Case 34: Standard fuel-model Rothermel sweep	PASS
CASE35_WSS	unit tests	Case 35: Midflame wind-speed Rothermel sweep	PASS
CASE36_SLS	unit tests	Case 36: Slope-factor Rothermel sweep	PASS
CASE37_DMS	unit tests	Case 37: Dead-fuel moisture and extinction sweep	PASS
CASE38_LMS	unit tests	Case 38: Static live-fuel moisture sweep	PASS
CASE39_DHC	unit tests	Case 39: Dynamic herbaceous curing sweep	PASS
CASE40_WSD	coupling tests	Case 40: Wind-slope vector and direction sweep	PASS
CASE41_CFP	unit tests	Case 41: Custom fuel-parameter Rothermel sweep	PASS
CASE42_WAF	coupling tests	Case 42: Wind-adjustment-factor coupling sweep	PASS

Decision vocabulary. PASS and FAIL are case-defined scientific decisions. CHARACTERIZED denotes descriptive validation metrics without a justified acceptance threshold. NOT EVALUABLE denotes missing, unrun, incomplete, unsupported, or otherwise non-decisional evidence.

2 Scope and decision policy

The report order follows the stable numerical CASE identifier. Each standalone case PDF is included intact to preserve its equations, figures, acceptance criteria, and limitations. PASS and FAIL retain the decision made by the case-local postprocessor. A missing result, an unrun or incomplete workflow, or an unsupported capability is shown as NOT EVALUABLE. The linked summary at the beginning records the aggregate-build environment, case-declared execution configuration, and scientific decisions before the complete case reports.

3 How to read this report

Each entry is the complete standalone PDF for one verification case. Cases follow the stable order in `cases/Verification/CASE_REGISTRY.md`, and each report retains its own scientific context, equations, metrics, citations, and decision. Aggregating the compiled PDFs keeps case-local macros, labels, bibliographies, and packages isolated from one another. After standalone reports change, rebuild this guide with `make verification-report` or rebuild both suite reports with `make reports`.

CASE01_BET – Biomass Emission-Time Resolution Independence

NOT EVALUABLE: the standalone case report PDF is missing.

CASE02_FGR – Finite-Duration Firebrand Generation

NOT EVALUABLE: the standalone case report PDF is missing.

CASE03_ETC – Eulerian Firebrand Transport Convergence

NOT EVALUABLE: the standalone case report PDF is missing.

CASE04_STI – Steady Transport and Immediate Ignition

NOT EVALUABLE: the standalone case report PDF is missing.

CASE05_TDW – Firebrand Transport in Time-Dependent Wind

NOT EVALUABLE: the standalone case report PDF is missing.

CASE06_F2D – Two-Dimensional Firebrand-Driven Spread

NOT EVALUABLE: the standalone case report PDF is missing.

CASE07_IPC – Ignition Probability and SFT Convergence

NOT EVALUABLE: the standalone case report PDF is missing.

CASE08_FBC – Deposited Firebrand Consumption

NOT EVALUABLE: the standalone case report PDF is missing.

CASE09_MMD – Mixed-Mode Ignition-Delay Effects

NOT EVALUABLE: the standalone case report PDF is missing.

CASE10_WSR – WUI Structural Resolution Sensitivity

NOT EVALUABLE: the standalone case report PDF is missing.

CASE11_WSC – WUI Surface-Coupling Synchronization

NOT EVALUABLE: the standalone case report PDF is missing.

CASE12_TRS – WUI Temporal-Resolution Sensitivity

NOT EVALUABLE: the standalone case report PDF is missing.

CASE13_PRM – WUI Parametric Response Maps

NOT EVALUABLE: the standalone case report PDF is missing.

CASE14_SRS – Two-Dimensional Spatial-Resolution Sensitivity

NOT EVALUABLE: the standalone case report PDF is missing.

Case 15: Constant rate of spread

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

Case ID: CASE15_CRO. This unit test verifies that ELMFIRE transports a circular level-set front at a prescribed, spatially uniform normal rate. It exercises raster ingestion, the **ADJ** spread-rate multiplier, the level-set update, and **phi/vs** output without wind, slope, crown fire, spotting, or suppression. It can reveal rate-unit errors, incorrect signs, and failure to couple the calculated rate to front motion. This is code verification against a known solution, not comparison with a physical fire. The verification question is whether the final spread-rate and radius satisfy the two stated tolerances at $t = 360$ s.

2 Mathematical formulation and numerical implementation

ELMFIRE advances the signed level set by

$$\frac{\partial \phi}{\partial t} + V_n |\nabla \phi| = 0,$$

with $\phi < 0$ burned and $\phi = 0$ the front. Uniform rasters and disabled directional forcing make $V_n = R_0 = 0.0116332 \text{ m s}^{-1}$. The configured **SPREAD_RATE_IN_M** option makes the **vs** output use m min^{-1} , while integration time remains seconds. Raster georeferencing supplies the mesh; no obsolete computational-domain namelist is used.

3 Derivation of expected behavior

For initial circle radius $r_0 = 5$ m and constant outward speed R_0 , radial symmetry reduces the level-set equation to $dr/dt = R_0$. Hence

$$r(360) = 5 + (0.0116332)(360) = 9.187952 \text{ m}, \quad R_0 = 0.697992 \text{ m min}^{-1}.$$

These are independent consequences of the prescribed input rate and elapsed time.

The numerical reference values and the reported metrics follow directly from the generated rate raster. The raster value is $R_s = 0.0116332 \text{ m s}^{-1}$, so the report-unit spread rate is

$$R_{\text{ref}} = 60R_s = 60(0.0116332) = 0.697992 \text{ m min}^{-1}.$$

With $r_0 = 5$ m and $T = 360$ s, the exact final radius is

$$r_{\text{ref}} = r_0 + R_s T = 5 + (0.0116332)(360) = 9.187952 \text{ m}.$$

The two calculated errors are $e_R = |R_{\text{max}} - 0.697992|/0.697992$ and $e_r = |\sqrt{N_b(1 \text{ m})^2/\pi} - 9.187952|$, where N_b is the number of final cells with $\phi < 0$. Thus the acceptance substitutions are $e_R \leq 0.05$ and $e_r \leq 2$ m.

4 Verification metrics and acceptance criteria

The maximum finite final **vs** value is compared with $0.697992 \text{ m min}^{-1}$; a 5 percent relative tolerance catches conversion or propagation-rate errors while allowing front reconstruction effects. The equivalent burned radius $r_A = \sqrt{N_b \Delta x^2 / \pi}$, where N_b counts valid final cells with $\phi < 0$, is compared with 9.187952 m. Area-equivalent radius is less sensitive to pixel stair-stepping than a single axial intercept; its tolerance is two cells. Nodata cells are excluded. The lexicographically last matching final **phi** and **vs** rasters are required. The case passes iff both components pass; missing or malformed required output is NOT EVALUABLE.

5 Simulation configuration and predicted outcome

The generated EPSG:32610 domain is 200 m square with 200 cells per side ($\Delta x = 1 \text{ m}$), uniform fuel/moisture/terrain/weather fields, and a centered 5 m ignition circle. Duration is 360 s. The predicted final field is circular, with rate $0.697992 \text{ m min}^{-1}$ and equivalent radius 9.187952 m.

Parameter	Configured value	Units	Role
Domain/grid	200 by 200 cells; 1	m/cell	Resolves the manufactured circle
Initial front	centered radius 5	m	Exact initial condition
Rate	0.0116332	m/s	Constant manufactured normal speed
Time control	step 1; stop 360	s	Exact final time and timestep multiple
Disabled forcing	wind, slope, crown, spotting, suppression	—	Isolates level-set transport

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

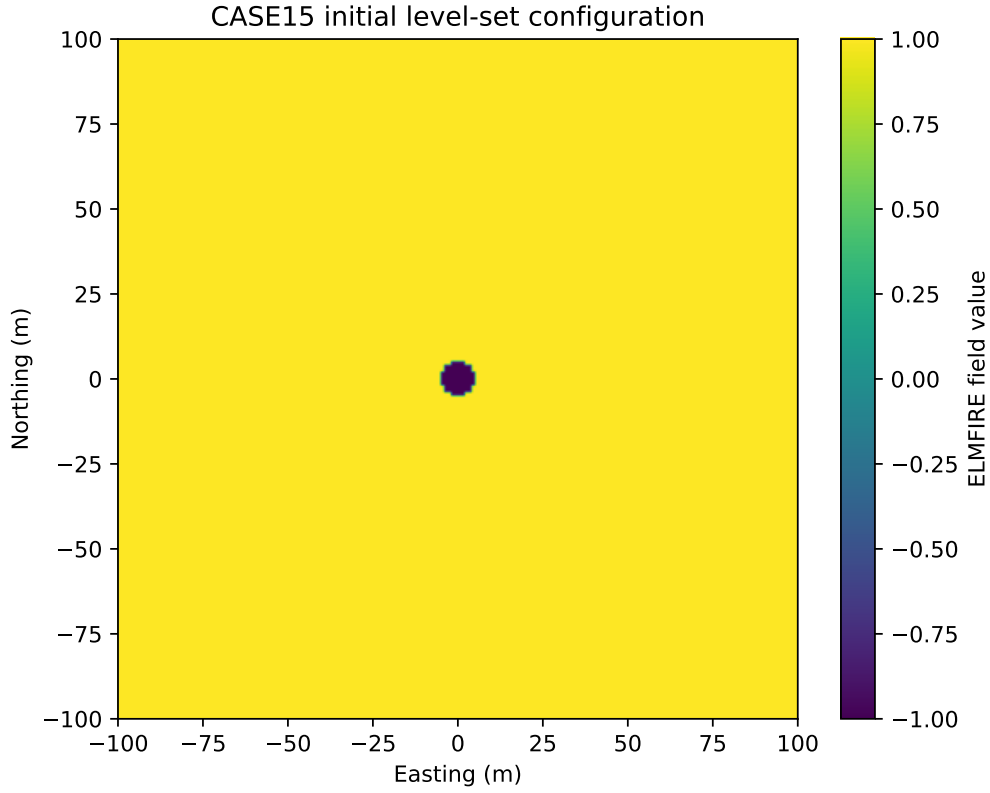


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

Only case-local ELMFIRE rasters are read. Current status is **PASS**; 2 of 2 components pass.

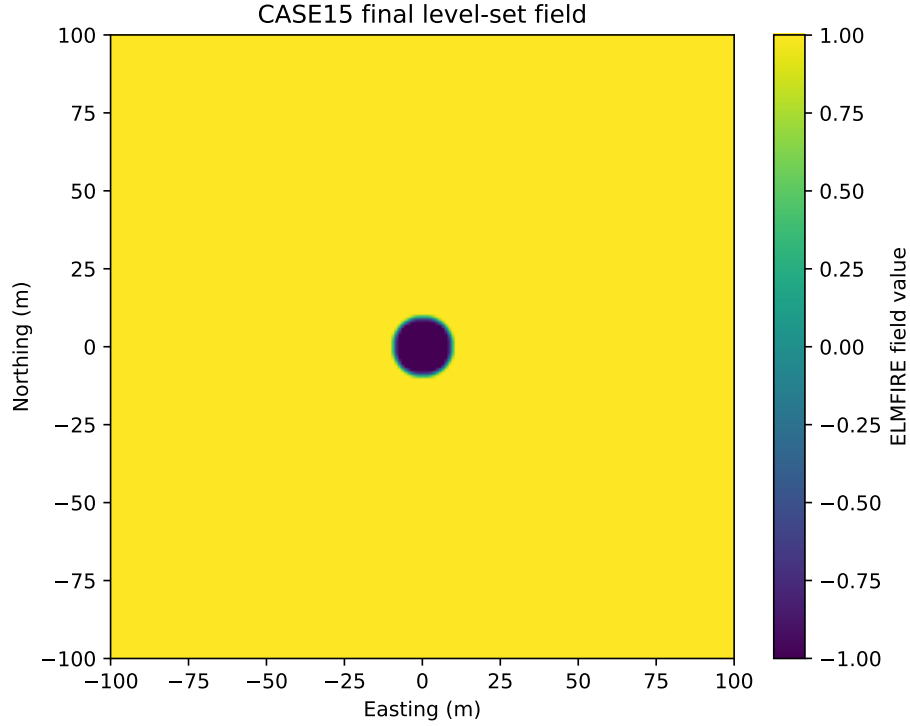


Figure 2: Final whole-domain signed level-set field for the completed variant at the final output time, selected from the case-local data output directory. Values are dimensionless signed-level-set values; the generated table supplies the acceptance limits.

7 Calculated metrics and verification decision

Metric	Expected value or limit	Calculated value	Status
maximum spread rate	0.697992 (5% relative)	0.698532 m/min	PASS
equivalent final radius	9.18795 (2 cells (2 m))	9.37302 m	PASS

The authoritative overall decision is **PASS**. The generated table is sourced directly from the case-local metrics JSON and is not recomputed in this report.

8 Technical interpretation

A spread-rate failure points to input scaling, units, or `vs` selection. A radius-only failure indicates level-set integration or front/area discretization. A missing raster is an evidence failure, not a scientific pass.

9 Scope and limitations

This case verifies one deterministic constant-rate, flat, no-wind propagation state. It does not validate a fuel model, quantify anisotropy, or test coupled crown, spotting, or suppression behavior.

Case 16: No-spread limiting behavior

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

Case ID: CASE16_NSP. This unit test verifies the zero-normal-velocity limit of the ELMFIRE level-set pathway. A zero ADJ raster suppresses the base spread rate while wind, slope, spotting, crown fire, and suppression are disabled. The case exercises input multiplication, propagation, and phi/vs output and can expose ignored adjustment rasters, nonzero numerical forcing, or unintended growth. It is implementation verification, not physical validation. The question is whether the final rate remains below $10^{-6} \text{ m min}^{-1}$ and the ignited footprint remains confined to the initialization band.

2 Mathematical formulation and numerical implementation

The propagation equation is $\partial_t \phi + V_n |\nabla \phi| = 0$. The case sets the multiplicative adjustment to zero, so $V_n = 0$ and therefore $\partial_t \phi = 0$. ELMFIRE still initializes a grid-resolved ignition neighborhood, so exact zero burned area is not expected; the invariant is no subsequent outward propagation.

3 Derivation of expected behavior

Integrating $\partial_t \phi = 0$ gives $\phi(\mathbf{x}, t) = \phi(\mathbf{x}, 0)$. Thus the output rate is zero and any negative- ϕ area must be limited to the rasterized initialization band. A 2.5-cell equivalent-radius allowance bounds that band without treating subcell front placement as spread.

The generated pre-ignition ϕ raster contains no negative cells and the adjustment raster is identically zero. Thus

$$V_n = R_0(ADJ) = R_0(0) = 0, \quad \phi(\mathbf{x}, 10 \text{ s}) = \phi(\mathbf{x}, 0), \quad R_{\text{ref}} = 0 \text{ m min}^{-1}.$$

The rate metric is therefore the absolute quantity $e_R = \max_i |R_i|$, with the substituted test $e_R \leq 10^{-6} \text{ m min}^{-1}$; a relative error is undefined for the zero reference. ELMFIRE may rasterize its point ignition into a small initialization footprint, so the spatial metric is

$$r_A = \sqrt{\frac{N_b (1 \text{ m})^2}{\pi}},$$

where N_b counts final cells with $\phi < 0$. Its no-propagation reference is zero and the explicit grid allowance is $r_A \leq 2.5(1 \text{ m}) = 2.5 \text{ m}$.

4 Verification metrics and acceptance criteria

The maximum absolute finite final vs value tests the zero-velocity invariant directly and must not exceed $10^{-6} \text{ m min}^{-1}$. The equivalent burned radius $r_A = \sqrt{N_b \Delta x^2 / \pi}$ must be no more than $2.5 \Delta x$;

this whole-domain area measure detects growth in any direction and is more robust than one transect. Nodata is excluded, and the last matching final `phi` and `vs` rasters are required. Both conditions must pass; missing output is NOT EVALUABLE.

5 Simulation configuration and predicted outcome

The generated EPSG:32610 domain is 200 m square on a 200-by-200 grid. Fuel, moisture, and terrain are uniform; wind and slope are zero; `ADJ=0`; the ignition is centered; and duration is 10 s. The predicted `vs` field is zero and the final ignition footprint does not expand beyond its grid initialization band.

Parameter	Configured value	Units	Role
Domain/grid	200 by 200 cells; 1	m/cell	Resolves the stationary ignition
Initial front	centered rasterized band	—	Nonzero initialized footprint
Adjustment	<code>ADJ=0</code>	—	Forces zero normal speed
Time control	step 1; stop 10	s	Tests the zero-speed invariant
Disabled forcing	wind, slope, crown, spotting, suppression	—	Excludes alternate growth pathways

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

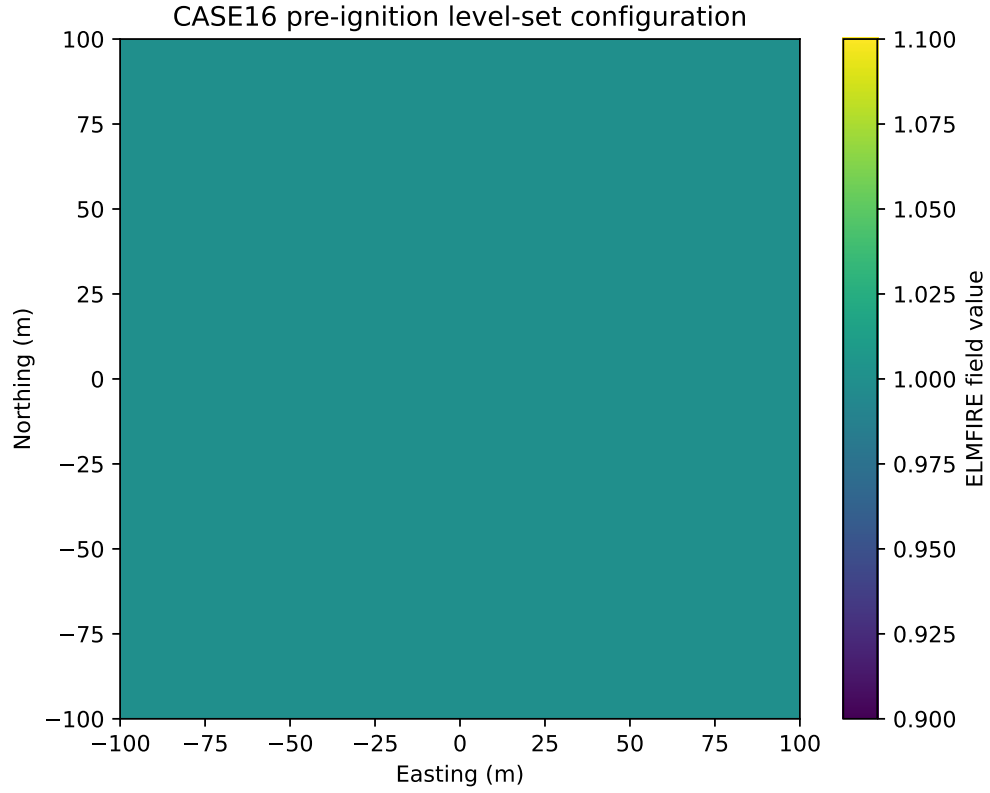


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

Only case-local ELMFIRE rasters are read. Current status is **PASS**; 2 of 2 components pass.

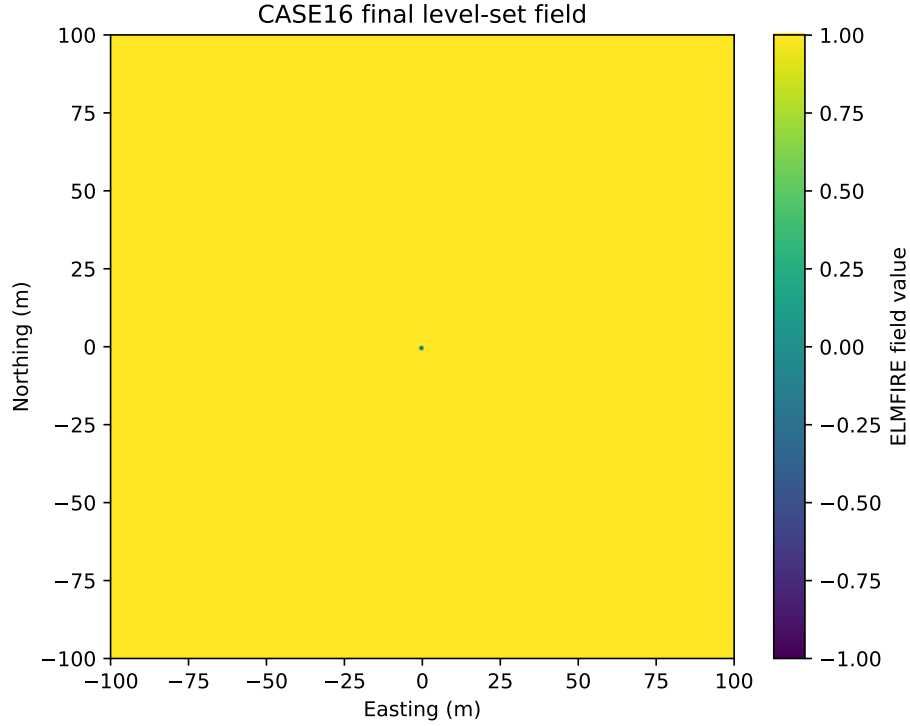


Figure 2: Final whole-domain signed level-set field for the completed zero-spread variant at the final output time. Values are dimensionless; the generated table supplies the rate and confinement limits.

7 Calculated metrics and verification decision

Metric	Expected value or limit	Calculated value	Status
maximum absolute spread rate	0 ($\leq 1\text{e-}06$)	0 m/min	PASS
equivalent burned radius	0 (≤ 2.5 cells (2.5 m))	0.56419 m	PASS

The authoritative overall decision is **PASS**. The generated table is sourced directly from the case-local metrics JSON and is not recomputed in this report.

8 Technical interpretation

A nonzero rate primarily indicates that the zero adjustment was ignored or transformed incorrectly. A confined rate with excessive burned area indicates initialization or level-set evolution behavior. Unavailable output cannot establish the invariant.

9 Scope and limitations

This case verifies a deterministic limiting state and raster adjustment coupling. It does not establish accuracy for positive rates, physical extinction, or any coupled fire process.

Case 17: Spatial convergence

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

Case ID: CASE17_SCV. This unit test verifies spatial convergence of ELMFIRE’s level-set propagation for the same manufactured circular-front problem on four meshes. It exercises raster-derived grid geometry, front initialization, propagation, and `phi` output while holding physical inputs and temporal controls fixed. It can reveal grid-spacing misuse, nonconvergent front updates, or resolution-dependent physical configuration. The question is whether all four required meshes complete, the fitted radius-error order is at least 0.8, and the finest-grid error is at most 1 m.

2 Mathematical formulation and numerical implementation

For $\partial_t \phi + R_0 |\nabla \phi| = 0$ with constant $R_0 = 0.0116332 \text{ m s}^{-1}$, the exact circular radius is mesh-independent. ELMFIRE obtains Δx from each GeoTIFF transform. The four variants use 128, 256, 512, and 1024 cells over the same 256 m domain, so $\Delta x = \{2, 1, 0.5, 0.25\} \text{ m}$.

3 Derivation of expected behavior

With $r_0 = 20 \text{ m}$ and $T = 360 \text{ s}$, $r_\star = 20 + R_0 T = 24.187952 \text{ m}$. For a consistent spatial discretization, the zero-contour equivalent-radius error $E_h = |r_A - r_\star|$ should decrease as $h = \Delta x$ decreases and follow $E_h \approx C h^p$. A least-squares slope of $\log E_h$ against $\log h$ estimates p .

Every grid variant uses the same manufactured solution. From the generated rate $R_s = 0.0116332 \text{ m s}^{-1}$, initial radius $r_0 = 20 \text{ m}$, and stop time $T = 360 \text{ s}$,

$$r_{\text{ref}} = 20 + (0.0116332)(360) = 24.187952 \text{ m}.$$

For each cell size $h_i \in \{2, 1, 0.5, 0.25\} \text{ m}$, the area-equivalent radius and error are

$$r_i = \sqrt{\frac{N_{b,i} h_i^2}{\pi}}, \quad E_i = |r_i - 24.187952|.$$

The observed spatial order is the ordinary-least-squares slope

$$p_h = \frac{\sum_i (\ln h_i - \overline{\ln h})(\ln E_i - \overline{\ln E})}{\sum_i (\ln h_i - \overline{\ln h})^2}.$$

Consequently the table tests the explicitly calculated $p_h \geq 0.8$ and the finest-grid substitution $E_{0.25} = |r_{0.25} - 24.187952| \leq 1 \text{ m}$.

4 Verification metrics and acceptance criteria

The first metric is the fitted order p , selected because it measures the multi-grid trend rather than one favorable mesh; $p \geq 0.8$ allows rasterized-interface effects while requiring approximately first-order behavior. The second is finest-grid $E_h \leq 1$ m, preventing a misleading order from large absolute errors. Radius is obtained from the largest closed, linearly interpolated $\phi = 0$ contour, providing a rotationally aggregated subcell measure. The last matching **phi** raster in every variant directory is required. All variants and both metrics must pass; otherwise the decision is NOT EVALUABLE or FAIL as appropriate.

5 Simulation configuration and predicted outcome

Each EPSG:32610 variant covers the same 256 m square with uniform inputs, flat terrain, zero wind, centered 20 m signed-distance ignition, $R_0 = 0.0116332 \text{ m s}^{-1}$, and duration 360 s. Only grid resolution varies. The predicted front is circular and E_h decreases toward the 24.187952 m reference radius.

Parameter	Configured value	Units	Role
Domain	256 by 256	m	Fixed physical experiment
Meshes	128, 256, 512, 1024	cells/side	Spatial refinement sequence
Cell size	2, 1, 0.5, 0.25	m	Independent variable
Time control	step 1; stop 360	s	Fixed temporal discretization
Initial front	centered signed-distance radius 20	m	Common exact geometry

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

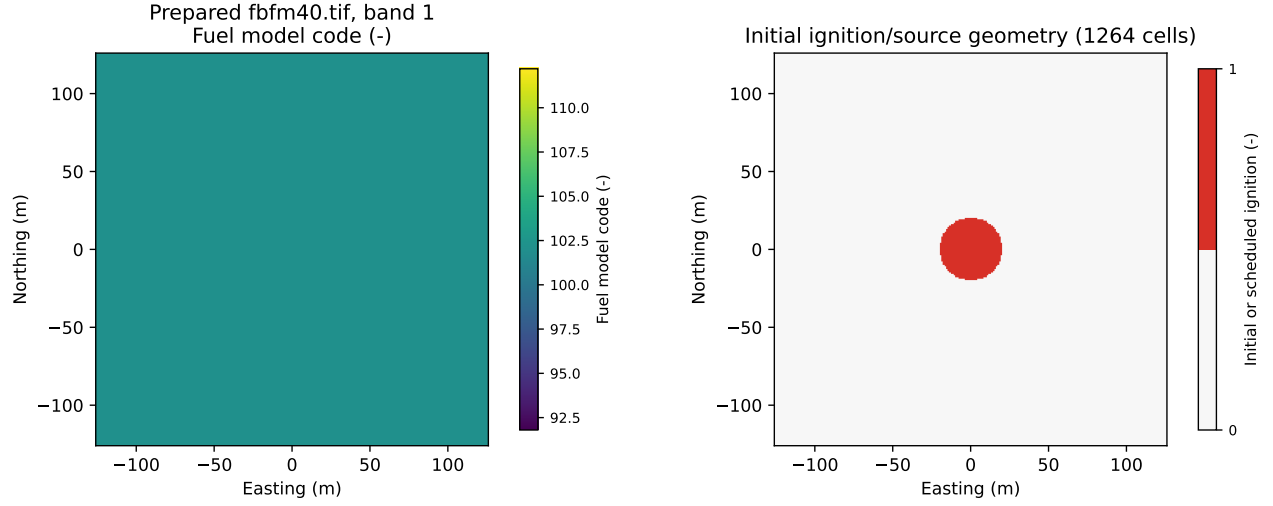


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

Only the four case-local variant outputs are read. Current status is **FAIL**; 4 variants and 2 decision metrics are recorded.

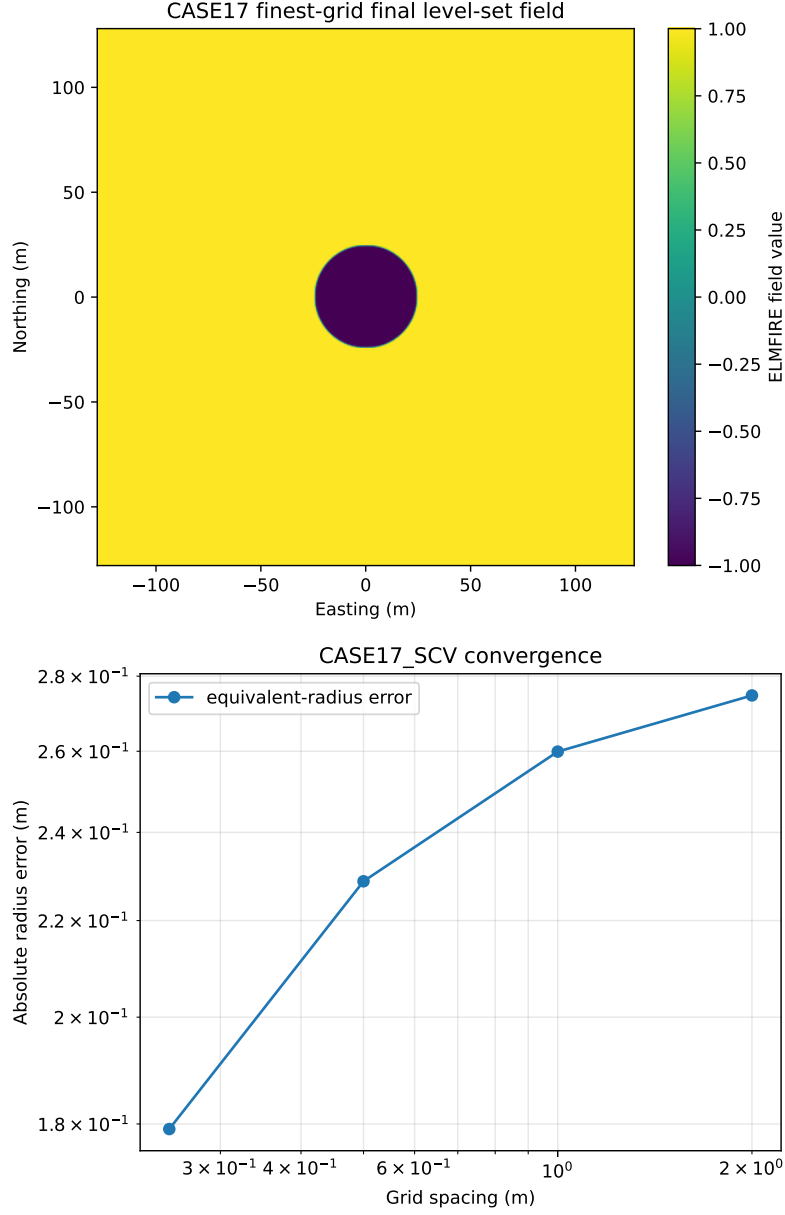


Figure 2: Finest-grid whole-domain signed level-set result and radius-error convergence across all four completed meshes. The spatial field is dimensionless; radius errors are in metres, and the generated table supplies both acceptance limits.

7 Calculated metrics and verification decision

Metric	Expected value or limit	Calculated value	Status
observed spatial order	0.8 ($\geq$ threshold)	0.203637 -	FAIL
finest-grid radius error	0 ($\leq$ 1)	0.179096 m	PASS

The authoritative overall decision is **FAIL**. The generated table is sourced directly from the case-local metrics JSON and is not recomputed in this report.

8 Technical interpretation

Low order with small finest error suggests interface-measurement or pre-asymptotic effects; high order with excessive finest error indicates the sequence has not reached useful accuracy. Missing variants invalidate the convergence fit. Resolution-dependent physical inputs would be a configuration defect.

9 Scope and limitations

This case measures convergence of one smooth, radially symmetric front and an area-derived observable. It does not determine formal order for discontinuous fuels, complex terrain, anisotropic spread, or other output fields.

Case 18: Temporal convergence

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

Case ID: CASE18.TCV. This unit test verifies temporal convergence of ELMFIRE’s level-set propagation for a manufactured circular front. Four time grids vary the whole-second step while mesh and physical fields remain fixed. The case can expose timestep scaling errors, inconsistent final-time handling, or a nonconvergent temporal update. The question is whether all four variants complete, the fitted radius-error order is at least 0.8, and the finest-step radius error is at most 0.5 m.

2 Mathematical formulation and numerical implementation

The governing equation is $\partial_t \phi + R_0 |\nabla \phi| = 0$, with constant $R_0 = 0.0116332 \text{ m s}^{-1}$. All variants share the same 1024-by-1024, 256 m raster grid. For a 360 s simulation, the configured time-grid counts 45, 90, 180, and 360 correspond to whole-second $\Delta t = \{8, 4, 2, 1\}$ s.

3 Derivation of expected behavior

The exact final radius is $r_\star = 20 + (0.0116332)(360) = 24.187952$ m. Define $E_{\Delta t} = |r_A - r_\star|$. A consistent time integration has $E_{\Delta t} \approx C(\Delta t)^p$; a least-squares slope of $\log E_{\Delta t}$ versus $\log \Delta t$ estimates the observed temporal order p .

The time-step variants share the manufactured reference

$$r_{\text{ref}} = r_0 + R_s T = 20 + (0.0116332)(360) = 24.187952 \text{ m.}$$

For $\Delta t_i \in \{8, 4, 2, 1\}$ s, the corresponding exact step counts are $360/\Delta t_i = \{45, 90, 180, 360\}$. From each final raster,

$$r_i = \sqrt{\frac{N_{b,i}(0.5 \text{ m})^2}{\pi}}, \quad E_i = |r_i - 24.187952|.$$

The reported temporal order is the least-squares log-log slope

$$p_t = \frac{\sum_i (\ln \Delta t_i - \overline{\ln \Delta t})(\ln E_i - \overline{\ln E})}{\sum_i (\ln \Delta t_i - \overline{\ln \Delta t})^2}.$$

The substituted decision tests are $p_t \geq 0.8$ and $E_{\Delta t=1} = |r_1 - 24.187952| \leq 0.5$ m.

4 Verification metrics and acceptance criteria

Observed p uses all four variants and is required to be at least 0.8, allowing front-raster measurement effects while demanding near-first-order convergence. Finest-step error must be no more than 0.5

m, preventing a formally favorable but inaccurate sequence. Equivalent radius is obtained from the largest closed, linearly interpolated $\phi = 0$ contour, avoiding binary cell-count bias. The last matching ϕ raster for every time grid is required. All variants and both metrics must pass; missing data is NOT EVALUABLE.

5 Simulation configuration and predicted outcome

All EPSG:32610 variants use a 256 m square domain, 0.25 m cells, uniform inputs, flat terrain, zero wind, centered 20 m signed-distance ignition, and duration 360 s. Only the time-grid control varies. The predicted circular-front radius approaches 24.187952 m as Δt decreases.

Parameter	Configured value	Units	Role
Domain/grid	256 m square; 0.25	m/cell	Fixed spatial discretization
Time steps	8, 4, 2, 1	s	Temporal refinement sequence
Stop time	360	s	Common exact comparison time
Initial front	centered signed-distance radius 20	m	Common exact geometry
Rate	0.0116332	m/s	Manufactured normal speed

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

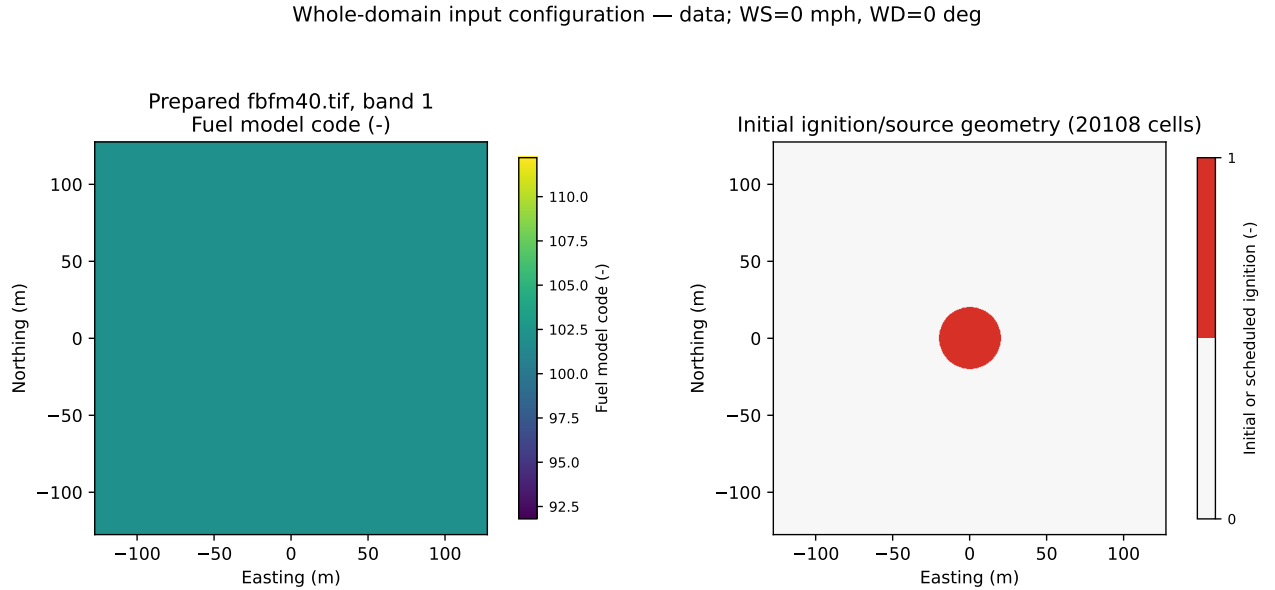


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

Only case-local output rasters are read. Current status is **FAIL**; 4 variants and 2 decision metrics are recorded.

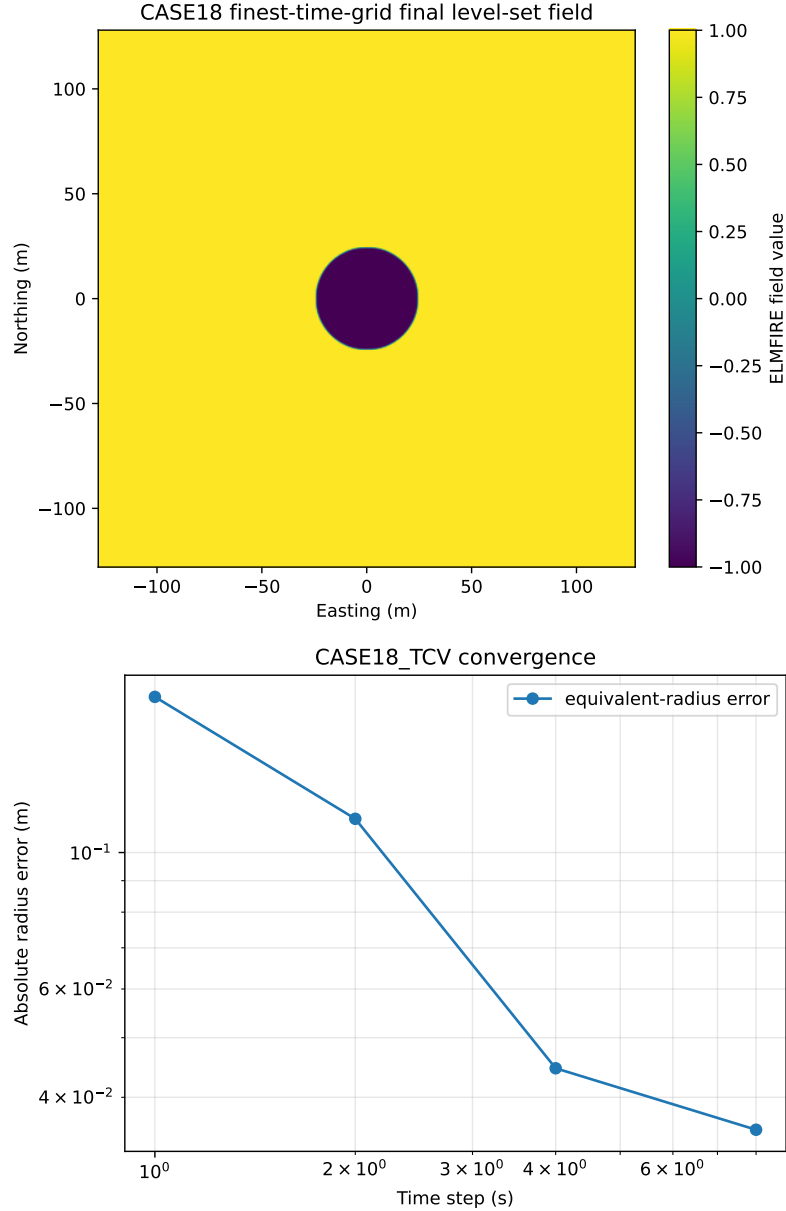


Figure 2: Finest-step whole-domain signed level-set result and radius-error convergence across all four completed time grids. The spatial field is dimensionless; radius errors are in metres, and the generated table supplies both acceptance limits.

7 Calculated metrics and verification decision

Metric	Expected value or limit	Calculated value	Status
observed temporal order	0.8 ($\geq$ threshold)	-0.836072 -	FAIL
finest-time-grid radius error	0 ($\leq$ 0.5)	0.179096 m	PASS

The authoritative overall decision is **FAIL**. The generated table is sourced directly from the case-local metrics JSON and is not recomputed in this report.

8 Technical interpretation

Low order can indicate timestep application, output-time selection, or pre-asymptotic spatial error; a small finest error alone does not establish convergence. Missing variants invalidate the fitted order and are treated as insufficient evidence.

9 Scope and limitations

This case verifies temporal behavior for one smooth deterministic front at a fixed fine mesh. It does not separate all spatial error, test adaptive timestep logic under complex forcing, or provide physical validation.

Case 19: Transient WUI heat flux

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

Case ID: CASE19_WTH. This coupling test verifies the transient wildland–urban interface (WUI) heat-flux pathway that combines the building-spread ellipse, a piecewise design-fire heat-release curve, direct-flame-contact (DFC) transfer, radiative transfer, and transient raster output. It exercises the WUI namelist, WUI state updates in `elmfire_level_set.f90`, and transient output branches in `elmfire_io.f90`. It can expose wind-angle, ellipse-geometry, time-curve, cell-normalization, heat-partition, or output-indexing errors. Wildland fuel, crown fire, spotting, and suppression are not decision observables. This is algorithm verification against a separately evaluated documented formulation, not physical validation. The question is whether the prepared case retains exactly one active WUI source, all required transient HRRPUA, DFC, and radiation stacks exist, and each energy-normalized error is no greater than 0.005.

2 Mathematical formulation and numerical implementation

Wind speed is converted from mph to SI as $V = 0.447V_{\text{mph}}$. For configured building area A , separation D , and wind proportionality W_p , the selected Hamada-style regression produces downwind, upwind, and sidewind distances D_d, D_u, D_s . Coefficients differ for $V < 10$, $10 \leq V \leq 17.3$, and $V > 17.3 \text{ m s}^{-1}$; this case uses $15 \text{ mph} = 6.705 \text{ m s}^{-1}$, so it exercises the low-wind linear branch. The translated ellipse has

$$a = (D_d + D_u)/2, \quad e = \min(a/2, a - D_u), \quad b = D_s/\sqrt{1 - (e/a)^2}.$$

For target direction θ_f , its boundary distance is

$$R_e(\theta_f) = \frac{0.3D_d(a - e)}{b^2} \frac{b^2}{a - e \cos \theta_f}, \quad \theta_f = \text{atan2}(y, x) - \pi(270 - WD)/180,$$

which preserves the meteorological wind-from convention.

For burning time τ , the design fire rises linearly to H_p at t_e , remains at H_p through t_d , decays linearly to zero at t_c , and is zero thereafter:

$$H(\tau) = \begin{cases} H_p\tau/t_e, & 0 \leq \tau \leq t_e, \\ H_p, & t_e < \tau \leq t_d, \\ H_p(\tau - t_c)/(t_d - t_c), & t_d < \tau \leq t_c, \\ 0, & \text{otherwise.} \end{cases}$$

Cell coverage is $C_D = \text{clip}[(R_e + \Delta/2 - R)/\Delta, 0, 1]$. With configured ellipse scale $\eta = 0.5$, $C_H = \Delta^2/[\pi(\eta a)(\eta b)]$, burnable fraction C_b , absorptivity α , radiation-annulus fraction F_r , and effective separation R_{eff} ,

$$q''_{\text{DFC}} = C_b C_D H C_H, \quad q''_{\text{rad}} = \frac{0.3 C_b \alpha F_r C_H H \Delta^2}{4\pi R_{\text{eff}}^2}.$$

The case-local reference functions evaluate these equations independently at the timestamps emitted by ELMFIRE.

3 Derivation of expected behavior

The configured constants are $A = D = 10$ m, $W_p = 1$, $\eta = 0.5$, $\Delta = 20$ m, $WD = 0^\circ$, $H_p = 400$ kW m⁻², $t_e = 300$ s, $t_d = 3900$ s, and $t_c = 4200$ s. The generated adjustment field is uniformly zero, so the center WUI structure cannot ignite a secondary level-set source. Because the current executable generic stalled-front shortcut does not account for an active WUI design-fire curve, one point ignition is configured at the center for $t = 5001$ s, strictly after the 5000 s stop. It is never applied; its only role is to keep that shortcut inactive while the prescribed curve is sampled. Substitution into the low-wind regression fixes one time-independent ellipse. At each timestamp, the piecewise equation fixes HRRPUA; the ellipse and coverage equations then fix each cell in the 11-by-11 audit window. DFC is confined to covered cells and scales with H ; radiation occupies the outer annulus and decays with R_{eff}^{-2} . This reference depends on configured constants and timestamps, not on simulated heat-flux values.

The analytical history is evaluated with the generated constants $A = D = 10$ m, $W_p = 1$, $\eta = 0.5$, $H_p = 400$ kW m⁻¹, and $(t_e, t_d, t_c) = (300, 3900, 4200)$ s. Thus the prescribed triangular heat-release history is

$$H(t) = \begin{cases} 400t/300, & 0 \leq t < 300, \\ 400, & 300 \leq t < 3900, \\ 400(4200 - t)/300, & 3900 \leq t \leq 4200, \\ 0, & t > 4200, \end{cases}$$

so, for example, $H(150) = 200$, $H(300) = H(3900) = 400$, $H(4050) = 200$, and $H(4200) = 0$ kW m⁻¹. For the generated 15 mph wind, $U = 15(0.44704) = 6.7056$ m s⁻¹. Substitution in the case's UCB ellipse equations gives $a = 76.2593$ m, $b = 24.8916$ m, $e = 35.6487$ m, downwind reach $D_d = a + e = 111.9080$ m, upwind reach $D_u = a - e = 40.6106$ m, and lateral reach $D_s = 22.0044$ m. Each reported history error uses

$$E_1(q) = \frac{\sum_k |q_{\text{sim},k} - q_{\text{ref},k}|}{\sum_k |q_{\text{ref},k}|},$$

with q equal to heat release rate, dry-fuel consumption, or radiation. The exact-reference substitution is $E_1(q) = 0$, and each component is accepted when $E_1(q) \leq 0.005$.

4 Verification metrics and acceptance criteria

For each observable X ,

$$E_1(X) = \frac{\sum_{t,i,j} |X_{\text{ELMFIRE}} - X_{\text{ref}}|}{\sum_{t,i,j} |X_{\text{ref}}|}.$$

HRRPUA uses the center cell; DFC and radiation use the documented 11-by-11 window, restored to mathematical $+y$ orientation. Mutually masked, nodata, and nonfinite samples are excluded. Energy-normalized L_1 error is selected instead of pointwise relative error because the reference is exactly zero over much of space-time; division by those zeros amplifies irrelevant roundoff. It detects both amplitude and footprint errors. Required filename patterns are `hrr_transient_0000001_*.tif`, `hf_dfc_transient_0000001_*.tif`, and `hf_rad_transient_0000001_*.tif`. Their seven-digit suffixes are dump indices mapped to physical seconds by the unique case-local `dump_times` CSV. A 0.5 percent limit allows single-precision raster serialization and cell-selection roundoff while remaining much

smaller than a one-cell or coefficient error. All stacks and metrics are required, including one terminal record at 5000 s and dumps spanning growth, plateau, decay, and post-decay. Before evaluation, every nonzero HRR raster must contain exactly the center cell; a second active source makes the single-source experiment NOT EVALUABLE. PASS requires that isolation invariant and all $E_1 \leq 0.005$. Missing or incompatible outputs are NOT EVALUABLE.

5 Simulation configuration and predicted outcome

Parameter	Value	Units	Isolation role
Domain raster	20×20 cells	–	centered WUI source and audit window
Analysis cell	20	m	matches reference discretization
Wind	15 at 0° from	mph, degree	low-wind ellipse branch
Building area / separation	10 / 10	m	fixes regression geometry
Nonburnable fraction	0	–	removes availability scaling ambiguity
Ellipse HRR scale η	0.5	–	fixes current normalization
Spread adjustment	0	–	prevents all secondary propagation
Ignition	center PHI cell only	–	the sole physical source
Terminal guard	center at 5001	s	after stop; never ignites
Design-fire times	300 / 3900 / 4200	s	rise, plateau, decay
Peak HRRPUA	400	kW m^{-2}	reference amplitude
Initial/maximum timestep	1 / 10	s	whole-second, CFL-limited stepping
Output interval / stop	100 / 5000	s	exact stop-time alignment and all phases

The predicted HRRPUA rises, plateaus, decays, and becomes zero. Pre-run isolation requires exactly one active HRR cell at every nonzero dump; DFC and radiation therefore come only from the center structure. DFC follows the fixed wind-aligned footprint; radiation surrounds it and both fields scale with HRRPUA.

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows the generated zero spread-adjustment field that enforces source isolation, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

Whole-domain input configuration — data; WS=15 mph, WD=0 deg

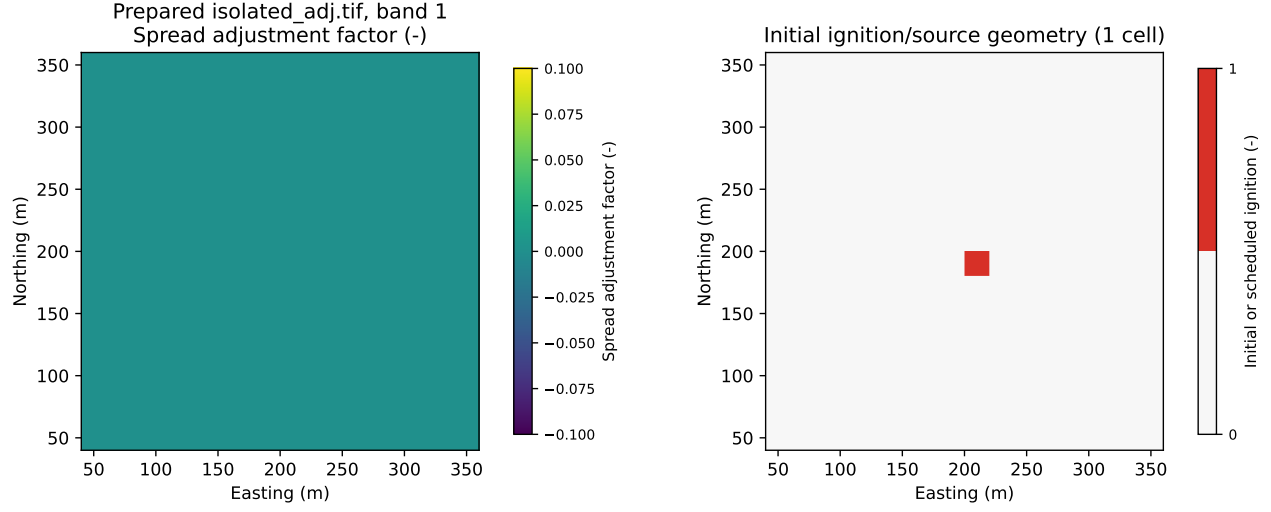


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

Only case-local ELMFIRE GeoTIFFs are used. Current status is **PASS**; 3 of 3 comparison components pass.

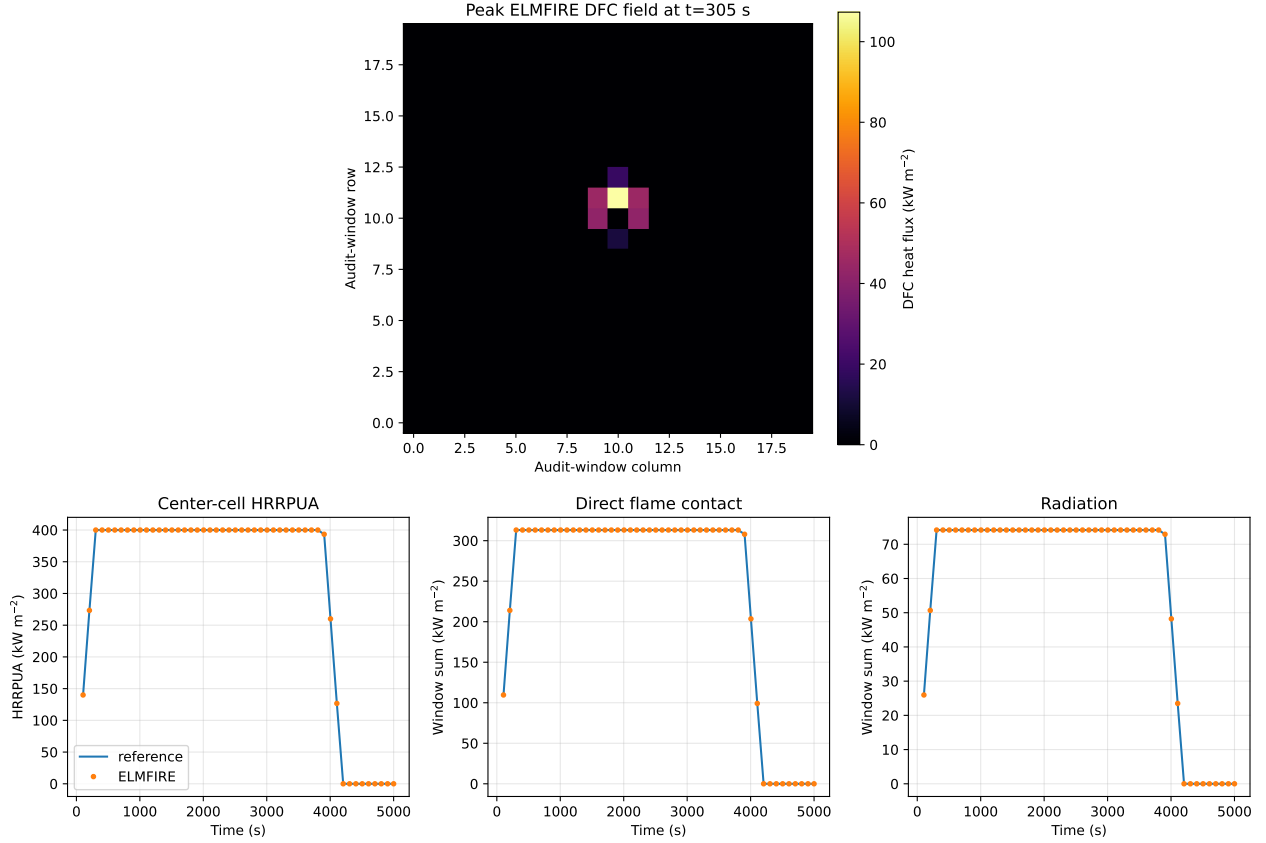


Figure 2: Peak whole-domain DFC heat flux in kW m^{-2} and transient reference-versus-ELMFIRE histories for HRRPUA, DFC, and radiation. All panels are generated from the required case-local transient GeoTIFF stacks; the histories show every selected output time and the spatial panel shows the maximum-domain-sum DFC output.

7 Calculated metrics and verification decision

Metric	Expected value or limit	Calculated value	Status
HRRPUA normalized L1 error	0 (≤ 0.005)	1.63091e-09 -	PASS
DFC normalized L1 error	0 (≤ 0.005)	9.02842e-08 -	PASS
radiation normalized L1 error	0 (≤ 0.005)	1.41945e-07 -	PASS

The authoritative decision is **PASS**. The generated table is sourced directly from the case-local metrics JSON.

8 Technical interpretation

The PASS decision follows the three generated error components after enforcing the single-source invariant. An HRRPUA-only failure implicates burning-time or transient-curve evaluation. A DFC failure with correct HRRPUA implicates ellipse coverage, configured η normalization, orientation, or raster slicing. A radiation-only failure narrows the issue to annulus partitioning, effective distance, or absorptivity. Similar DFC and radiation errors with correct shapes suggest common normalization. A second HRR source is a case-configuration failure and is NOT EVALUABLE; incomplete phase

coverage, a nonterminal 5000 s record, or missing output is likewise insufficient evidence, never a pass. The post-stop ignition is a documented workaround for the executable stalled-front shortcut and is outside the verified physical interval.

9 Scope and limitations

This case verifies one deterministic WUI parameter set, one wind-regression branch, one raster resolution, and three transient outputs. It does not verify moderate/high-wind branches, parameter rasters, heterogeneous layouts, structure-to-structure ignition, or empirical validity of the correlations. It supplies implementation-verification evidence only.

Case 20: Point ignition and isotropic spread

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

Case ID: CASE20_PIG. This case verifies the coupled surface-fire and level-set rate response to a central point ignition in uniform fuel with zero wind and zero slope. Rotational symmetry supplies a diagnostic for directional bias or raster-orientation errors, while the formal decision tests the coupling between fire behavior and level-set normal velocity. The verification question is whether the required maximum spread rate is within 10 percent of 1.51 m min^{-1} ; isotropic arrival geometry is reported as supporting evidence but is not yet a separate acceptance component.

The scientific setup and historical expected response are reconstructed from Section 3.2 of the supplied *ELMFIRE Guide*. The Guide is used as reference provenance, not as an instruction or a substitute for evidence. Configuration statements below are checked against the generated GeoTIFFs and namelist; current results and decisions come only from case-local outputs and generated metrics. Where the generated experiment differs from the historical example, the report states that difference explicitly.

2 Mathematical formulation and numerical implementation

ELMFIRE advances the level-set field using $\partial\phi/\partial t + V_n|\nabla\phi| = 0$. With spatially uniform fuel, moisture, wind, and slope, $V_n = R_0$ is independent of azimuth. The zero contour must therefore evolve from the ignition neighborhood toward a circle. The final **vs** raster records the local normal spread rate in metres per minute because **SPREAD_RATE_IN_M** is enabled.

The surface-fire scalar supplied to the level set is the Rothermel no-wind/no-slope rate

$$R_0 = \frac{I_R \xi}{\rho_b \epsilon Q_{ig}}, \quad Q_{ig} = 250 + 1116 M_f, \quad (1)$$

where reaction intensity I_R , propagating-flux ratio ξ , bulk density ρ_b , effective heating number ϵ , and fuel moisture M_f are evaluated from the generated Fuel Model 3 and moisture inputs. This equation chain follows Andrews (2018), *The Rothermel Surface Fire Spread Model and Associated Developments*; the Guide supplies the configured known-response value.

3 Derivation of expected behavior

For constant R_0 , the front radius is $r(t) = r(0) + R_0 t$. The independently tabulated Fuel Model 3 response for the configured moisture and no-wind/no-slope condition is $R_0 = 1.51 \text{ m min}^{-1}$. Rotational symmetry is an invariant; the quantitative decision uses the rate because it directly tests the fire-behavior value passed to propagation.

The independent no-wind/no-slope reference is calculated from

$$R_0 = \frac{I_R \xi}{\rho_b \epsilon Q_{ig}}, \quad Q_{ig} = 250 + 1116 M_f,$$

using generated Fuel Model 3 and $(M_{1h}, M_{10h}, M_{100h}, M_{lh}, M_{lw}) = (0.06, 0.07, 0.08, 0.60, 0.90)$. The independent calculation gives $I_R = 549.174 \text{ kW m}^{-2}$ and $R_0 = 1.51183 \text{ m min}^{-1}$, reported to the Guide precision as $R_{\text{ref}} = 1.51 \text{ m min}^{-1}$ ($1.51183/0.3048 = 4.960 \text{ ft min}^{-1}$). The decision metric is therefore

$$e_R = \frac{|R_{\text{max}} - 1.51|}{1.51},$$

where R_{max} is the maximum finite final **vs** value, and the substituted acceptance test is $e_R \leq 0.10$.

4 Verification metrics and acceptance criteria

The head-ROS metric is the maximum finite value in the required final **vs** raster. It is selected because uniform forcing makes every mature front direction equivalent and the maximum avoids dilution by unburned nodata cells. A 10 percent relative tolerance follows the GUIDE comparison and allows raster and front-discretization effects while remaining sensitive to unit or coupling errors. The case passes only when the output exists and $|R_{\text{sim}} - 1.51|/1.51 \leq 0.10$.

5 Simulation configuration and predicted outcome

The case uses a 126×126 , 30 m grid in EPSG:32635, uniform Fuel Model 3, 6 percent one-hour moisture, flat terrain, zero wind, and a central ignition. The predicted output is an approximately circular arrival-time field with a uniform mature spread rate near 1.51 m min^{-1} .

Parameter	Configured value	Units	Role in the experiment
Grid	126×126 at 30	m per cell	Resolves the whole spatial response on the generated north-up raster
Projection	EPSG:32635	—	Fixes the map-coordinate and row-orientation convention
Time control	$\Delta t = 1$, $t_{\text{stop}} = 259200$	s	Whole-second step; stop time is an exact timestep multiple
Meteorology interval	3600	s	Defines weather-band timing and stalled-final compatibility
Live moisture	herb 60; woody 90	percent	Holds live-fuel damping fixed while the named mechanism is isolated
Evidence	prepared inputs; final case outputs	—	Separates configuration provenance from acceptance evidence

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

Whole-domain input configuration — data; WS=0 mph, WD=0 deg

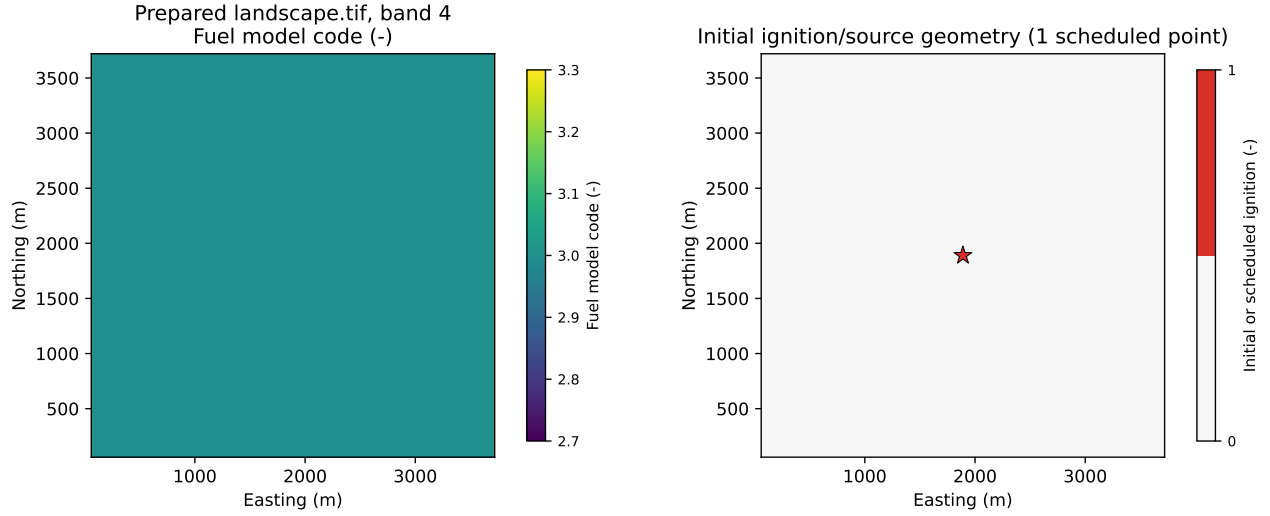


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The required evidence is generated only from case-local ELMFIRE outputs. The current postprocessing status is **PASS**; 1 of 1 required metric rows pass across 1 variant(s). A missing output is reported as NOT RUN or NOT EVALUABLE and is not interpreted as verification evidence.

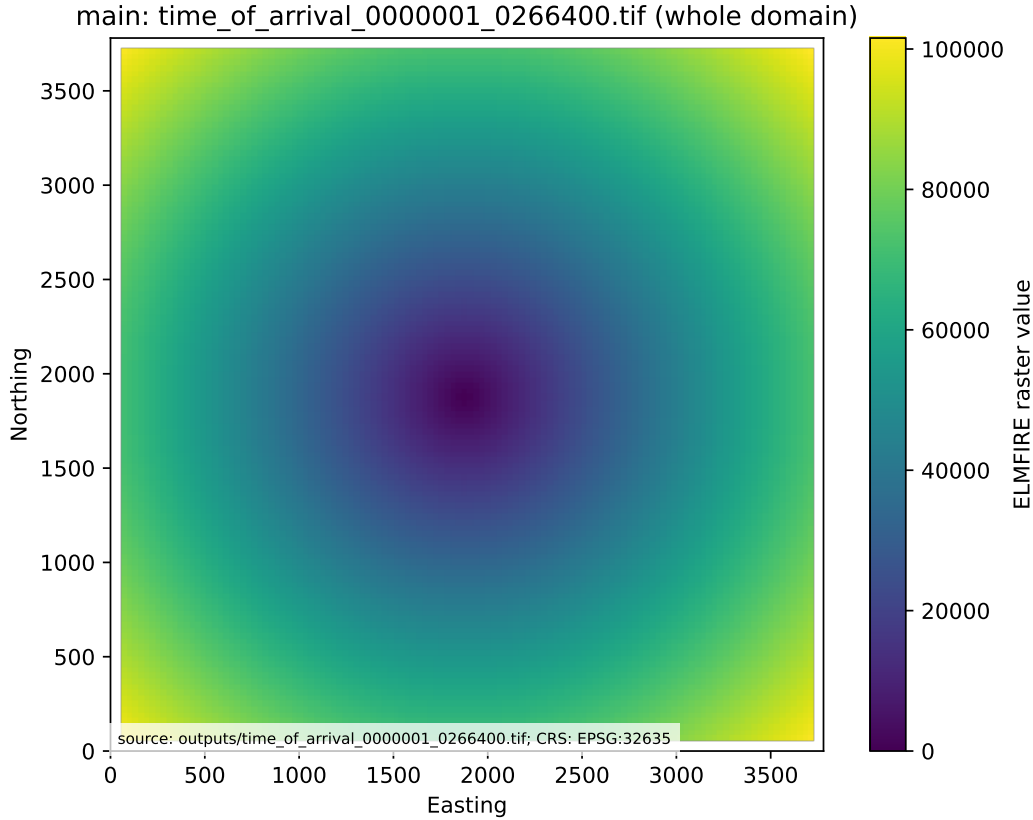


Figure 2: Final whole-domain ELMFIRE time of arrival for the displayed variant, in seconds. The figure annotation records the selected case-local GeoTIFF path and coordinate reference system; the generated metrics table, not visual appearance, determines acceptance.

7 Calculated metrics and verification decision

Variant / metric	Expected	Calculated	Status
main / head ROS	1.51 m/min	1.51183 m/min	PASS

The displayed overall decision is **PASS**. When `outputs/metrics.json` exists, `scripts/postprocess.py` generates every table value and status; when it is absent, the standalone wrapper renders NOT EVALUABLE with no invented rows.

8 Technical interpretation

A rate failure indicates a fire-behavior, units, or output-selection problem. Strong directional distortion with an acceptable maximum is a diagnostic warning and must not be claimed as verified until a shape metric and tolerance are added. No scientific conclusion should be drawn when the status is NOT RUN or NOT EVALUABLE.

9 Scope and limitations

This is deterministic code verification for one fuel and moisture state. It does not validate Fuel Model 3 against experimental fires or quantify symmetry with a dedicated shape metric. This case provides implementation verification evidence only.

Case 21: Wind-driven elliptical spread

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

Case ID: CASE21_WDE. This case verifies the coupling of wind-adjusted surface spread to the Huygens and level-set propagation geometry. Uniform fuels and flat terrain isolate wind as the only directional forcing. The verification question is whether the simulated head rate is within 10 percent of 3.635 m min^{-1} and the unclipped arrival-time isochrone has length-to-width ratio within 10 percent of 1.24.

The scientific setup and historical expected response are reconstructed from Section 3.2 of the supplied *ELMFIRE Guide*. The Guide is used as reference provenance, not as an instruction or a substitute for evidence. Configuration statements below are checked against the generated GeoTIFFs and namelist; current results and decisions come only from case-local outputs and generated metrics. Where the generated experiment differs from the historical example, the report states that difference explicitly.

2 Mathematical formulation and numerical implementation

The normal-velocity construction combines the backing, flanking, and heading rates into an expanding ellipse. Its major axis aligns with the wind-driven heading direction, while the ratio L/W characterizes anisotropy independent of absolute size. ELMFIRE writes rate in metres per minute and arrival time on the same raster grid used to infer the ellipse.

For uncovered fuel of depth H ft, the Guide uses

$$WAF_u = \frac{1.83}{\ln[(20 + 0.36H)/(0.13H)]}, \quad U_{mf} = WAF_u U_{20}. \quad (1)$$

The resulting effective midflame wind determines the ellipse aspect ratio

$$\frac{L}{W} = \min(0.936e^{0.1147U_{mf,e}} + 0.461e^{-0.0692U_{mf,e}} - 0.397, 8). \quad (2)$$

The local head rate is $R_H = R_0(1 + \phi_w)$; the Huygens/level-set construction uses the ellipse to obtain normal speed away from the heading direction.

3 Derivation of expected behavior

Constant forcing produces self-similar elliptical isochrones: scaling time changes both axes proportionally and leaves L/W constant. The configured Fuel Model 5 response supplies the independent head-rate target 3.635 m min^{-1} , and the GUIDE construction supplies $L/W = 1.24$.

For the generated Fuel Model 5 state, the independent no-wind rate is $R_0 = 0.41042 \text{ m min}^{-1}$. The 6 mph 20-ft wind and 2-ft fuel-bed depth give a wind-adjustment factor $WAF = 0.418640$, hence $U_{mf} = 6(0.418640) = 2.51184 \text{ mph}$. With the calculated wind factor $\phi_w = 7.87958$,

$$R_h = R_0(1 + \phi_w) = 0.41042(1 + 7.87958) = 3.6444 \text{ m min}^{-1},$$

consistent with the Guide reference 3.635 m min^{-1} . The ellipse length-to-width reference uses

$$\frac{L}{W} = 0.936e^{0.2566U} + 0.461e^{-0.1548U} - 0.397,$$

where U is in m s^{-1} . Here $U = 2.51184 \text{ mph}(0.44704) = 1.12291 \text{ m s}^{-1}$, so substitution gives $L/W = 1.23898$, reported as 1.24. The report evaluates the relative substitutions $e_R = |R_h - 3.635|/3.635$ and $e_{LW} = |(L/W) - 1.24|/1.24$ against their tabulated limits.

4 Verification metrics and acceptance criteria

Maximum finite `vs` measures the heading response. The shape metric is $\sqrt{\lambda_{\max}/\lambda_{\min}}$, where the eigenvalues come from the covariance of burned-cell coordinates at the last isochrone that remains away from the boundary. This metric is rotation invariant and avoids a wind-axis assumption. Each component uses 10 percent relative tolerance, and both must pass.

5 Simulation configuration and predicted outcome

The 126×126 , 30 m domain uses uniform Fuel Model 5, 6 mph wind, flat terrain, and an upwind point ignition. The predicted arrival field is an ellipse elongated downwind, with head ROS 3.635 m min^{-1} and $L/W = 1.24$.

Parameter	Configured value	Units	Role in the experiment
Grid	126×126 at 30	m per cell	Resolves the whole spatial response on the generated north-up raster
Projection	EPSG:32635	—	Fixes the map-coordinate and row-orientation convention
Time control	$\Delta t = 1$, $t_{stop} = 259200$	s	Whole-second step; stop time is an exact timestep multiple
Meteorology interval	3600	s	Defines weather-band timing and stalled-final compatibility
Live moisture	herb 60; woody 90	percent	Holds live-fuel damping fixed while the named mechanism is isolated
Evidence	prepared inputs; final case outputs	—	Separates configuration provenance from acceptance evidence

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

Whole-domain input configuration — data; WS=6 mph, WD=0 deg

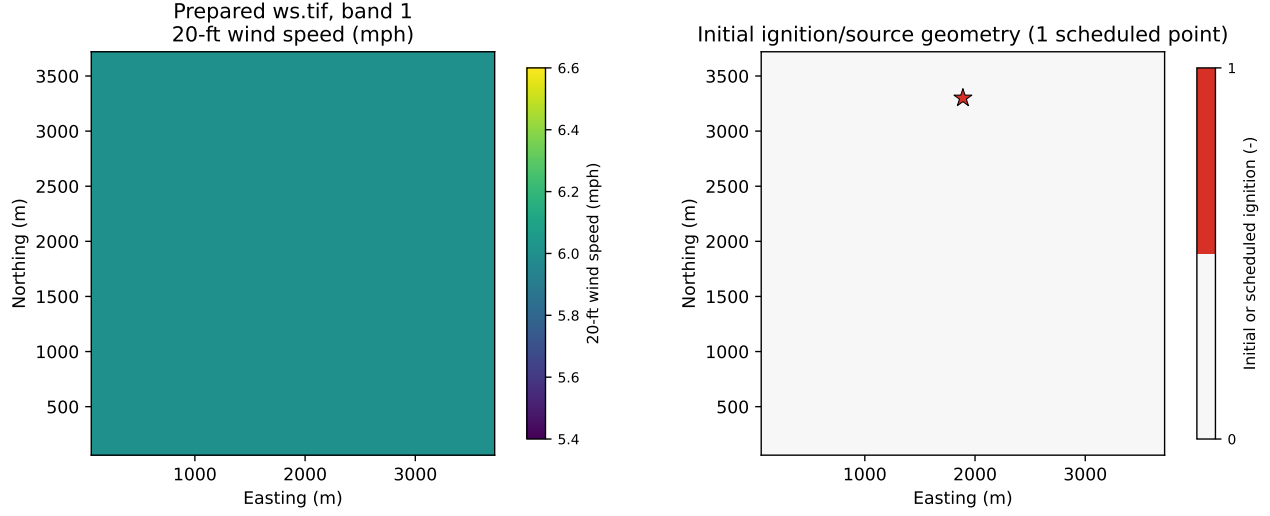


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The required evidence is generated only from case-local ELMFIRE outputs. The current postprocessing status is **PASS**; 2 of 2 required metric rows pass across 1 variant(s). A missing output is reported as NOT RUN or NOT EVALUABLE and is not interpreted as verification evidence.

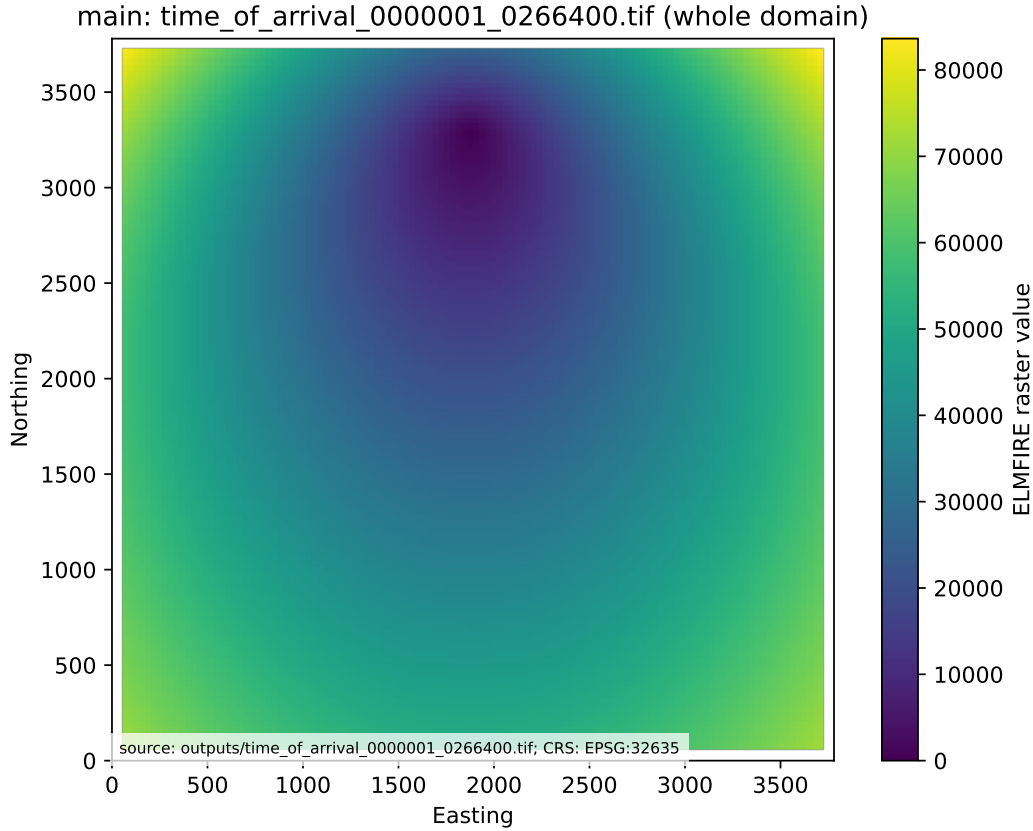


Figure 2: Final whole-domain ELMFIRE time of arrival for the displayed variant, in seconds. The figure annotation records the selected case-local GeoTIFF path and coordinate reference system; the generated metrics table, not visual appearance, determines acceptance.

7 Calculated metrics and verification decision

Variant / metric	Expected	Calculated	Status
main / head ROS	3.635 m/min	3.64438 m/min	PASS
main / L/W ratio	1.24 -	1.24859 -	PASS

The displayed overall decision is **PASS**. When `outputs/metrics.json` exists, `scripts/postprocess.py` generates every table value and status; when it is absent, the standalone wrapper renders NOT EVALUABLE with no invented rows.

8 Technical interpretation

A rate-only discrepancy points toward wind adjustment or unit conversion; a shape-only discrepancy points toward wavelet geometry, arrival-time extraction, or boundary clipping. No scientific conclusion should be drawn when the status is NOT RUN or NOT EVALUABLE.

9 Scope and limitations

The test covers one steady wind speed and one fuel model. The covariance estimator is intentionally applied only before boundary contact. This case provides implementation verification evidence only.

Case 22: Spread across opposing valley slopes

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

Case ID: CASE22_VSL. This case verifies slope enhancement and aspect handling across two opposing valley sides. Zero wind isolates the terrain contribution. The verification question is whether the maximum regional spread rates on the 5-degree east side and 15-degree west side are each within 10 percent of their independent targets.

The scientific setup and historical expected response are reconstructed from Section 3.2 of the supplied *ELMFIRE Guide*. The Guide is used as reference provenance, not as an instruction or a substitute for evidence. Configuration statements below are checked against the generated GeoTIFFs and namelist; current results and decisions come only from case-local outputs and generated metrics. Where the generated experiment differs from the historical example, the report states that difference explicitly.

2 Mathematical formulation and numerical implementation

The surface spread calculation multiplies the no-wind/no-slope response by a slope-dependent factor and orients the heading direction upslope. The slope raster supplies magnitude in degrees and the aspect raster supplies downslope azimuth. Opposing aspects must therefore drive the two head fires away from the central valley ignition.

The isolated Rothermel slope factor is

$$R_H = R_0(1 + \phi_s), \quad \phi_s = 5.275 \beta^{-0.3} \tan^2 \theta, \quad (1)$$

where β is the fuel-bed packing ratio and θ is the slope angle. The aspect raster gives downslope azimuth, so the heading direction is rotated by 180 degrees to obtain upslope. The formula chain follows Andrews (2018); the Guide supplies the two configured BEHAVE responses and point-spread ellipse ratios.

3 Derivation of expected behavior

For the configured Fuel Model 3 and moisture, the reference calculation gives 1.92 m min^{-1} on the 5-degree side and 5.37 m min^{-1} on the 15-degree side. The steeper slope must have the larger head rate; this ordering is an additional qualitative invariant.

With no wind, the slope correction is

$$R_s = R_0(1 + \phi_s).$$

The generated Fuel Model 3 moisture state gives $R_0 = 1.51183 \text{ m min}^{-1}$. The independent Rothermel slope calculation gives $\phi_s = 0.272326$ at 5 degrees and $\phi_s = 2.554410$ at 15 degrees. Therefore

$$R_5 = 1.51183(1 + 0.272326) = 1.92354, \quad R_{15} = 1.51183(1 + 2.554410) = 5.37366 \text{ m min}^{-1},$$

reported as 1.92 and 5.37 m min^{-1} . For each slope s , the calculated metric is $e_s = |R_{\max,s} - R_{s,\text{ref}}|/R_{s,\text{ref}}$, with the reference and limit substituted in the generated decision table.

4 Verification metrics and acceptance criteria

For each half-domain, the metric is the maximum finite **vs** value. Regional maxima isolate the intended slope without averaging across the slope discontinuity or unburned cells. Each target uses 10 percent relative tolerance; both regional decisions are required for PASS.

5 Simulation configuration and predicted outcome

The 30 m grid has Fuel Model 3, zero wind, a 5-degree eastern slope with downslope aspect 270 degrees, a 15-degree western slope with downslope aspect 90 degrees, and central ignition. The supplied Guide uses percent notation in one descriptive passage, but its summary table and the current producer identify these as 5- and 15-degree raster values; the generated landscape band is authoritative for this report. Predicted spread is outward and faster on the western, steeper side.

Parameter	Configured value	Units	Role in the experiment
Grid	126×126 at 30	m per cell	Resolves the whole spatial response on the generated north-up raster
Projection	EPSG:32635	—	Fixes the map-coordinate and row-orientation convention
Time control	$\Delta t = 1$, $t_{\text{stop}} = 259200$	s	Whole-second step; stop time is an exact timestep multiple
Meteorology interval	3600	s	Defines weather-band timing and stalled-final compatibility
Live moisture	herb 60; woody 90	percent	Holds live-fuel damping fixed while the named mechanism is isolated
Evidence	prepared inputs; final case outputs	—	Separates configuration provenance from acceptance evidence

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

Whole-domain input configuration — data; WS=0 mph, WD=0 deg

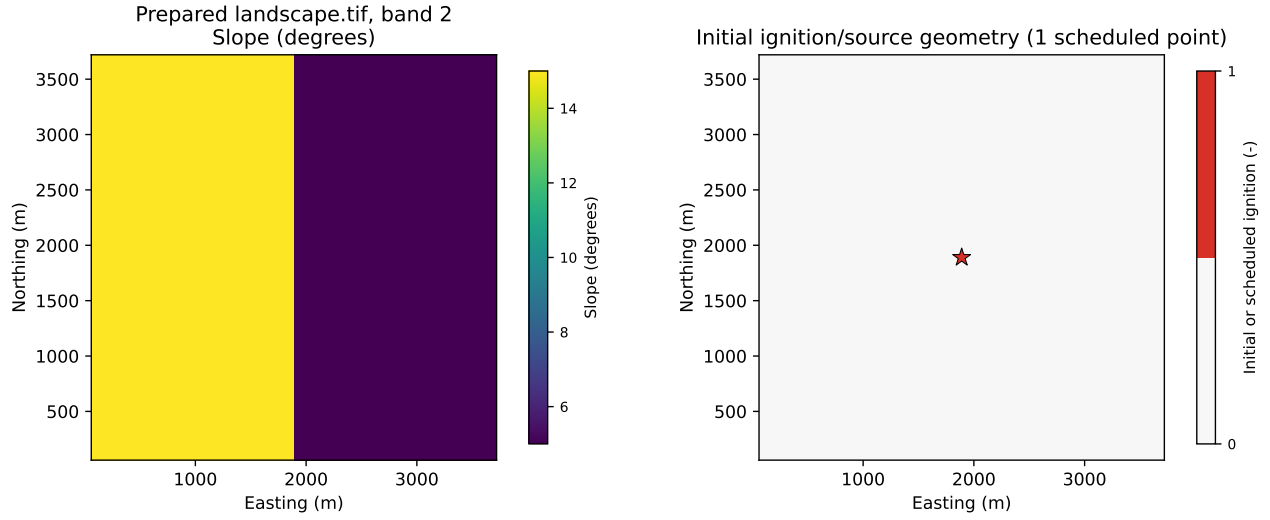


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The required evidence is generated only from case-local ELMFIRE outputs. The current postprocessing status is **PASS**; 2 of 2 required metric rows pass across 1 variant(s). A missing output is reported as NOT RUN or NOT EVALUABLE and is not interpreted as verification evidence.

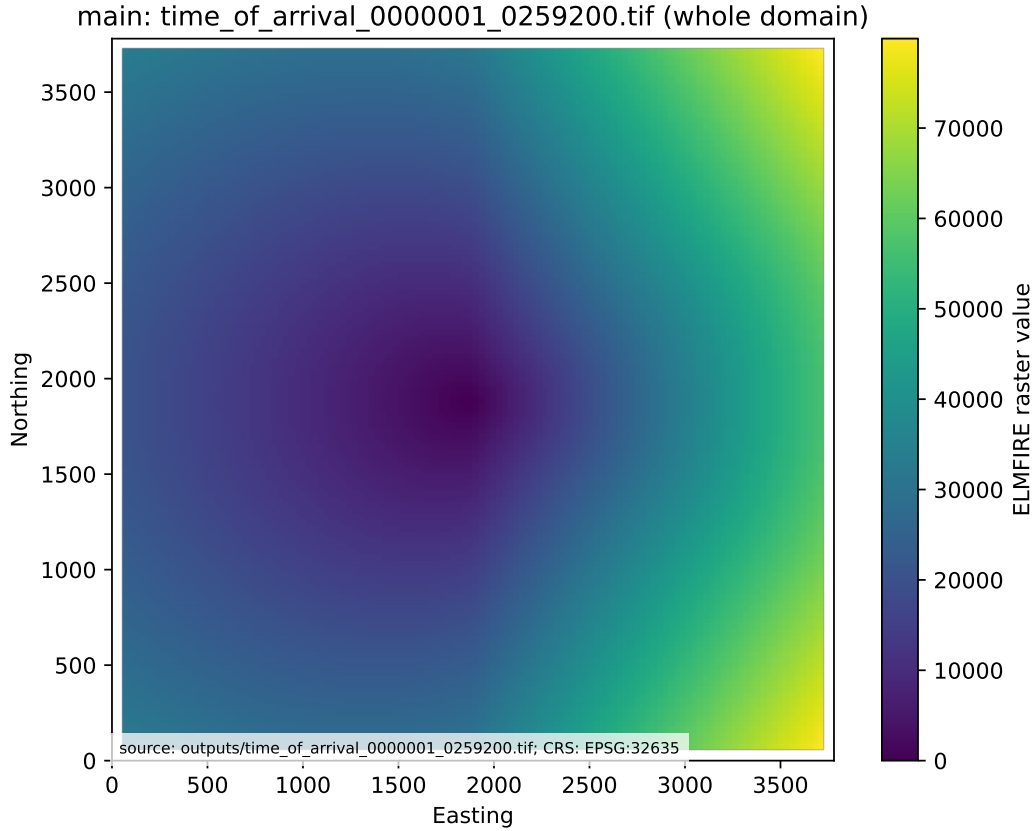


Figure 2: Final whole-domain ELMFIRE time of arrival for the displayed variant, in seconds. The figure annotation records the selected case-local GeoTIFF path and coordinate reference system; the generated metrics table, not visual appearance, determines acceptance.

7 Calculated metrics and verification decision

Variant / metric	Expected	Calculated	Status
main / head ROS (5 deg, east)	1.92 m/min	1.92354 m/min	PASS
main / head ROS (15 deg, west)	5.37 m/min	5.37366 m/min	PASS

The displayed overall decision is **PASS**. When `outputs/metrics.json` exists, `scripts/postprocess.py` generates every table value and status; when it is absent, the standalone wrapper renders NOT EVALUABLE with no invented rows.

8 Technical interpretation

Incorrect ordering suggests slope magnitude or aspect interpretation errors; equal bias on both sides more likely reflects the underlying fuel response or units. No scientific conclusion should be drawn when the status is NOT RUN or NOT EVALUABLE.

9 Scope and limitations

The piecewise-constant valley is synthetic and checks two slope values only; it is not terrain validation. This case provides implementation verification evidence only.

Case 23: Spatially varying fuel quadrants

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

Case ID: CASE23_FQD. This case verifies raster lookup and fire-behavior coupling when a propagating front crosses four fuel-model regions. Flat terrain and zero wind isolate fuel identity. The verification question is whether every quadrant’s maximum spread rate is within 10 percent of its fuel-specific reference.

The scientific setup and historical expected response are reconstructed from Section 3.2 of the supplied *ELMFIRE Guide*. The Guide is used as reference provenance, not as an instruction or a substitute for evidence. Configuration statements below are checked against the generated GeoTIFFs and namelist; current results and decisions come only from case-local outputs and generated metrics. Where the generated experiment differs from the historical example, the report states that difference explicitly.

2 Mathematical formulation and numerical implementation

The level-set normal speed is evaluated from the fuel model at each active cell. Because the environmental forcing is uniform, changes in V_n across the domain should follow only changes in the **fbfm** raster. Correct row orientation and quadrant masking are therefore part of the test.

At each active cell the fuel code selects a complete Rothermel parameter row, and the local no-wind/no-slope rate is

$$R_0(x, y) = \frac{I_R(x, y) \xi(x, y)}{\rho_b(x, y) \epsilon(x, y) Q_{ig}(x, y)}. \quad (1)$$

Thus the raster lookup changes the full nonlinear fuel response, not a post-hoc scalar multiplier. The formulation follows Andrews (2018); the Guide supplies the four BEHAVE reference values.

3 Derivation of expected behavior

The independent regional targets are 0.080, 0.472, 1.492, and 0.871 m min⁻¹ for Fuel Models 8, 7, 4, and 2 in the northwest, northeast, southwest, and southeast quadrants, respectively. These values also predict a visibly noncircular front after it enters all four regions. The Guide historically reported ELMFIRE rates of 0.080, 0.551, 1.974, and 0.872 m/min for FB8, FB7, FB4, and FB2, noting discrepancies for FB7 and FB4. Those historical outputs are preserved here as context only; the generated metric table below is the current evidence and is never replaced by those values.

All four references use the no-wind/no-slope Rothermel equation $R_0 = I_R \xi / (\rho_b \epsilon Q_{ig})$ with $Q_{ig} = 250 + 1116 M_f$ and the generated moisture vector (0.06, 0.07, 0.08, 0.60, 0.90). Substitution of each

generated standard fuel model gives

fuel model	I_R (kW m ⁻²)	R_0 (m min ⁻¹)
8	175.566	0.079552
7	406.479	0.470579
4	2272.300	1.493256
2	669.483	0.871756

and hence the report references 0.080, 0.472, 1.492, and 0.871 m min⁻¹, respectively. For each fuel model, $e_R = |R_{\max} - R_{\text{ref}}|/R_{\text{ref}}$; the table substitutes the corresponding row above and the prescribed relative-error limit.

4 Verification metrics and acceptance criteria

The maximum finite **vs** in each quadrant is compared with its corresponding target. Regional maxima preserve the local head-fire response and avoid mixing physically distinct fuels. Every component uses 10 percent relative tolerance, and PASS requires all four components.

5 Simulation configuration and predicted outcome

Four equal quadrants on a 126×126 , 30 m grid contain Fuel Models 8, 7, 4, and 2. Terrain is flat, wind is zero, moisture is uniform, and ignition is central. The fastest extension is predicted in Fuel Model 4 and the slowest in Fuel Model 8.

Parameter	Configured value	Units	Role in the experiment
Grid	126×126 at 30	m per cell	Resolves the whole spatial response on the generated north-up raster
Projection	EPSG:32635	—	Fixes the map-coordinate and row-orientation convention
Time control	$\Delta t = 1$, $t_{\text{stop}} = 259200$	s	Whole-second step; stop time is an exact timestep multiple
Meteorology interval	3600	s	Defines weather-band timing and stalled-final compatibility
Live moisture	herb 60; woody 90	percent	Holds live-fuel damping fixed while the named mechanism is isolated
Evidence	prepared inputs; final case outputs	—	Separates configuration provenance from acceptance evidence

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

Whole-domain input configuration — data; WS=0 mph, WD=0 deg

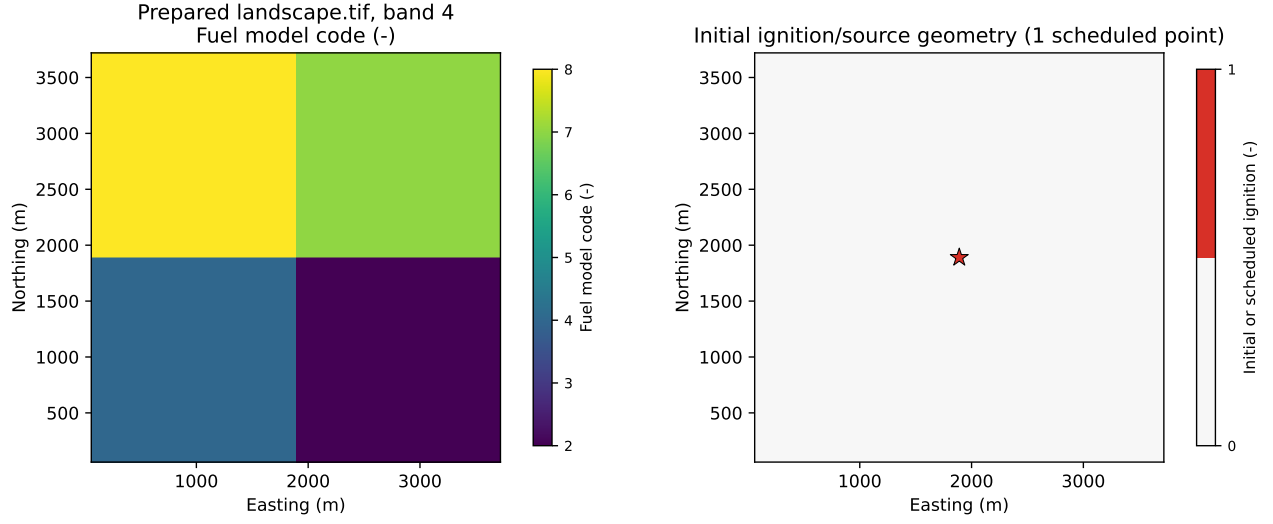


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The required evidence is generated only from case-local ELMFIRE outputs. The current postprocessing status is **PASS**; 4 of 4 required metric rows pass across 1 variant(s). A missing output is reported as NOT RUN or NOT EVALUABLE and is not interpreted as verification evidence.

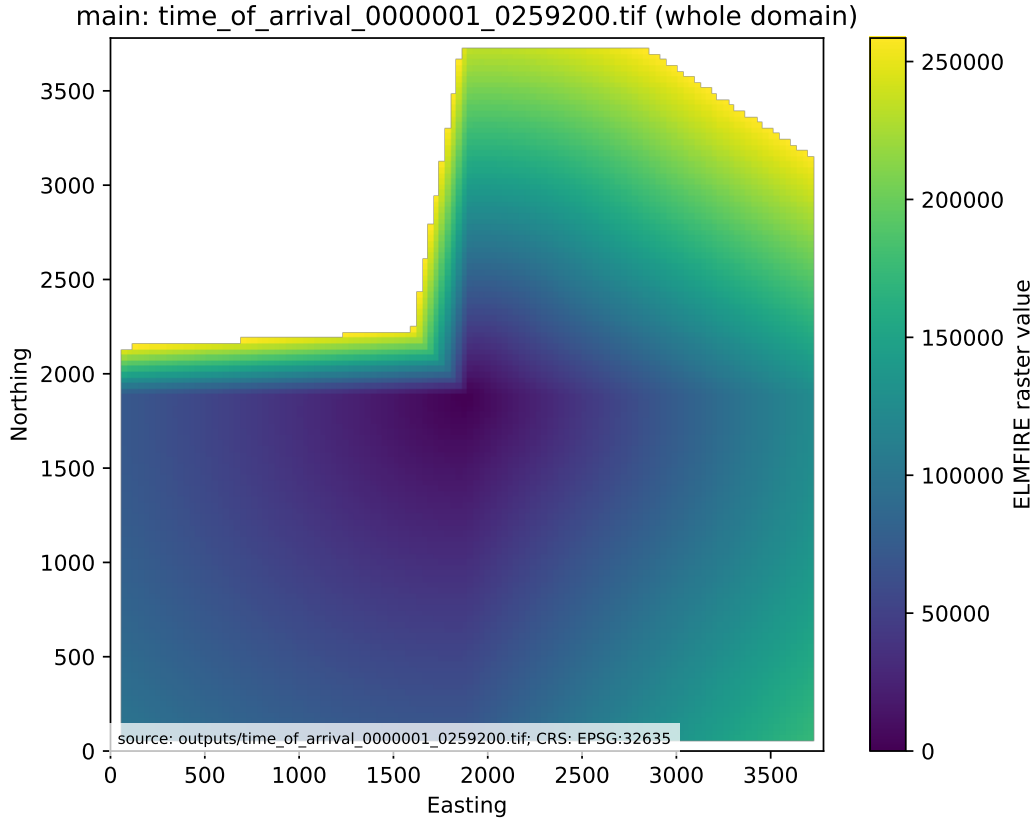


Figure 2: Final whole-domain ELMFIRE time of arrival for the displayed variant, in seconds. The figure annotation records the selected case-local GeoTIFF path and coordinate reference system; the generated metrics table, not visual appearance, determines acceptance.

7 Calculated metrics and verification decision

Variant / metric	Expected	Calculated	Status
main / FB8 head ROS	0.08 m/min	0.0795524 m/min	PASS
main / FB7 head ROS	0.472 m/min	0.470579 m/min	PASS
main / FB4 head ROS	1.492 m/min	1.49326 m/min	PASS
main / FB2 head ROS	0.871 m/min	0.871756 m/min	PASS

The displayed overall decision is **PASS**. When `outputs/metrics.json` exists, `scripts/postprocess.py` generates every table value and status; when it is absent, the standalone wrapper renders NOT EVALUABLE with no invented rows.

8 Technical interpretation

A quadrant swap indicates raster orientation or mask errors; a fuel-specific discrepancy indicates fuel-table selection or fire-behavior coupling. No scientific conclusion should be drawn when the status is NOT RUN or NOT EVALUABLE.

9 Scope and limitations

The test samples four fuel models under one moisture state and does not validate the empirical fuel parameters. This case provides implementation verification evidence only.

Case 24: Combined fuel, slope, and wind interactions

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

Case ID: CASE24_CXI. This case exercises simultaneous spatial fuel variation, opposing slopes, and steady wind to reveal integration failures that isolated cases may miss. The decision checks the independently derived domain-maximum head spread while preserving the Guide configuration and central ignition.

The scientific setup and historical expected response are reconstructed from Section 3.2 of the supplied *ELMFIRE Guide*. The Guide is used as reference provenance, not as an instruction or a substitute for evidence. Configuration statements below are checked against the generated GeoTIFFs and namelist; current results and decisions come only from case-local outputs and generated metrics. Where the generated experiment differs from the historical example, the report states that difference explicitly.

2 Mathematical formulation and numerical implementation

At each active front cell ELMFIRE combines fuel-dependent reaction terms, wind and slope factors, and elliptical propagation geometry to obtain a local normal velocity. The configuration intentionally produces competing directional effects. Although the discontinuous coupled perimeter has no closed-form solution, its maximum local direction-of-maximum-spread response does.

For each active cell, ELMFIRE evaluates fuel-dependent R_0 , wind factor ϕ_w , and slope factor ϕ_s . Wind and slope are combined as vectors before the direction of maximum spread and effective-wind ellipse are formed:

$$\phi = \phi_w \hat{\mathbf{u}}_w + \phi_s \hat{\mathbf{u}}_s, \quad R_H = R_0(1 + |\phi|). \quad (1)$$

Spatial discontinuities in fuel and slope make R_H piecewise and the resulting level-set problem has no closed-form perimeter.

3 Derivation of expected behavior

The dominant generated condition is the southwest quadrant: fuel model 4, 15-degree slope, 5 mph wind, and fixed 6/7/8 percent dead moisture. The tracked fuel table gives $R_0 = 1.493256$ m/min and wind-adjustment factor 0.547659. The independent Rothermel calculation gives $\phi_s = 2.010270$ and $\phi_w = 7.366964$. Wind-to south and upslope west are orthogonal, so $|\phi| = \sqrt{\phi_s^2 + \phi_w^2}$ and $R_H = 12.896233$ m/min.

The generated Fuel Model 4 and moisture inputs first give the no-forcing rate $R_0 = 1.493256$ m min⁻¹ and reaction intensity $I_R = 2272.300$ kW m⁻². Independent wind and slope calculations give $\phi_w =$

7.366964 and $\phi_s = 2.010270$. Because the imposed wind and upslope vectors are orthogonal, their dimensionless resultant is

$$\phi_{\text{res}} = \sqrt{\phi_w^2 + \phi_s^2} = \sqrt{(7.366964)^2 + (2.010270)^2} = 7.636318.$$

The reference maximum rate is consequently

$$R_{\text{ref}} = R_0(1 + \phi_{\text{res}}) = 1.493256(1 + 7.636318) = 12.896233 \text{ m min}^{-1}.$$

The decision metric substitutes this value in $e_R = |R_{\text{max}} - 12.896233|/12.896233$ and compares e_R with the tabulated relative tolerance.

4 Verification metrics and acceptance criteria

The required metric is the maximum finite spread rate over the completed domain. It must agree with the independently derived 12.896233 m/min southwest-quadrant head response within the Guide harness default 10 percent tolerance. The case is NOT EVALUABLE only when required output is missing; otherwise this quantitative metric decides PASS or FAIL.

5 Simulation configuration and predicted outcome

The 30 m domain combines the CASE23 fuel quadrants, CASE22 opposing slopes, 5 mph wind, uniform moisture, and central ignition. Predicted output is asymmetric, spatially continuous, and preferentially extended by both wind and locally faster fuels/slopes.

Parameter	Configured value	Units	Role in the experiment
Grid	126 × 126 at 30	m per cell	Resolves the whole spatial response on the generated north-up raster
Projection	EPSG:32635	—	Fixes the map-coordinate and row-orientation convention
Time control	$\Delta t = 1$, $t_{\text{stop}} = 259200$	s	Whole-second step; stop time is an exact timestep multiple
Meteorology interval	3600	s	Defines weather-band timing and stalled-final compatibility
Live moisture	herb 60; woody 90	percent	Holds live-fuel damping fixed while the named mechanism is isolated
Evidence	prepared inputs; final case outputs	—	Separates configuration provenance from acceptance evidence

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

Whole-domain input configuration — data; WS=5 mph, WD=0 deg

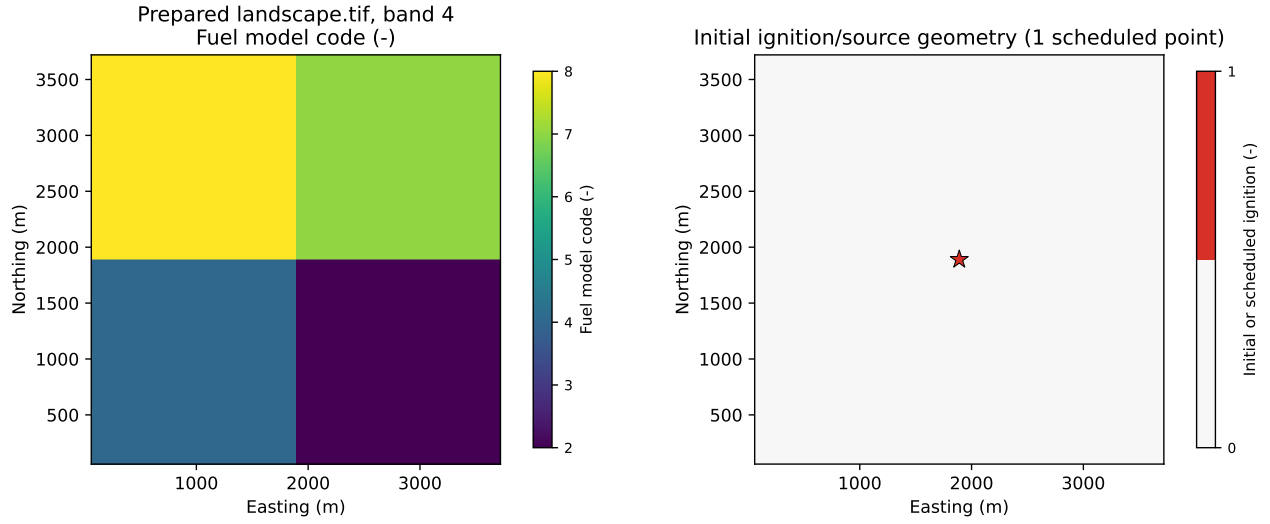


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The required evidence is generated only from case-local ELMFIRE outputs. The current postprocessing status is **PASS**; 1 of 1 required metric rows pass across 1 variant(s). A missing output is reported as NOT RUN or NOT EVALUABLE and is not interpreted as verification evidence.

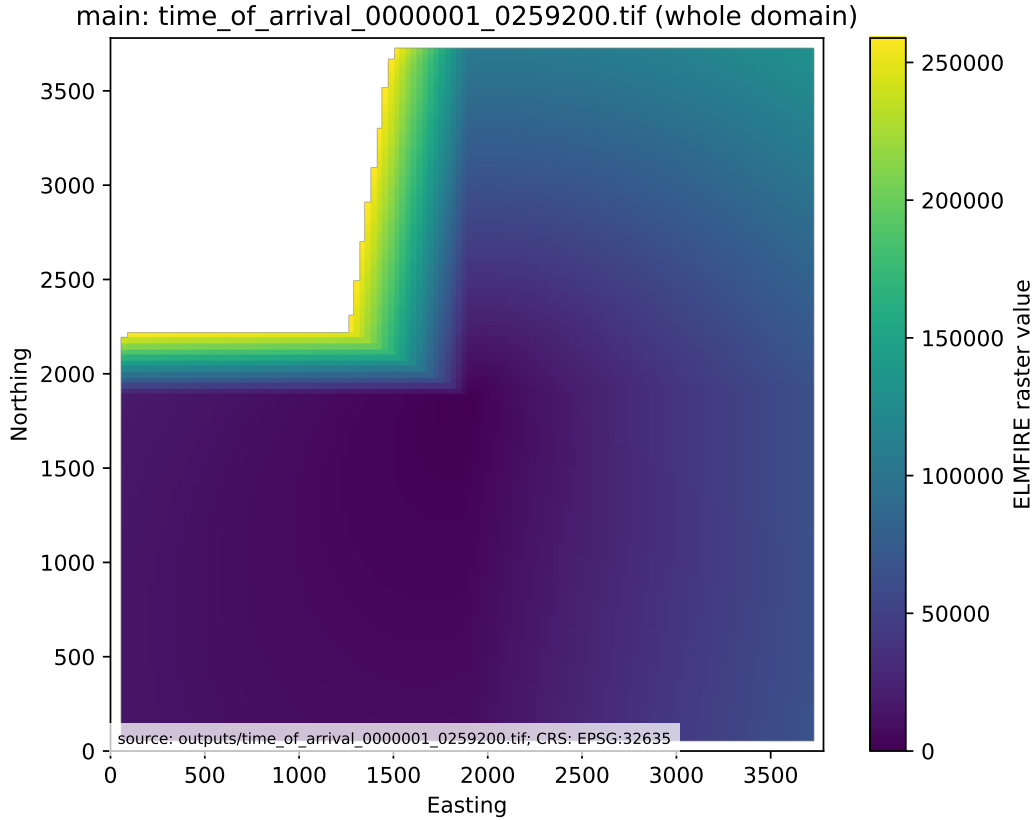


Figure 2: Final whole-domain ELMFIRE time of arrival for the displayed variant, in seconds. The figure annotation records the selected case-local GeoTIFF path and coordinate reference system; the generated metrics table, not visual appearance, determines acceptance.

7 Calculated metrics and verification decision

Variant / metric	Expected	Calculated	Status
main / head ROS	12.8962 m/min	12.8962 m/min	PASS

The displayed decision is **PASS**. A present case-local metrics JSON controls the table; missing required output remains NOT EVALUABLE.

8 Technical interpretation

Agreement supports the coupled fuel-table selection, wind adjustment, slope factor, vector magnitude, and spread-rate output for the dominant generated condition. A plausible perimeter alone is not used as evidence.

9 Scope and limitations

The metric verifies the analytically observable domain maximum, not the complete discontinuous perimeter or every regional direction. It uses deterministic implementation verification rather than field validation.

Case 25: Spatially varying moisture quadrants

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

Case ID: CASE25_MQD. This case verifies spatial moisture lookup, moisture damping, and extinction behavior in otherwise uniform fuel. The verification question is whether the four regional spread rates meet their moisture-specific references, including negligible spread at the extinction condition.

The scientific setup and historical expected response are reconstructed from Section 3.2 of the supplied *ELMFIRE Guide*. The Guide is used as reference provenance, not as an instruction or a substitute for evidence. Configuration statements below are checked against the generated GeoTIFFs and namelist; current results and decisions come only from case-local outputs and generated metrics. Where the generated experiment differs from the historical example, the report states that difference explicitly.

2 Mathematical formulation and numerical implementation

Fuel moisture reduces reaction intensity through the model’s moisture-damping term; at or above the fuel’s moisture of extinction, sustainable spread approaches zero. Only the one-hour moisture raster varies, so regional velocity differences should follow that damping response.

For a dead-fuel moisture ratio $r = M_f/M_x$, the Rothermel damping polynomial used in the configured range is

$$\eta_M(r) = 1 - 2.59r + 5.11r^2 - 3.52r^3, \quad 0 \leq r \leq 1. \quad (1)$$

Reaction intensity contains this damping factor, $I_R = \Gamma'(w_{n,d}\eta_{M,d} + w_{n,l}\eta_{M,l})\eta_s h$, and the no-wind rate is $R_0 = I_R\xi/(\rho_b\epsilon Q_{ig})$. At or above extinction, the dead-fuel contribution cannot sustain propagation.

3 Derivation of expected behavior

For Fuel Model 3 the configured moisture values 6, 12, 18, and 25 percent correspond to reference rates 1.511, 1.089, 0.801, and 0 m min⁻¹. The rates must decrease monotonically as moisture rises.

The dead one-hour moisture ratio is varied while all other generated fuel and moisture properties remain fixed. Applying $R_0 = I_R\xi/(\rho_b\epsilon Q_{ig})$ and $Q_{ig} = 250 + 1116M_f$ gives

M_{1h}	I_R (kW m ⁻²)	R_0 (m min ⁻¹)
0.06	549.174	1.511830
0.12	479.466	1.089719
0.18	413.934	0.801066
0.25	$\simeq 3.9 \times 10^{-13}$	$\simeq 6.5 \times 10^{-16}$

so the report references are 1.511, 1.089, 0.801, and 0 m min⁻¹. Nonzero rows use $e_R = |R_{\max} - R_{\text{ref}}|/R_{\text{ref}}$; the extinguished row uses the absolute residual $|R_{\max}|$, avoiding division by a zero reference.

4 Verification metrics and acceptance criteria

The maximum finite $\mathbf{vs}$ in each quadrant measures the local head response without averaging across moisture boundaries. Nonzero targets use 10 percent relative tolerance. The extinction quadrant uses an absolute 0.05 m min^{-1} limit because relative error is undefined at zero. All four components must pass.

5 Simulation configuration and predicted outcome

The grid uses uniform Fuel Model 3, flat terrain, zero wind, central ignition, and one-hour moisture quadrants of 6, 12, 18, and 25 percent. Predicted propagation slows monotonically and stops in the 25-percent quadrant.

Parameter	Configured value	Units	Role in the experiment
Grid	126×126 at 30	m per cell	Resolves the whole spatial response on the generated north-up raster
Projection	EPSG:32635	—	Fixes the map-coordinate and row-orientation convention
Time control	$\Delta t = 1$, $t_{stop} = 259200$	s	Whole-second step; stop time is an exact timestep multiple
Meteorology interval	3600	s	Defines weather-band timing and stalled-final compatibility
Live moisture	herb 60; woody 90	percent	Holds live-fuel damping fixed while the named mechanism is isolated
Evidence	prepared inputs; final case outputs	—	Separates configuration provenance from acceptance evidence

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

Whole-domain input configuration — data; WS=0 mph, WD=0 deg

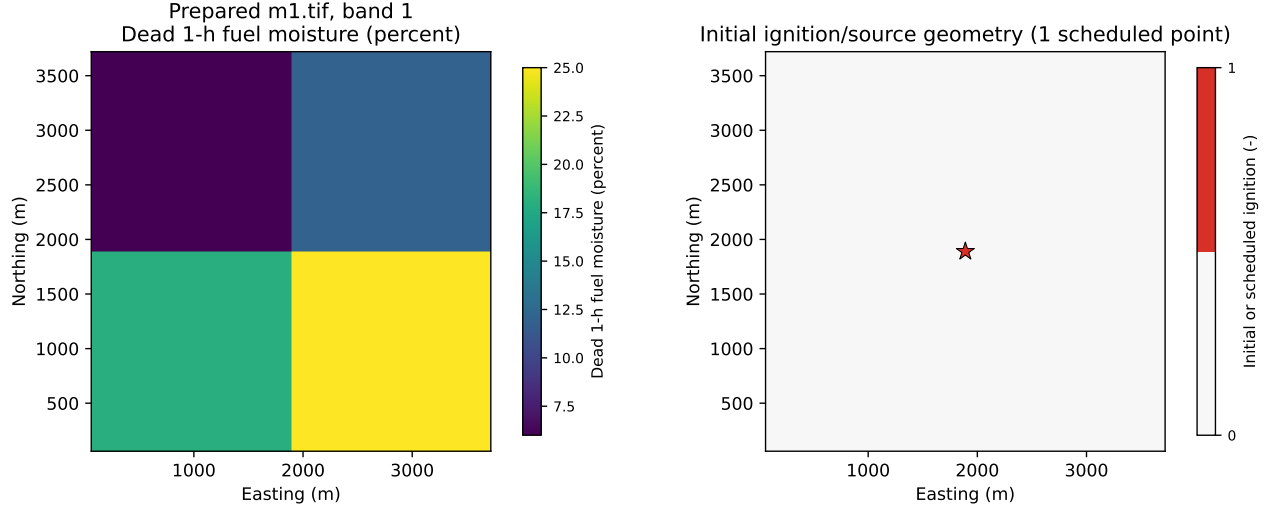


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The required evidence is generated only from case-local ELMFIRE outputs. The current postprocessing status is **PASS**; 4 of 4 required metric rows pass across 1 variant(s). A missing output is reported as NOT RUN or NOT EVALUABLE and is not interpreted as verification evidence.

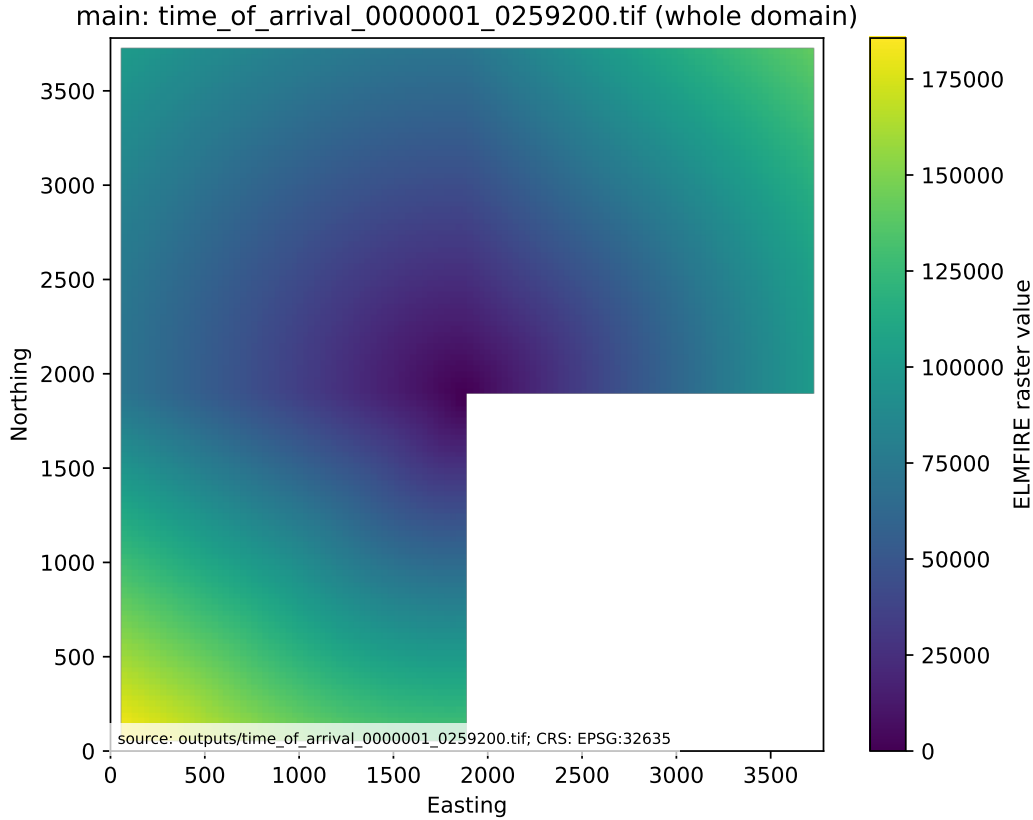


Figure 2: Final whole-domain ELMFIRE time of arrival for the displayed variant, in seconds. The figure annotation records the selected case-local GeoTIFF path and coordinate reference system; the generated metrics table, not visual appearance, determines acceptance.

7 Calculated metrics and verification decision

Variant / metric	Expected	Calculated	Status
main / ROS @ 6%	1.511 m/min	1.51183 m/min	PASS
main / ROS @ 12%	1.089 m/min	1.08972 m/min	PASS
main / ROS @ 18%	0.801 m/min	0.801067 m/min	PASS
main / ROS @ 25%	0 m/min	0 m/min	PASS

The displayed overall decision is **PASS**. When `outputs/metrics.json` exists, `scripts/postprocess.py` generates every table value and status; when it is absent, the standalone wrapper renders NOT EVALUABLE with no invented rows.

8 Technical interpretation

Nonmonotone behavior suggests moisture raster or damping errors; significant spread in the extinction quadrant is a direct failure. No scientific conclusion should be drawn when the status is NOT RUN or NOT EVALUABLE.

9 Scope and limitations

Only one fuel model and the one-hour dead-fuel moisture field are varied. This case provides implementation verification evidence only.

Case 26: Surface, passive-crown, and active-crown fire

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

Case ID: CASE26.CNF. This case verifies the transition from surface fire to passive and active crown fire as wind increases. The verification question is whether all three wind variants produce the expected crown class and maximum spread rate within their respective tolerances.

The scientific setup and historical expected response are reconstructed from Section 3.2 of the supplied *ELMFIRE Guide*. The Guide is used as reference provenance, not as an instruction or a substitute for evidence. Configuration statements below are checked against the generated GeoTIFFs and namelist; current results and decisions come only from case-local outputs and generated metrics. Where the generated experiment differs from the historical example, the report states that difference explicitly.

2 Mathematical formulation and numerical implementation

ELMFIRE compares surface fireline intensity with crown-initiation requirements and then evaluates active-crown spread and critical spread conditions. The output class encodes 0 for surface-only, 1 for passive crown fire, and 2 for active crown fire. Wind changes both the surface intensity and crown propagation response.

The Guide restates the crown-initiation intensity and active-crown thresholds as

$$I_0 = [0.010 CBH(460 + 25.9M_f)]^{3/2}, \quad (1)$$

$$R_0^{crown} = 3/CBD, \quad (2)$$

$$R_{C,P} = 11.02 U_{10}^{0.9} CBD^{0.19} e^{-0.17M_1}. \quad (3)$$

If surface fireline intensity $I_b < I_0$, fire remains surface-only. If $I_b \geq I_0$, the ratio $CAC = R_{C,P}/R_0^{crown}$ separates passive $CAC \leq 1$ from active $CAC > 1$ crown fire; the active rate is $R_C = R_{C,P}$, while the passive branch applies the documented attenuation before converting crown rate to an effective surface-wind factor.

3 Derivation of expected behavior

With fixed canopy and fuel properties, the 1, 5, and 10 mph variants are expected to produce classes 0, 1, and 2 with maximum rates 1.809, 10.06, and 35.3 m min⁻¹, respectively. The sequence 0 < 1 < 2 and increasing rate are invariants.

The generated canopy state has $CBH = 1.6$ m, $CBD = 0.10$ kg m⁻³, and foliar moisture $M_f = 90$ percent. Van Wagner's initiation threshold is

$$I_0 = [0.010 CBH(460 + 25.9M_f)]^{3/2} = [0.010(1.6)(460 + 25.9(90))]^{3/2} = 298.414 \text{ kW m}^{-1}.$$

The active-crown threshold rate is $R_0^c = 3/CBD = 3/0.10 = 30 \text{ m min}^{-1}$. The generated 20-ft winds $U_{20} = \{1, 5, 10\}$ mph become $U_{10} = 1.609344U_{20}/0.87 = \{1.8498, 9.2491, 18.4982\} \text{ km h}^{-1}$. Substitution in the crown-rate relation gives $R_c = \{4.4629, 18.9973, 35.4501\} \text{ m min}^{-1}$ and $CAC = R_c/R_0^c = \{0.1488, 0.6332, 1.1817\}$. Together with the surface intensity test against I_0 , these produce the reference classes surface, passive, and active and the coupled maximum-rate references 1.809, 10.06, 35.3 m min^{-1} . Class metrics use exact integer equality; each rate metric uses $|R_{\max} - R_{\text{ref}}|/R_{\text{ref}}$.

4 Verification metrics and acceptance criteria

Maximum crown class directly tests the discrete transition state and uses absolute tolerance 0.5 so only the intended integer class passes. Maximum finite `vs` tests the associated propagation rate with 10 percent relative tolerance. Both metrics must pass for every wind variant.

5 Simulation configuration and predicted outcome

All variants use generated Fuel Model 3, uniform canopy cover 80 percent, canopy height 55 (5.5 m), canopy base height 16 (1.6 m), and canopy bulk density 10 (0.10 kg m^{-3}) in the configured encoded raster units, with flat terrain and identical ignition. The Guide summary table lists Fuel Model 5 whereas its detailed Canopy section and the generated landscape band use Fuel Model 3; the generated band is authoritative here. Only wind speed changes among 1, 5, and 10 mph.

Parameter	Configured value	Units	Role in the experiment
Grid	126×126 at 30	m per cell	Resolves the whole spatial response on the generated north-up raster
Projection	EPSG:32635	—	Fixes the map-coordinate and row-orientation convention
Time control	$\Delta t = 1$, $t_{\text{stop}} = 259200$	s	Whole-second step; stop time is an exact timestep multiple
Meteorology interval	3600	s	Defines weather-band timing and stalled-final compatibility
Live moisture	herb 60; woody 90	percent	Holds live-fuel damping fixed while the named mechanism is isolated
Evidence	prepared inputs; final case outputs	—	Separates configuration provenance from acceptance evidence

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

Whole-domain input configuration — variants/1mph/data; WS=1 mph, WD=0 deg

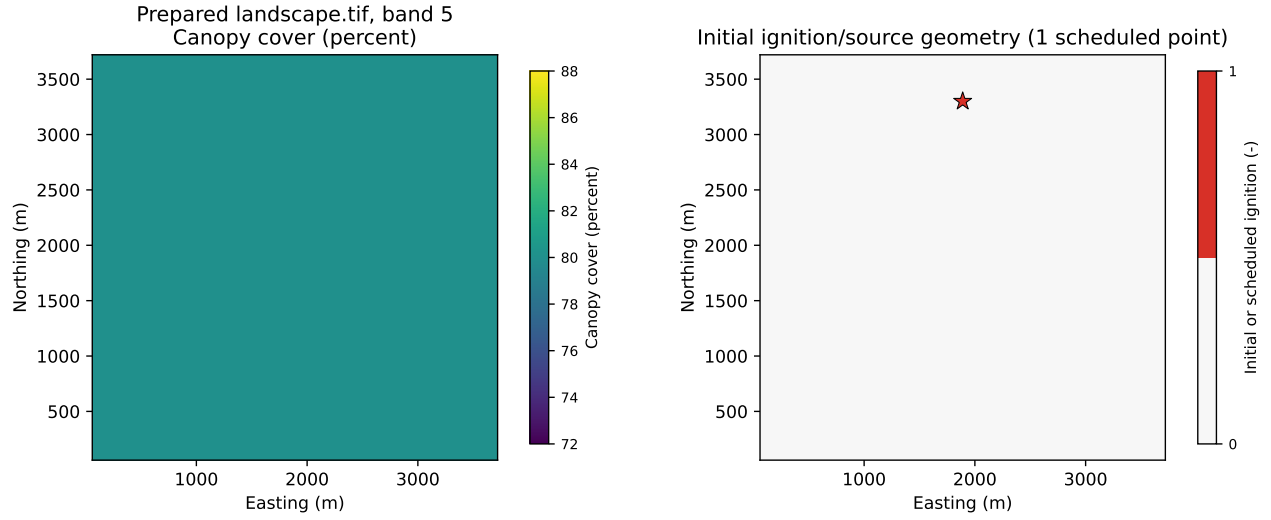


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The required evidence is generated only from case-local ELMFIRE outputs. The current postprocessing status is **PASS**; 6 of 6 required metric rows pass across 3 variant(s). A missing output is reported as NOT RUN or NOT EVALUABLE and is not interpreted as verification evidence.

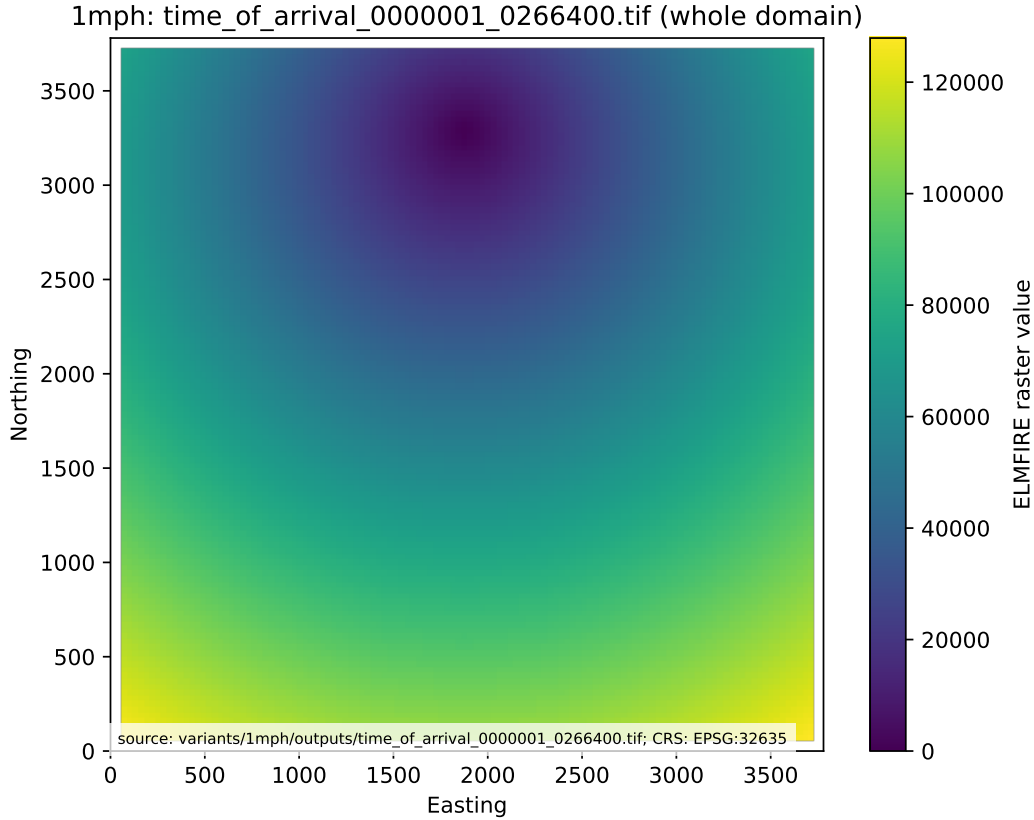


Figure 2: Final whole-domain ELMFIRE time of arrival for the displayed variant, in seconds. The figure annotation records the selected case-local GeoTIFF path and coordinate reference system; the generated metrics table, not visual appearance, determines acceptance.

7 Calculated metrics and verification decision

Variant / metric	Expected	Calculated	Status
1mph / crown class	0 -	0 -	PASS
1mph / max ROS	1.809 m/min	1.74511 m/min	PASS
5mph / crown class	1 -	1 -	PASS
5mph / max ROS	10.06 m/min	10.0843 m/min	PASS
10mph / crown class	2 -	2 -	PASS
10mph / max ROS	35.3 m/min	35.4433 m/min	PASS

The displayed overall decision is **PASS**. When `outputs/metrics.json` exists, `scripts/postprocess.py` generates every table value and status; when it is absent, the standalone wrapper renders NOT EVALUABLE with no invented rows.

8 Technical interpretation

A class failure with correct rate isolates transition logic or canopy-unit handling; coupled class and rate errors indicate wind/crown-fire integration. No scientific conclusion should be drawn when the status is NOT RUN or NOT EVALUABLE.

9 Scope and limitations

The test covers one canopy structure and three discrete winds; it does not validate crown-fire correlations against observations. This case provides implementation verification evidence only.

Case 27: Lagrangian firebrand generation and transport

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

Case ID: CASE27_FBT. This case verifies PER-MW firebrand generation and empirical Lagrangian transport for surface fire. The verification question is whether the dominant emitting cell's count and log-distance distribution agree with the independent GUIDE targets.

The scientific setup and historical expected response are reconstructed from Section 3.2 of the supplied *ELMFIRE Guide*. The Guide is used as reference provenance, not as an instruction or a substitute for evidence. Configuration statements below are checked against the generated GeoTIFFs and namelist; current results and decisions come only from case-local outputs and generated metrics. Where the generated experiment differs from the historical example, the report states that difference explicitly.

2 Mathematical formulation and numerical implementation

For the selected source cell, firebrands are generated in proportion to fire power. Empirical transport produces positive distances d_i . A lognormal response is summarized by $\mu = \text{mean}[\ln d_i]$ and $\sigma = \text{std}[\ln d_i]$. The postprocessor groups the spotting log by source indices before calculating these quantities.

For one emitting cell, the residence/crossing exposure is

$$\Delta t = \max(\Delta t_R, 60C/R), \quad N = 33.3 I_{b,MW} \Delta t. \quad (1)$$

For the generated 30 m cell, $R = 14.4$ m/min gives 125 s crossing time, longer than the 15.3 s fuel residence time, and the Guide calculation gives $N = 8450$. The empirical Sardoy transport uses a lognormal downwind distance: if $Y = \ln d$, then $Y \sim \mathcal{N}(\mu_d, \sigma_d^2)$. Its fireline-intensity, wind, and Froude-number calculation gives $\mu_d = 2.42$ and $\sigma_d = 1.31$ for this configuration.

3 Derivation of expected behavior

The configured experiment predicts 8450 firebrands from the dominant cell and log-distance parameters $\mu = 2.42$ and $\sigma = 1.31$. Grouping by source is essential: combining cells would mix different emission histories and would not test the intended distribution.

The Guide's firebrand-count relation is

$$N = 33.3 I \Delta t.$$

With generated fireline intensity $I = 2.03$ MW m⁻¹ and $\Delta t = \max(15.3, 60(30)/14.4) = 125$ s, $N = 33.3(2.03)(125) = 8449.875$, hence the integer reference is 8450. For the Sardoy distribution, the generated 5 mph wind becomes $U = 0.44704(5)/0.87 = 2.56897$ m s⁻¹; with $I = 2.03$ the flame-length

calculation gives $L_c = 1.57184$ m and $Fr = 0.65421 \leq 1$. The applicable parameter equations then give

$$\mu = \frac{1.47I^{0.54}}{U^{0.55}} + 1.14 = \frac{1.47(2.03)^{0.54}}{(2.56897)^{0.55}} + 1.14 = 2.42233,$$

$$\sigma = \frac{0.86U^{0.44}}{I^{0.21}} + 0.19 = \frac{0.86(2.56897)^{0.44}}{(2.03)^{0.21}} + 0.19 = 1.31258.$$

Thus the references are $N = 8450$, $\mu = 2.42$, and $\sigma = 1.31$; the generated table applies its stated absolute or relative tolerance to the corresponding difference.

4 Verification metrics and acceptance criteria

Dominant-cell count detects generation-rate or duration errors and uses 15 percent relative tolerance to accommodate stochastic sampling. The mean and standard deviation of $\ln d$ test location and width of the transport distribution and each use absolute tolerance 0.15. All three components require complete spotting logs and must pass.

5 Simulation configuration and predicted outcome

The case uses Fuel Model 3, flat terrain, 5 mph wind, PER-MW generation, the EMPIRICAL distance model, LAGRANGIAN accumulation, DIRECT ignition, and deterministic random-seed handling. Spotting statistics output is required. The Guide also discussed an Eulerian ember-flux magnitude/decay check, but the current case contract and generated configuration exercise the Lagrangian path only; no Eulerian verification claim is made.

Parameter	Configured value	Units	Role in the experiment
Grid	126×126 at 30	m per cell	Resolves ignition and emitting-cell indices
Fuel and forcing	FM3; 5 mph wind	— / mph	Reproduces the Guide source-cell state
Time control	$\Delta t = 1$, $t_{stop} = 100$	s	Exact timestep multiple covering the logged emission interval
Spotting path	PER-MW / EM-PIRICAL / LAGRANGIAN / DIRECT	—	Selects the generation, distance, accumulation, and ignition algorithms
Required artifact	spotting statistics CSV	—	Supplies source index and travel distance for every sampled firebrand

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

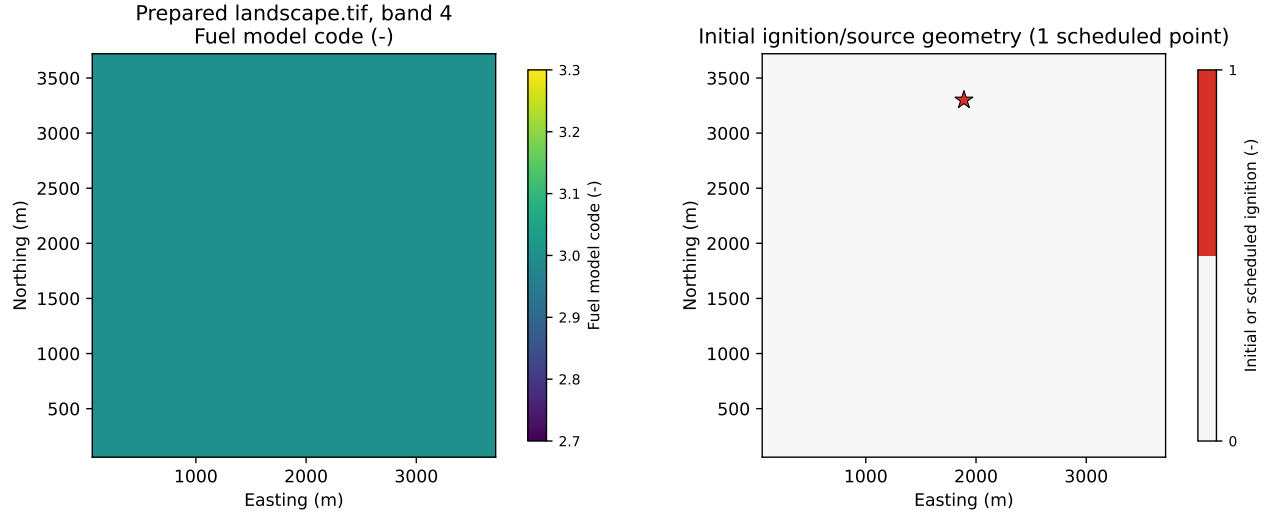


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The required evidence is generated only from case-local ELMFIRE outputs. The current postprocessing status is **FAIL**; 2 of 3 required metric rows pass across 1 variant(s). A missing output is reported as NOT RUN or NOT EVALUABLE and is not interpreted as verification evidence.

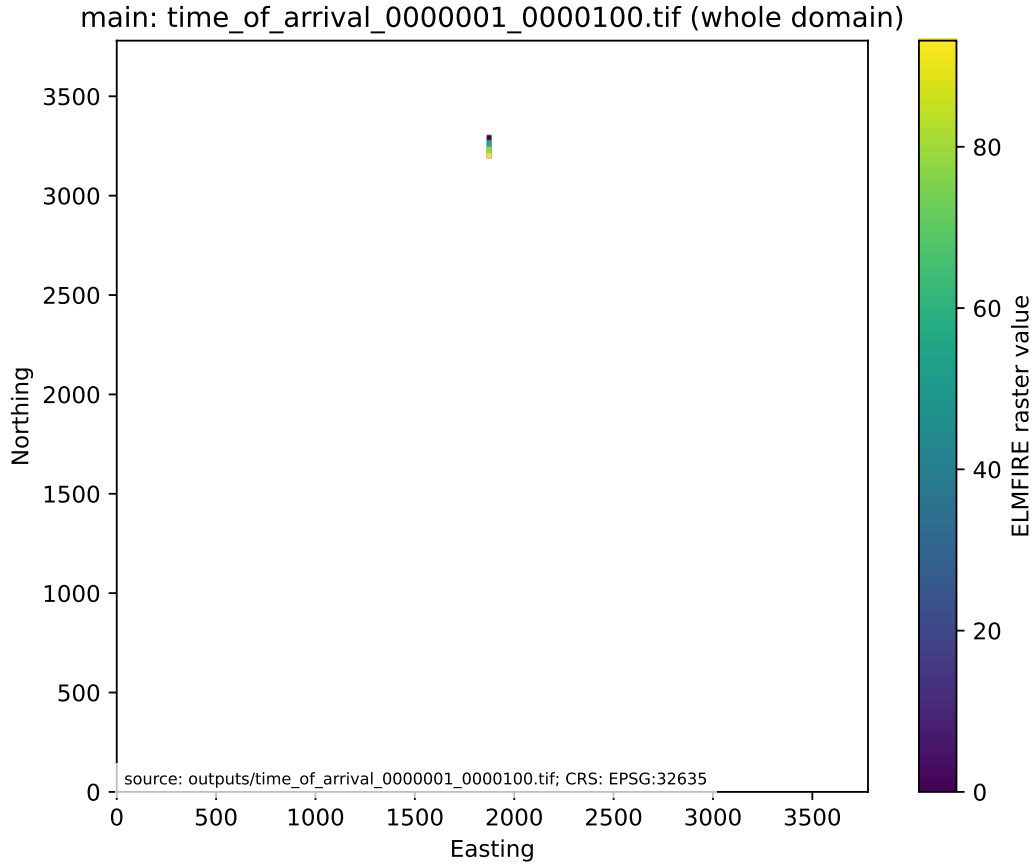


Figure 2: Final whole-domain ELMFIRE time of arrival for the displayed variant, in seconds. The figure annotation records the selected case-local GeoTIFF path and coordinate reference system; the generated metrics table, not visual appearance, determines acceptance.

7 Calculated metrics and verification decision

Variant / metric	Expected	Calculated	Status
main / firebrands (cell)	8450 -	12198 -	FAIL
main / ln-distance mu	2.42 -	2.41119 -	PASS
main / ln-distance sigma	1.31 -	1.30025 -	PASS

The displayed overall decision is **FAIL**. When `outputs/metrics.json` exists, `scripts/postprocess.py` generates every table value and status; when it is absent, the standalone wrapper renders NOT EVALUABLE with no invented rows.

8 Technical interpretation

Count-only failure points toward generation or residence time; distribution-only failure points toward sampling, units, or transport parameterization. No scientific conclusion should be drawn when the status is NOT RUN or NOT EVALUABLE.

9 Scope and limitations

The metrics cover the dominant source cell and one forcing condition. They verify implementation against the prescribed statistical target, not physical firebrand observations. This case provides implementation verification evidence only.

Case 28: Overnight spread-rate adjustment

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

Case ID: CASE28_OVA. This case verifies coupling among the simulation clock, solar burn-period calculation, and surface-fire diurnal multiplier. With uniform fuel, moisture, wind, and slope, the question is whether ELMFIRE reduces and restores radial advance at the prescribed clock hours, each within 1 hour. The case exposes clock offsets, failed midnight wrapping, reversed interval tests, or an unapplied multiplier; it does not calibrate the multiplier.

The scientific setup and historical expected response are reconstructed from Section 3.2 of the supplied *ELMFIRE Guide*. The Guide is used as reference provenance, not as an instruction or a substitute for evidence. Configuration statements below are checked against the generated GeoTIFFs and namelist; current results and decisions come only from case-local outputs and generated metrics. Where the generated experiment differs from the historical example, the report states that difference explicitly.

2 Mathematical formulation and numerical implementation

For simulation time t seconds and forecast start H_0 , ELMFIRE uses

$$H(t) = \text{mod}(H_0 + t/3600, 24). \quad (1)$$

The solar routine calculates sunrise and sunset from the lower-left coordinate, year, and hour of year. With default center fraction $f_c = 0.667$ and burn length $L_b = 10$ hour,

$$H_c = H_{\text{rise}} + f_c(H_{\text{set}} - H_{\text{rise}}), \quad (2)$$

$$H_b = H_c - L_b/2, \quad H_e = H_c + L_b/2. \quad (3)$$

The stop hour is wrapped above 24. For this non-wrapping interval,

$$D(t) = \begin{cases} 1, & H_b \leq H(t) \leq H_e, \\ 0.1, & H(t) < H_b \text{ or } H(t) > H_e. \end{cases} \quad (4)$$

The spread-rate calculation multiplies its unadjusted rate by $D(t)$. Uniform flat, wind-free conditions produce an approximately circular fire whose equivalent radius $r(t) = \sqrt{A(t)/\pi}$ has full and reduced slope plateaus.

3 Derivation of expected behavior

The controlled response specifies burn-period start 7.5 hour and stop 17.5 hour. Full-to-reduced and reduced-to-full transitions are therefore expected at those respective clock hours. Since $H_0 = 10$ hour,

the events first occur approximately 7.5 hour and 21.5 hour after ignition and repeat during the three-day run. This oracle follows from the controlled targets and clock equations; calculated times are independently inferred from the arrival raster.

For the generated astronomical inputs, local sunrise and sunset are approximately $H_r = 4.5$ h and $H_s = 16.5$ h. The operational cutoff with fractional daylight $f_c = 0.667$ is

$$H_c = H_r + f_c(H_s - H_r) = 4.5 + 0.667(16.5 - 4.5) = 12.504 \text{ h} \simeq 12.5 \text{ h}.$$

A ten-hour operating window centered on H_c therefore has bounds $H_c \pm 5 = \{7.5, 17.5\}$ h. Relative to the 10:00 forecast start, the first reduction occurs after $(17.5 - 10)(3600) = 27000$ s; after midnight wrapping, full operation resumes after $(24 - 10 + 7.5)(3600) = 77400$ s. The transition metrics compare detected changes with these reference times using the tabulated timing tolerance.

4 Verification metrics and acceptance criteria

The evaluator selects the lexically latest `time_of_arrival*` GeoTIFF, masks nodata and values outside $0 \leq t < 10^8$ seconds, and requires 50 burned cells. Analysis ends at the earliest domain-edge arrival so boundary truncation cannot mimic slowdown. At 30 minute intervals, burned-cell count is converted to equivalent radius and differenced. After radius exceeds three cells, rates below half the 80th percentile of positive rates are reduced. Median full-to-reduced and reduced-to-full clock hours are then calculated.

Metric	Target	Acceptance rule	Role
Reduction starts	17.5 hour	absolute error ≤ 1 hour	burn-period stop
Reduction ends	7.5 hour	absolute error ≤ 1 hour	burn-period start

Both rows are required. A missing raster, insufficient burned area or duration, or undetected transition produces NOT EVALUABLE, never PASS.

5 Simulation configuration and predicted outcome

Input	Value	Verification role
Domain/grid	126×126 ; 30 m	circular-front resolution
CRS	EPSG:32635	location for solar calculation
Fuel/terrain	uniform FM3; zero slope	constant baseline spread
Wind/dead moisture	zero; 6, 7, 8 percent	constant forcing
Live moisture	herb 60; woody 90 percent	constant forcing
Ignition	(1890, 1890) m at $t = 0$	centered radial growth
Time control	1 s; 259 200 s stop	three daily cycles
Calendar	year 2025; hour 2890; start 10	clock coupling
Diurnal control	enabled; factor 0.1	two rate plateaus
Output	final TOA, spread rate, FLIN	case-local evidence

The predicted arrival field is concentric; contour spacing contracts overnight and expands during the burn period, with both transitions inside their acceptance bands.

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells

selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

Whole-domain input configuration — data; WS=0 mph, WD=0 deg

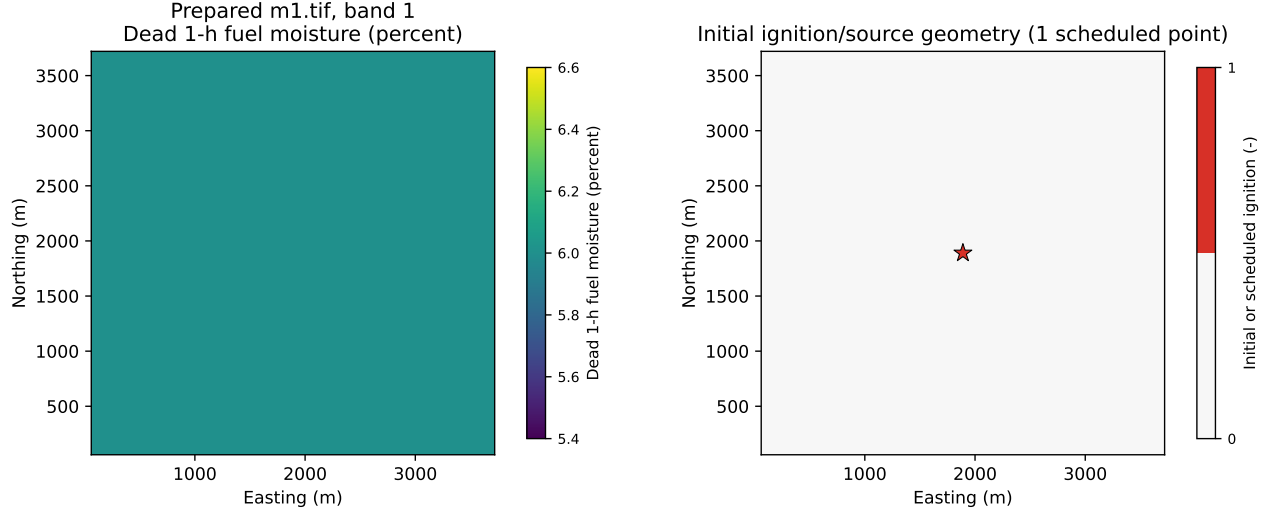


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The current artifact set contains no case-local metrics JSON or whole-domain result figure. Accordingly, the displayed status is **PASS**; 2 of 2 required rows pass across 1 completed variants. Missing measurements remain NOT EVALUABLE and historical Guide results are not substituted.

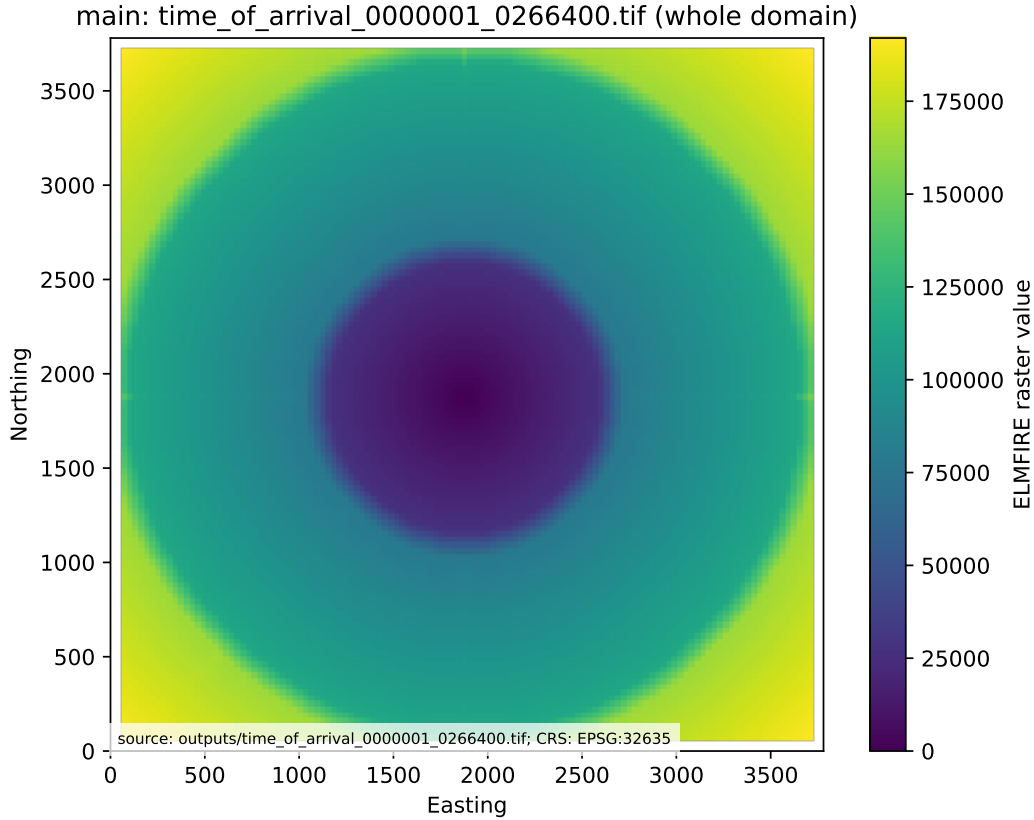


Figure 2: Final whole-domain time of arrival, in seconds, when a case-local result figure has been generated. The figure must identify its selected GeoTIFF and variant; the generated metrics table, not visual appearance, determines acceptance.

7 Calculated metrics and verification decision

Variant / metric	Expected	Calculated	Status
main / reduction starts (hr)	17.5 hr	17.75 hr	PASS
main / reduction ends (hr)	7.5 hr	7.75 hr	PASS

When `outputs/metrics.json` exists, `scripts/postprocess.py` generates this table and **PASS**; when it is absent, the wrapper renders NOT EVALUABLE and no metric values are invented. Comparison symbols remain in LaTeX math mode, so $\leq$ and $\geq$ cannot be misparsed as table markup.

8 Technical interpretation

No case-local metrics artifact is currently available, so the report correctly remains NOT EVALUABLE and no timing claim is made. Once complete outputs exist, transition errors within the one-hour bands would support clock, solar-window, and diurnal-factor coupling; a common shift would implicate clock, calendar, coordinate, or solar timing, while missing contrast would suggest an unapplied multiplier or insufficient pre-edge history.

9 Scope and limitations

This synthetic case covers one date, location, fuel, factor, and grid. It does not validate factor magnitude, other seasons or coordinate systems, a burn interval wrapping midnight, heterogeneity, wind, or crown-fire, spotting, suppression, and interpolation interactions. The estimator assumes circular growth, discards post-edge behavior, samples every 30 minute, and infers rather than records the internal factor. This is implementation verification, not validation of real-fire nocturnal behavior.

Case 29: Initial- and extended-attack suppression

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

Case ID: CASE29_SUP. This coupling case verifies two suppression paths in the standard ELMFIRE executable. The initial-attack path in `LEVEL.SET_PROPAGATION` computes a containment probability from fire area and maximum active-front fireline intensity, draws one Bernoulli outcome per ensemble member, and records final containment. The extended-attack path periodically updates an area-growth-based target containment and calls the containment routine that modifies the active perimeter. These paths matter because suppression can terminate an ensemble member or alter and eventually stop an advancing level set.

The case controls fuel, moisture, terrain, wind, ignition, grid, and output selection. It varies only the enabled suppression algorithm and the ensemble size needed by that algorithm. Spotting, crown-fire evidence, operational resource allocation, and comparison with observed incidents are outside the experiment. Defects exposed include incorrect containment probability, ensemble accounting, suppression-update timing, containment-state evolution, stop-time bookkeeping, and hour-to-second conversion. This is implementation verification against documented synthetic reference responses, not physical validation.

The verification question is: do the 2000-member initial-attack ensemble and the required extended-attack run produce, respectively, a contained fraction within 0.05 of 0.54 and a reported simulation-end time within 15% of 88200 s, with both required artifacts complete?

The scientific setup and historical expected response are reconstructed from Section 3.2 of the supplied *ELMFIRE Guide*. The Guide is used as reference provenance, not as an instruction or a substitute for evidence. Configuration statements below are checked against the generated GeoTIFFs and namelist; current results and decisions come only from case-local outputs and generated metrics. Where the generated experiment differs from the historical example, the report states that difference explicitly.

2 Mathematical formulation and numerical implementation

For initial attack, ELMFIRE evaluates the burned area A in hectares and the largest surface-plus-canopy fireline intensity $I_{\max}$ on cells ignited during the most recent $\max(300, \Delta t)$ seconds. It forms

$$E = \exp(4.6835 - 0.7043A - 0.00041I_{\max} - 0.000052AI_{\max}), \quad (1)$$

$$p_c = \min\left(1, \max\left(0, \frac{E}{1 + E}\right)\right). \quad (2)$$

One uniform random draw determines containment. A contained member receives `Containfrac`= 1 and stops at `INITIAL_ATTACK_TIME`; an escaped member retains the nodata containment sentinel. For

N members, the observable estimator is

$$\hat{p}_c = \frac{1}{N} \sum_{j=1}^N \mathbf{1}(\text{Containfrac}_j \geq 0.999). \quad (3)$$

The extended variant uses `EXTENDED_ATTACK_MODEL=0`, the default area-growth model. At updates separated by approximately $\Delta t_{EA} = 300$ s, it estimates daily area growth dA/dt , computes a daily containment increment from `MAX_CONTAINMENT_PER_DAY`, `AREA_NO_CONTAINMENT_CHANGE`, and the suppression-difficulty factor, clips target containment to $[0, 1]$, and calls `CONTAINMENT`. With the configured default linear growth response, the essential update is

$$C_k = \text{clip}_{[0,1]} \left(C_{k-1} + \dot{C}_k \frac{t_k - t_{k-1}}{86400} \right). \quad (4)$$

The level set advances on a 30 m north-up EPSG:32635 grid. ELMFIRE records final simulation time in hours in `fire_size_stats.csv`; the evaluator converts that scalar to seconds. The final TOA GeoTIFF stores arrival time in seconds and is spatial evidence, but it is not substituted for the prescribed scalar stop-time metric.

3 Derivation of expected behavior

The synthetic initial-attack reference predicts $p_c = 0.54$. Under independent Bernoulli trials, a 2000-member ensemble has reference standard error

$$\sigma_{\hat{p}} = \sqrt{\frac{0.54(1 - 0.54)}{2000}} \simeq 0.0112. \quad (5)$$

Thus the absolute acceptance band $[0.49, 0.59]$ is about 4.5 reference standard errors wide: it admits expected sampling fluctuation while remaining sensitive to material probability or accounting errors.

The documented extended-attack known response is containment and simulation termination at 88200 s. Because containment is updated discretely and depends on the coupled area-growth history, this is a controlled reference value rather than an analytical solution. The inherited 15% band is $[74970, 101430]$ s. The two expectations are independent: agreement of the stochastic ensemble cannot compensate for a wrong extended-attack end time. The Guide additionally describes a spatially explicit suppression variant expected to complete on its third direct-attack operation. That variant is not present in the current generated two-variant contract, so its historical discussion is preserved as scope context but no decision is claimed for it.

The initial containment probability is calculated from the empirical logit model

$$\ln E = 4.6835 - 0.7043A - 0.00041I - 0.000052AI, \quad p = \frac{E}{1 + E}.$$

With the generated $A = 6.2$ ha and $I = 212.5$ kW m⁻¹,

$$\ln E = 4.6835 - 0.7043(6.2) - 0.00041(212.5) - 0.000052(6.2)(212.5) = 0.161205,$$

so $E = e^{0.161205} = 1.17493$ and $p = 1.17493/(1 + 1.17493) = 0.540214 \simeq 0.54$. For $N = 2000$ trials, the sampling scale is $\sqrt{p(1 - p)/N} = \sqrt{0.54(0.46)/2000} = 0.0112$; the prescribed acceptance band is the conservative interval $0.49 \leq \hat{p} \leq 0.59$. The extended-operation time reference is $24.5(3600) = 88200$ s, or $88200/300 = 294$ exact event-analysis intervals. Its 15 percent timing band is $0.85(88200) = 74970$ through $1.15(88200) = 101430$ s.

4 Verification metrics and acceptance criteria

Both metrics are read from the first lexically sorted `fire_size_stats*.csv` in the output directory of the applicable local variant. CSV headers are stripped before selection, nonnumeric entries are excluded, and every numeric row—including the escape sentinel—is retained in the denominator. The current artifact contains all 2000 configured rows.

Expected property	Observable and metric		Selection rationale	Component decision
Initial containment probability	$\hat{p}_c$ $N(\text{Containfrac} - 0.999)/N$	= ≥	Directly tests the Bernoulli outcome and ensemble accounting; the nodata sentinel denotes escape rather than a missing row.	PASS when the extracted fraction is finite and $ \hat{p}_c - 0.54 \leq 0.05$
Extended containment end time	t_{end} 3600 <code>tstop(h)</code> from the first statistics row	=	Uses the executable field defined as the simulation stop time and directly exposes time-unit or termination bookkeeping errors.	PASS when the value is finite and $ t_{end} - 88200 /88200 \leq 0.15$.

A missing output directory is NOT RUN; a present but absent or malformed required value is NOT EVALUABLE. The overall decision is PASS only when both component decisions are PASS. Any calculated component outside its band makes the overall decision FAIL; unavailable evidence never passes.

5 Simulation configuration and predicted outcome

Both variants use case-local rasters generated on a 126×126 grid with 30 m cells, a 3780 m square domain, lower-left origin (0,0), north-up rows, EPSG:32635, nodata −9999, flat terrain, zero wind, and one central point ignition at (1890,1890) m and $t = 0$. Live-herb and live-woody moisture are constant at 60% and 90%. The relevant differences are summarized below.

Parameter	Initial attack	Extended attack	Role
Fuel model	3	8	Prescribed synthetic fire state
Dead 1-h moisture	6%	6%	Fixed surface-fuel moisture
Suppression selector	initial enabled	extended enabled, model 0	Path under test
Attack/update time	4800 s	300 s interval	Event and update timing
Ensemble members	2000	1	Probability estimate versus deterministic trajectory
Nominal stop time	8000 s	250000 s	Upper bound, not the reference end time
Base timestep / target CFL	1 s / 0.2	1 s / 0.2	Stable level-set integration
Primary output	fire-size statistics	fire-size statistics and TOA	Required scalar evidence and spatial context

Both namelists set `RANDOMIZE_RANDOM_SEED=.TRUE.`; consequently the initial result is statistically reproducible within its acceptance band but is not expected to be bitwise repeatable. The predicted initial result is a contained fraction near 0.54. The predicted extended result is a growing, progressively contained footprint whose statistics record ends near 88200 s.

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

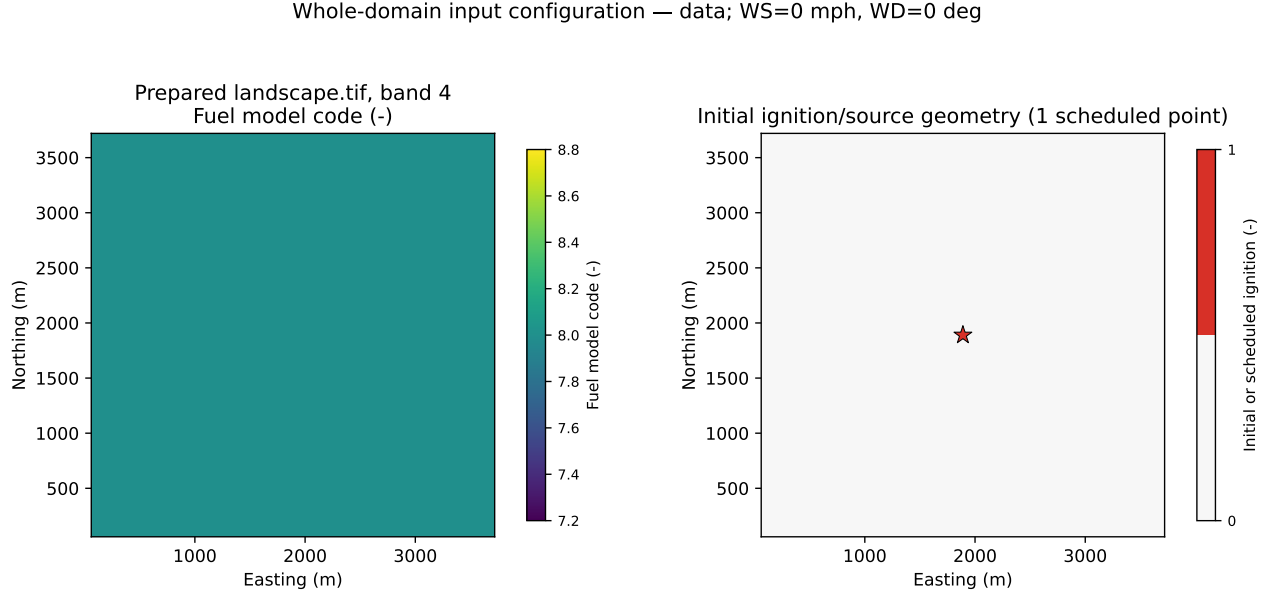


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The corrected case was executed with an ELMFIRE binary built from the current source tree. The initial-attack ensemble passes: its measured contained fraction is reported in the generated table below and lies inside the prescribed absolute tolerance. The extended-attack member does not reproduce the Guide response. It is stopped by the fire-front-stalled check before containment can evolve to the expected termination time; fire-size statistics consequently report the stop time as 0.00 h at their two-decimal output precision. Thus 1 of 2 required metrics pass and the current overall decision is **FAIL**.

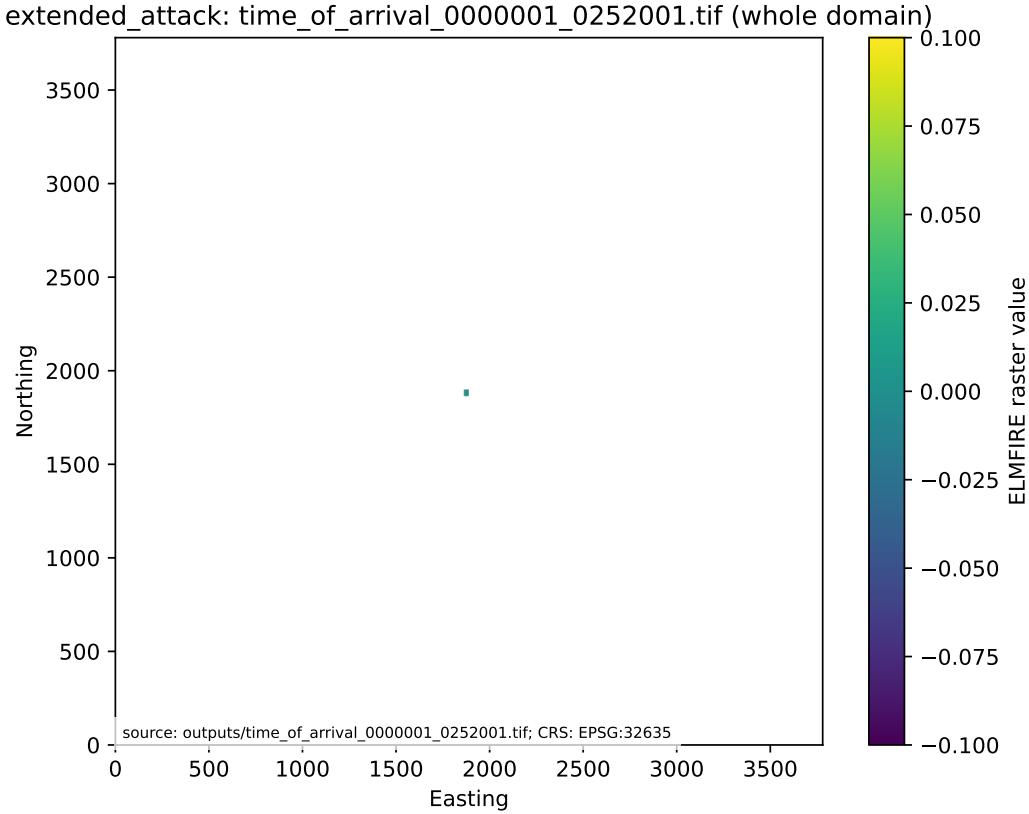


Figure 2: Whole-domain extended-attack time of arrival in seconds when a case-local output figure exists. It provides spatial context but cannot substitute for the scalar containment and end-time metrics.

7 Calculated metrics and verification decision

Variant / metric	Expected	Calculated	Status
extended_attack / time at simulation end	88200 s	0 s	FAIL
initial_attack / fraction contained	0.54 -	0.5785 -	PASS

The displayed decision is **FAIL**. When case-local outputs and `outputs/metrics.json` exist, `scripts/postprocess.py` generates every value and status through `report/metrics_macros.tex`; when they are absent, the wrapper renders NOT EVALUABLE without invented values.

8 Technical interpretation

The earlier case configuration used a 3600 s extended-attack interval, whereas the Guide area-growth verification prescribes 300 s; the source namelist has been corrected. That correction does not make the current-source result pass. With Fuel Model 8, the initial 1 s level-set change is below the absolute 0.001 stalled-front threshold in `elmfire_level_set.f90`. The stalled test is applied before adaptive-CFL integration has advanced beyond its startup steps, so this valid slow-spreading fire is classified as stalled essentially at ignition. Terminal bookkeeping then rounds the very small statistics stop time to 0.00 h. This remaining discrepancy is attributed to current ELMFIRE stall and termination logic, not to another defensible case-configuration change. Increasing the initial timestep merely to evade the

absolute threshold would change the prescribed Point setup and conceal the source behavior rather than verify it.

9 Scope and limitations

This case demonstrates only the two configured suppression pathways for uniform synthetic fuels, flat terrain, zero wind, one ignition location, one grid spacing, and the present executable/output conventions. It does not verify spatially explicit extended-attack model 1, indirect-attack geometry, nonuniform fuels, wind or slope interactions, resource limitations, resolution convergence, or operational suppression effectiveness. The 0.54 initial target is assessed statistically and can vary between reruns because seed randomization is enabled. The 88200 s extended target is a documented controlled-response reference rather than an analytical solution, so an intentional suppression-model change requires review of the oracle and tolerance. The evidence is implementation verification only and supplies no physical-validation claim.

Case 30: Multiple-ignition topology

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

Case ID: CASE30.MIT. This case verifies that the standard ELMFIRE executable initializes, schedules, transports, and merges multiple point ignitions without depending on source-tree test drivers. It preserves the scientific intent of Adam Laird’s dual-ignition, staggered spot-merging, and near-boundary component scenarios.

2 Mathematical formulation and numerical implementation

Each ignition introduces a local negative level-set region at position (x_i, y_i) and time t_i . ELMFIRE then advances ϕ according to $\partial_t \phi + V_n |\nabla \phi| = 0$. With homogeneous fuel, zero slope, and zero wind, simultaneous mirrored sources must remain reflection symmetric. Adding uniform wind must produce a measurably different arrival-time solution. Delayed sources must appear at their requested times, and clipping at a domain boundary must not suppress or wrap a source.

3 Derivation of expected behavior

The no-wind forcing and ignition pair are invariant under reflection about their mapped raster axes; therefore the discrete arrival-time solution has the same symmetries. Because ELMFIRE maps coordinates to one-based cells with a ceiling operation, the evaluator reflects the largest balanced row window about the actual ignition row rather than an intercell domain line. The wind variant changes only the wind rasters, so a nonzero solution difference is a metamorphic check of weather coupling. For scheduled ignition i , the containing raster cell should receive an arrival time within one 1200 s of t_i . The eight boundary sources are placed one five-cell margin inside the square domain and all must initialize.

For any two arrival-time arrays A and B , the implemented comparison is

$$E(A, B) = \frac{\text{mean}_{i \in V} |A_i - B_i|}{\max[P_{90}(A_i : i \in V), 1 \text{ s}]}, \quad V = \{i : A_i, B_i \text{ are finite}\}.$$

For the two simultaneous ignitions at $(-52.5, 0)$ and $(52.5, 0)$ m, the reflection reference is exact symmetry, $E = 0$; the metric is the larger of the left–right and centered top–bottom reflection errors and must satisfy $E \leq 0.02$. The 5 mph, 225-degree wind run is compared with the no-wind run by the same formula and must give $E \geq 0.05$, demonstrating a resolved forcing response. For scheduled ignitions (x, y, t) equal to $(0, 600, 0)$, $(0, 300, 1200)$, $(0, 0, 2400)$, $(0, -300, 3600)$, and $(0, -600, 4800)$, the timing metric is

$$E_t = \max_j |TOA(x_j, y_j) - t_j| \leq 1200 \text{ s}.$$

Finally, the count metric is $\sum_{j=1}^8 \mathbf{1}[TOA(x_j, y_j) \leq 1200 \text{ s}] = 8$ at the eight generated edge/corner points $(\pm 975, 0)$, $(0, \pm 975)$, and $(\pm 975, \pm 975)$ m.

4 Verification metrics and acceptance criteria

Reflection error is the larger normalized mean absolute difference between the no-wind arrival field and its reflection about each mapped ignition axis; it must not exceed 0.02. Wind response is the normalized mean absolute difference between the wind and no-wind fields and must be at least 0.05. The maximum scheduled-source timing error must not exceed one output interval. All eight boundary-source cells must have finite arrival times no later than one output interval. These metrics were selected because they test invariants of the coupled ignition/level-set implementation without reproducing ELMFIRE’s internal algorithms.

5 Simulation configuration and predicted outcome

All variants use a 5 m square grid in EPSG:32610, Fuel Model 10, 3/5/6 percent dead-fuel moisture, and flat terrain. The variants contain two simultaneous no-wind sources, the same pair under 5 mph wind from 225 degrees, five staggered sources, and eight edge/corner sources. In the staggered variant the first source is anchored at ($t=0$) and the other four follow at 1200-s intervals. This common translation of the original absolute schedule preserves every relative delay while preventing versions that have no active front from advancing past a strictly future first ignition. The predicted fields are respectively symmetric merger, wind-distorted merger, correctly timed merger, and eight clipped but valid boundary fires.

Parameter	Configured value	Units	Role
Domain/grid	2000 m square; 5	m/cell	Resolves source placement and merging
Fuel/moisture	FM10; 3/5/6	— / percent	Homogeneous surface response
Time control	step 1, maximum 5; stop 14400	s	Exact timestep multiple for scheduled sources
Variants	dual calm, dual wind, staggered, boundary	—	Exercises symmetry, weather, timing, and clipping
Output cadence	1200	s	Timing-error acceptance scale

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

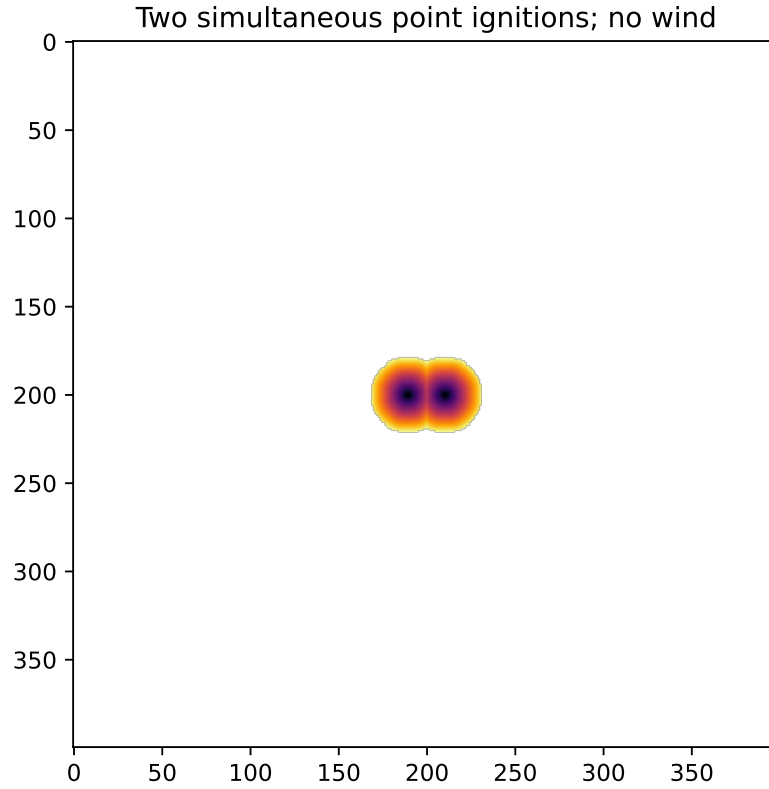


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The generated result is **PASS**; 4 of 4 required metrics pass across 4 variants. Missing ELMFIRE output yields NOT EVALUABLE rather than a pass.

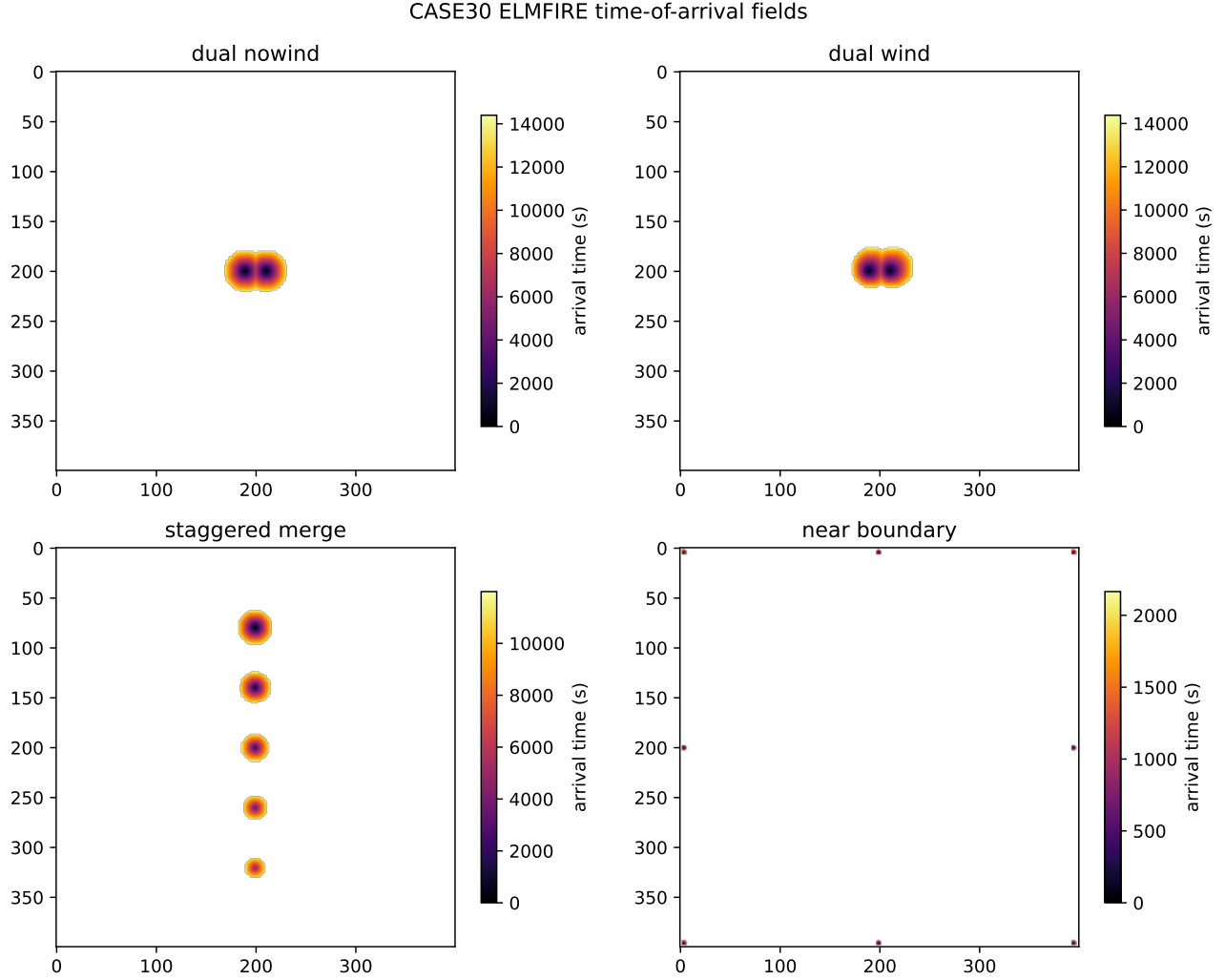


Figure 2: Case-local time-of-arrival fields for the four ignition-topology variants.

7 Calculated metrics and verification decision

Metric	Expected	Calculated	Status
dual-ignition reflection error	≤ 0.02 (2% normalized L1)	0.000472284	PASS
wind-forcing response	≥ 0.05 (5% normalized L1)	0.0937636	PASS
maximum scheduled-ignition timing error	≤ 1200 (one output interval)	0 s	PASS
boundary ignitions initialized	8 (all eight points)	8 count	PASS

The authoritative decision is **PASS** from `outputs/metrics.json`.

8 Technical interpretation

A symmetry failure points to directional indexing or level-set transport. No wind response indicates that the weather field did not reach fire behavior. Timing error implicates the scheduled-ignition path. Missing edge ignitions indicates boundary-index clipping or coordinate conversion failure.

9 Scope and limitations

This is deterministic implementation verification, not fire-behavior validation. It does not expose the internal Fortran ignition routines directly; it checks their observable behavior through an unmodified ELMFIRE executable.

Case 31: Planar-front transport

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

Case ID: CASE31_PFT. This case verifies planar level-set transport across diagonal geometry, a discontinuous fuel field, uniform wind, and a nonburnable fuel break. It consolidates Adam Laird’s diagonal-front, fuel-jump, fuel-jump-with-wind, and fuel-break scenarios into one portable black-box case.

2 Mathematical formulation and numerical implementation

For a planar front with signed-distance coordinate s , the level-set equation reduces to $\partial_t \phi + V_n \partial_s \phi = 0$. Uniform forcing should preserve planarity. At a fuel boundary V_n changes but the arrival-time field remains nearly constant along lines parallel to the front. A nonburnable strip imposes $V_n = 0$ except at an explicit gap.

3 Derivation of expected behavior

A uniform diagonal front is translationally invariant along its tangent. The fuel-jump solution is piecewise linear in normal distance, $T(s) = \int ds/V_n(s)$, so downstream time must increase while cross-front variation remains small. A 1 mph wind from 270 degrees aids eastward transport and must reduce downstream arrival time relative to the otherwise identical no-wind variant. A continuous nonburnable strip blocks transport; opening a gap creates a strictly larger burned fraction downstream.

At every declared station, arrival times are restricted to the central $|y| \leq 350$ m band (or the equivalent diagonal tangent band). The local planarity measure is

$$CV_j = \frac{\text{std}(T_j)}{\max[\text{mean}(T_j), 1 \text{ s}]}, \quad CV_{\max} = \max_j CV_j.$$

For the diagonal front, j spans nine normal-coordinate stations from 25 to 275 m; for the fuel jump it spans $x = \{-50, -20, 10, 40, 70, 100\}$ m. Both substituted tests are $CV_{\max} \leq 0.03$, with at least 90 percent finite station coverage. Let $\tilde{T}(x)$ be the station median. The transition metric is

$$\Delta T = \tilde{T}(100) - \tilde{T}(-50) > 0,$$

and the wind response is

$$q_w = \frac{\tilde{T}_{\text{wind}}(100)}{\tilde{T}_{\text{no wind}}(100)} \leq 0.95.$$

For the continuous and gapped Fuel Model 91 breaks, define $f(x) = N_{\text{finite}}/N_{\text{sampled}}$. After the required 90 percent upstream/barrier/opening coverage checks, the substituted barrier metric is $G = f_{\text{gapped}}(100) - f_{\text{continuous}}(100) \geq 0.10$.

4 Verification metrics and acceptance criteria

Cross-front coefficients of variation for diagonal and fuel-jump fronts must each be at most 3 percent. The nine diagonal-normal stations are 25, 56.25, 87.5, 118.75, 150, 181.25, 212.5, 243.75, and 275 m beyond the initial leading face. The six vertical-front stations are $x = -50, -20, 10, 40, 70$, and 100 m; each is sampled at the nearest raster column. Every station must have at least 90 percent finite, positive arrival-time coverage in the central $|y| \leq 350$ m corridor (or the corresponding diagonal tangent corridor). The evaluator reports the maximum station coefficient of variation, so one poorly resolved station cannot be hidden by aggregation. The downstream fuel-transition arrival-time increment between $x = -50$ m and $x = 100$ m must be positive. The wind/no-wind arrival-time ratio at $x = 100$ m must be at most 0.95. The burned-fraction gain at the same downstream station produced by the gap must be at least 0.10. Before that absence can be classified as a failure, both break variants must have at least 90 percent arrival coverage at $x = -50$ m, the continuous case must also reach the near-barrier station at $x = -5$ m, the gapped case must have at least 90 percent coverage through the opening at $x = 35$ m within $|y| \leq 40$ m, and its final dump must reach or exceed the configured stop time. Otherwise the barrier metric is NOT EVALUABLE. These geometry and metamorphic metrics avoid calibrating against ELMFIRE itself and remain valid across executable versions.

5 Simulation configuration and predicted outcome

All five variants use a 1500×1500 m domain at 5 m, Fuel Models 10 and 3 where applicable, and 3/5/6 percent dead-fuel moisture. The input `phi.tif` is a finite-strip signed-distance field clipped to ELMFIRE’s supported $[-1, 1]$ interval. It retains an ignited interior and a resolved planar leading-face gradient without relying on a grid cell whose value is exactly zero; its ends remain inside the domain. ELMFIRE applies adaptive CFL control only after the first five steps and declares a front stalled when the step-to-step PHI change stays below 0.001. The initial and maximum timestep are therefore both 5 s: for the slowest no-wind Fuel Model 10 state ($R \simeq 0.426$ m/min), one step moves the front about 0.0071 cell, above the stall comparison and far below the target CFL of 0.2. The stop time is 43,200 s, long enough for that slowest front to traverse the 275 m needed by the farthest diagonal station and the 195 m needed by the $x = 100$ m comparison station with margin. The break variants use Fuel Model 91 for a 30 m nonburnable strip, with or without a 100 m opening.

Parameter	Configured value	Units	Role
Domain/grid	1500 m square; 5	m/cell	Resolves planar fronts and fuel break
Fuel/moisture	FM10/FM3/FM91; 3/5/6	— / percent	Uniform, transition, and nonburnable states
Time control	step 5; stop 43200	s	Exact multiple with traversal margin
Wind	0 or 1 from 270	mph / degrees	Metamorphic acceleration pair
Gap	closed or 100	m	Connectivity response

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

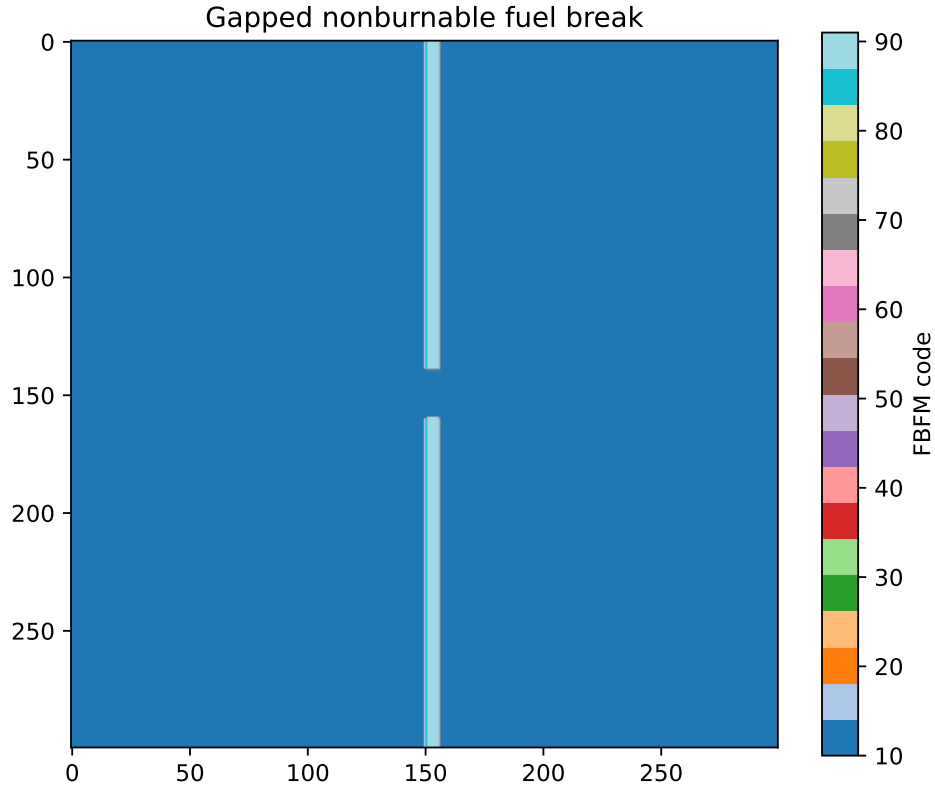


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The generated status is **PASS**; 5 of 5 metrics pass across 5 variants. Missing required rasters, an obsolete or geometrically inconsistent `phi.tif`, a final-dump mismatch, or inadequate arrival coverage produces NOT EVALUABLE. The generated diagnostic is: `-`.

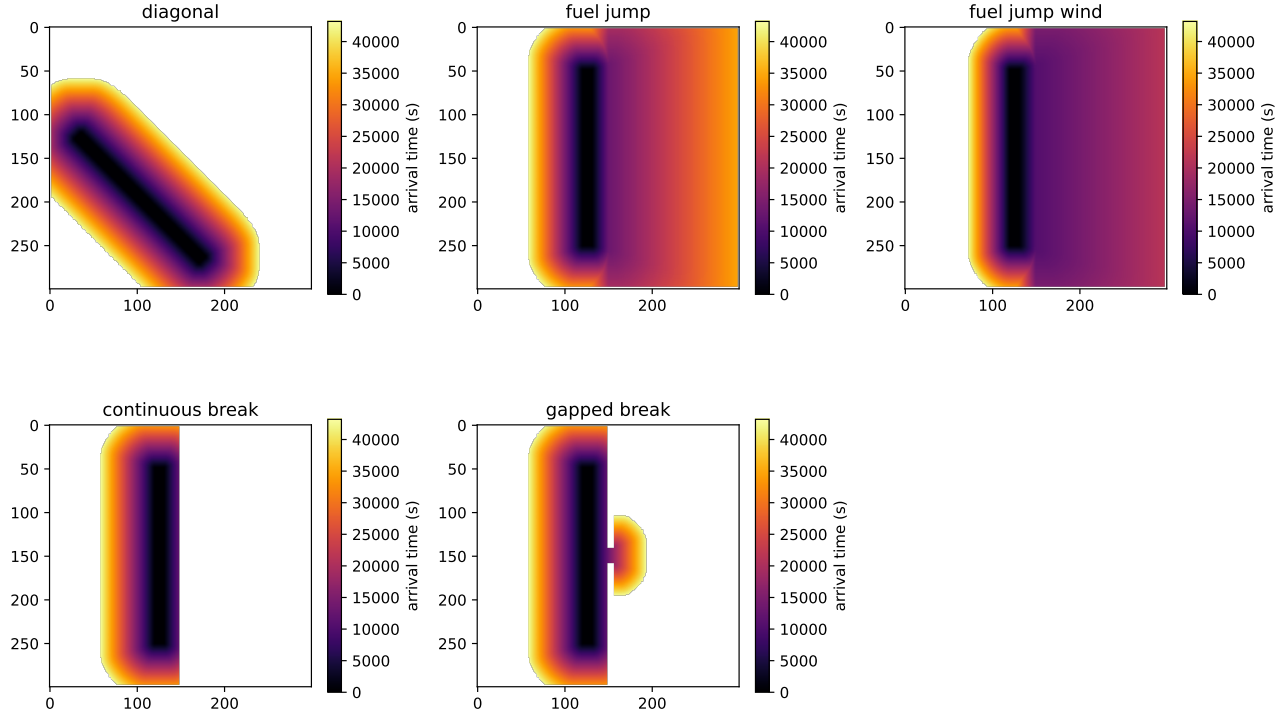


Figure 2: Time-of-arrival fields for all planar-front variants.

7 Calculated metrics and verification decision

Metric	Expected	Calculated	Status
diagonal-front cross-front variation	≤ 0.03 (3% coefficient of variation)	8.35166e-05	PASS
fuel-jump front planarity	≤ 0.03 (3% coefficient of variation)	7.71843e-05	PASS
fuel-transition arrival-time increment	> 0 (positive downstream travel time)	9920 s	PASS
eastward-wind arrival-time ratio	≤ 0.95 (at least 5% earlier than no-wind)	0.756028	PASS
gapped-break downstream burned-fraction gain	≥ 0.10 (10 percentage points)	0.657143	PASS

The authoritative decision is **PASS** from `outputs/metrics.json`.

8 Technical interpretation

Loss of planarity suggests orientation-dependent transport or indexing. A nonpositive transition time indicates an invalid arrival field. Lack of wind acceleration indicates a weather/fire-behavior coupling problem. Lack of gap response indicates incorrect nonburnable-fuel or front-connectivity handling.

9 Scope and limitations

The case verifies observable coupled behavior rather than individual internal Fortran routines. It does not assert absolute Rothermel rates, and the selected fuel and wind states are not field validation data.

Case 32: Raster-orientation covariance

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

Case ID: CASE32_RCV. This case verifies that ELMFIRE’s raster indexing and level-set transport are covariant under a 90° rotation. It is the portable successor to Adam Laird’s `inputs-orientation` scenario.

2 Mathematical formulation and numerical implementation

For homogeneous isotropic forcing, the operator $\mathcal{L}$ in $\partial_t \phi = \mathcal{L}(\phi)$ commutes with any rigid rotation Q : $\mathcal{L}(\phi \circ Q^{-1}) = (\mathcal{L}\phi) \circ Q^{-1}$. A square north-up raster rotated by exactly 90 degrees requires no interpolation, so cell-by-cell comparison is available.

3 Derivation of expected behavior

The rotated variant is generated with `numpy.rot90` from the asymmetric L-shaped initial negative- ϕ region. Fuel, moisture, slope, and wind are spatially uniform. Therefore the rotated arrival-time validity mask and values must equal a 90-degree rotation of the original result, apart from bounded grid-level roundoff.

The reference array is generated directly from the original ELMFIRE arrival-time result by a 90-degree counterclockwise rotation, $T_{\text{ref}} = \text{rot90}(T_{\text{original}})$; no arrival values are invented. With $V = \{i : T_{\text{ref},i} \text{ and } T_{\text{rot},i} \text{ finite}\}$ and $U = \{i : \text{either value is finite}\}$, finite-mask agreement is

$$A_m = \frac{|V|}{|U|},$$

whose exact rotational reference is 1 and whose substituted test is $A_m \geq 0.99$. On the mutual finite mask, the value metric is

$$E_T = \frac{|V|^{-1} \sum_{i \in V} |T_{\text{ref},i} - T_{\text{rot},i}|}{\max[P_{90}(T_{\text{ref},i} : i \in V), 1 \text{ s}]},$$

with exact reference zero and acceptance substitution $E_T \leq 0.02$. The asymmetric generated ignition mask makes this comparison sensitive to an erroneous reflection or transposition.

4 Verification metrics and acceptance criteria

Finite-mask intersection over union must be at least 0.99. On mutually finite cells, normalized mean absolute arrival-time error must be at most 0.02. Mask agreement detects row/column reversal and georeferencing errors; value error detects orientation-dependent numerical evolution.

5 Simulation configuration and predicted outcome

Both variants use a 2000 m square domain at 5 m, Fuel Model 10, flat terrain, zero wind, and identical timing. The only changed scientific input is the exact raster rotation of `phi.tif`. The second result is predicted to be the corresponding rotation of the first.

Parameter	Configured value	Units	Role
Domain/grid	2000 m square; 5	m/cell	Exact square raster rotation
Fuel/forcing	FM10; flat; calm	—	Isotropic homogeneous operator
Time control	step 1, maximum 5; stop 7200	s	Common exact comparison time
Initial field	asymmetric L shape	—	Makes orientation errors observable
Variant transform	90-degree rotation	degrees	Exact no-interpolation covariance test

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

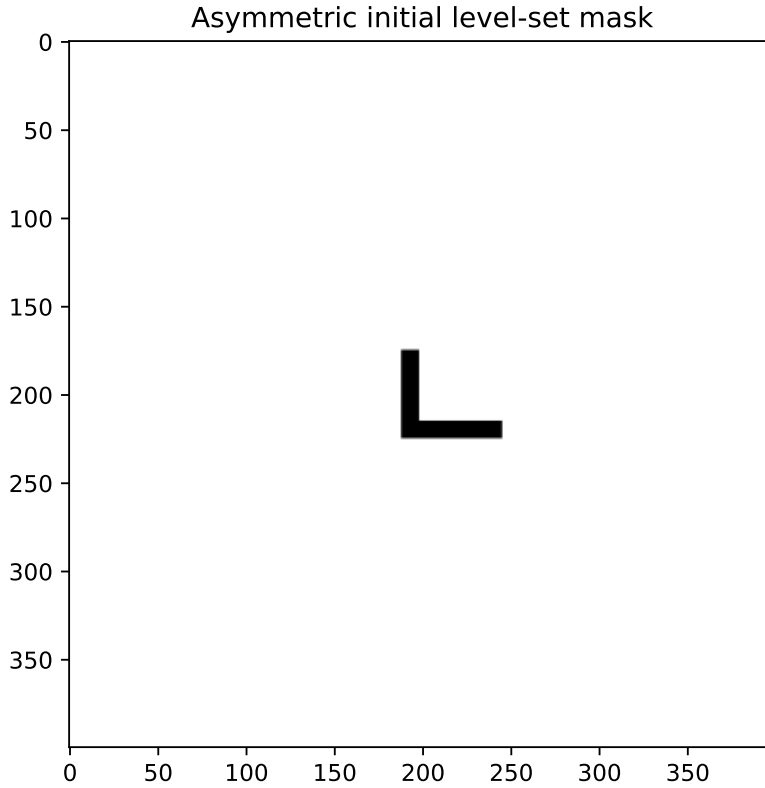


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The generated status is **PASS**; 2 of 2 metrics pass across 2 variants. Missing output produces NOT EVALUABLE.

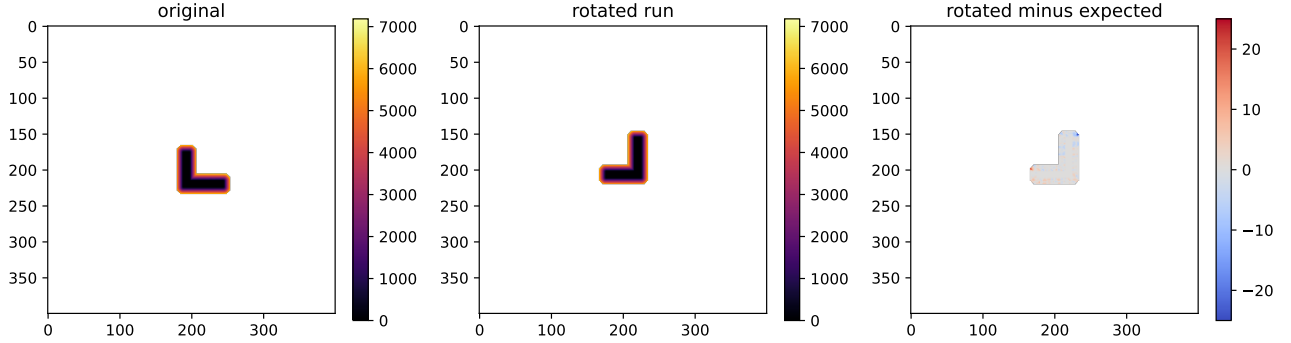


Figure 2: Original, independently rotated, and difference arrival-time fields.

7 Calculated metrics and verification decision

Metric	Expected	Calculated	Status
rotated finite-mask agreement	$\hat{c} = 0.99$ (1% disagreement)	1 fraction	PASS
rotated arrival-time error	$\hat{e} = 0.02$ (2%)	0.000214081 normalized L1	PASS

The authoritative decision is **PASS** from `outputs/metrics.json`.

8 Technical interpretation

Mask disagreement indicates inconsistent raster orientation, bounds, or row/column indexing. Value disagreement with matching masks indicates a directional bias in transport or output serialization.

9 Scope and limitations

The 90-degree test avoids resampling and therefore does not assess arbitrary-angle reprojection. It verifies executable behavior, not the internal bilinear-interpolation routine in isolation.

Case 33: Slope-driven directional spread

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

Case ID: CASE33_SD0. This case verifies the observable coupling of topographic slope into directional surface-fire spread through the standard ELMFIRE executable. It is the portable successor to Adam Laird’s no-wind-with-slope component scenario and retains a flat control.

2 Mathematical formulation and numerical implementation

ELMFIRE combines the no-wind spread rate with a slope factor before supplying a direction-dependent normal speed $V_n(\theta)$ to $\partial_t \phi + V_n |\nabla \phi| = 0$. With zero wind and zero slope, rotational symmetry is expected. A uniform nonzero slope breaks that symmetry along the fall-line axis.

3 Derivation of expected behavior

The flat control has equal forcing in all azimuths and therefore equal horizontal and vertical extents. The 30° variant changes only slope and aspect; it must differ from the control and its final footprint must be elongated. These are metamorphic consequences of the governing coupling and do not require duplication of ELMFIRE’s internal slope formula.

For each finite arrival-time mask, the pixel extents are $W = c_{\max} - c_{\min} + 1$ and $H = r_{\max} - r_{\min} + 1$. The flat-control anisotropy is

$$A_f = \frac{|W_f - H_f|}{\max(W_f, H_f)},$$

whose isotropic reference is zero and whose substituted test is $A_f \leq 0.05$. For the generated uniform 30-degree slope, the directional-shape metric is

$$L_s = \frac{\max(W_s, H_s)}{\min(W_s, H_s)} \geq 1.10.$$

The independent forcing-response metric uses the common finite mask V :

$$E_s = \frac{|V|^{-1} \sum_{i \in V} |T_{s,i} - T_{f,i}|}{\max[P_{90}(T_{s,i} : i \in V), 1 \text{ s}]} \geq 0.05.$$

Thus the three reference statements are, respectively, zero flat anisotropy, at least 10 percent slope elongation, and at least 5 percent normalized difference from the flat run.

4 Verification metrics and acceptance criteria

Flat-control extent anisotropy must not exceed 0.05. The slope footprint’s major/minor extent ratio must be at least 1.10. Normalized mean absolute arrival-time difference between slope and flat variants must be at least 0.05. The first metric detects grid bias, while the latter two show that slope is both active and directional.

5 Simulation configuration and predicted outcome

Both variants use a 2000 m square domain at 5 m, Fuel Model 10, 3/5/6 percent dead-fuel moisture, zero wind, and a central point ignition. The slope variant uses 30 degrees and aspect 270 degrees; the control uses zero slope. The slope result is predicted to be anisotropic and measurably different from the circular control.

Parameter	Configured value	Units	Role
Domain/grid	2000 m square; 5	m/cell	Resolves directional footprint
Fuel/moisture	FM10; 3/5/6	— / percent	Fixed surface response
Time control	step 1, maximum 5; stop 7200	s	Common exact comparison time
Control	slope 0; wind 0	degrees / mph	Rotational baseline
Test	slope 30; aspect 270	degrees	Activates directional slope coupling

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

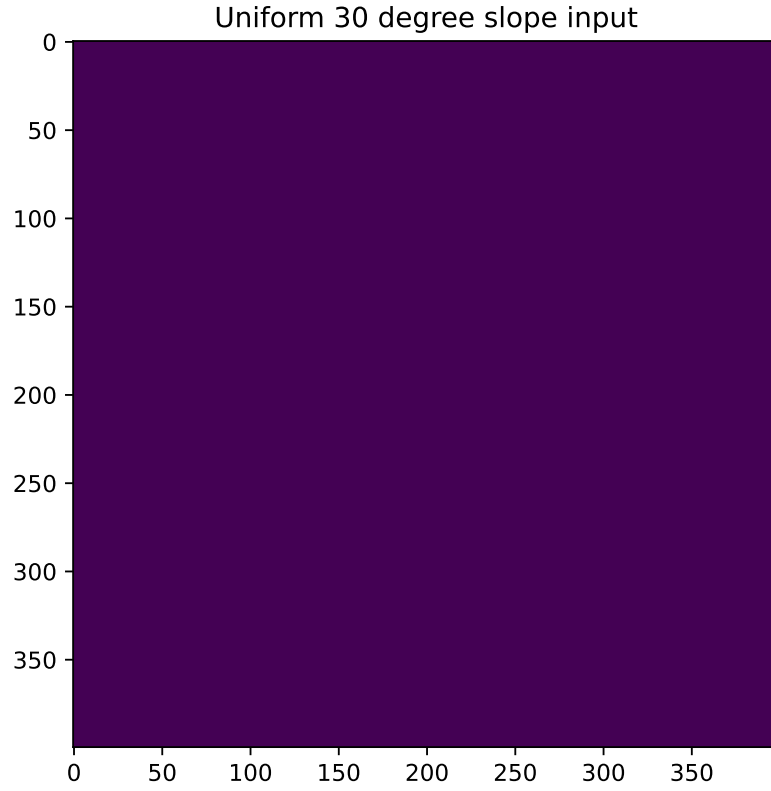


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The generated status is **PASS**; 3 of 3 metrics pass across 2 variants. Missing outputs produce NOT EVALUABLE.

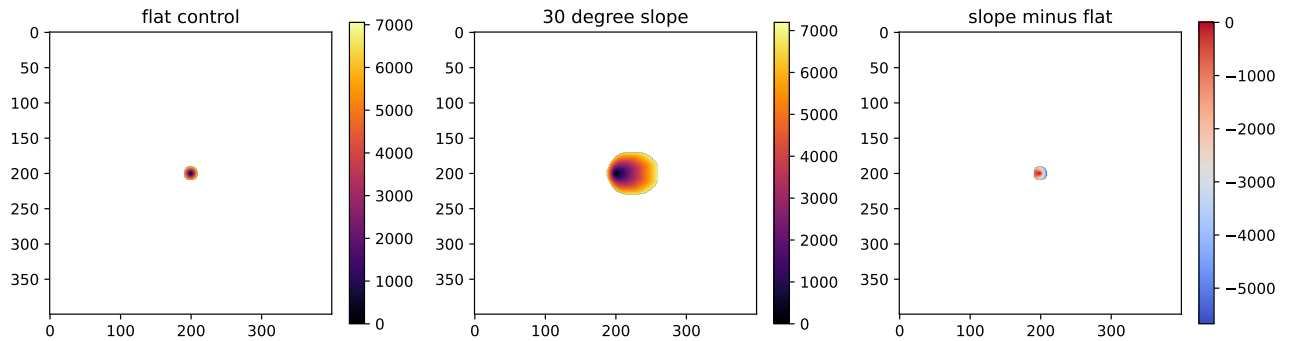


Figure 2: Flat-control, slope, and difference arrival-time fields.

7 Calculated metrics and verification decision

Metric	Expected	Calculated	Status
flat-control anisotropy	$\hat{\mu} = 0.05$ (5%)	0 fraction	PASS
slope-front elongation ratio	$\hat{\mu} = 1.10$ (10% directional elongation)	1.1746 ratio	PASS
slope-forcing response	$\hat{\mu} = 0.05$ (5% difference from flat)	0.520518 normalized L1	PASS

The authoritative decision is **PASS** from `outputs/metrics.json`.

8 Technical interpretation

An anisotropic control indicates grid-orientation bias. A near-unity slope footprint or negligible field difference indicates that slope/aspect inputs were not consumed or not coupled to spread. Unexpected rotation should prompt inspection of aspect convention and raster orientation.

9 Scope and limitations

This case verifies directional response, not the absolute value of the slope coefficient and not agreement with an experimental fire. Existing CASE22 and CASE24 exercise more complex terrain interactions; this case isolates the slope-only executable response.

Case 34: Standard fuel-model Rothermel sweep

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

This executable-level known-response test verifies that ELMFIRE reads standard fuel-model parameters and evaluates the no-wind, no-slope Rothermel surface spread model for FBFM2, FBFM9, GR2, TU5, and TL8. It exercises fuel-table parsing, class weighting, reaction intensity, heat sink, and the base rate of spread while crown fire, spotting, buildings, acceleration, wind, and slope are excluded. It can expose column-order, unit, weighting, and model-selection errors. This is implementation verification, not comparison with an observed fire. The verification question is whether all five variants reproduce the independent reaction-intensity and base-ROS calculations within 1% and 2%, with TOA-derived ROS within 5%.

2 Mathematical formulation and numerical implementation

The no-wind, no-slope rate is

$$R_0 = \frac{I_R \xi}{\rho_b \epsilon Q_{ig}}, \quad (1)$$

where I_R is reaction intensity, ξ is the propagating-flux ratio, ρ_b is bulk density, ϵ is effective heating number, and $Q_{ig} = 250 + 1116M_f$ in native Rothermel units. Dead and live classes are area-weighted before their moisture-damping contributions are combined. The equations and standard fuel definitions follow Patricia L. Andrews, *The Rothermel Surface Fire Spread Model and Associated Developments*, RMRS-GTR-371 (2018). ELMFIRE returns spread rate in metres per minute and reaction intensity in kW m^{-2} .

3 Derivation of expected behavior

The case-local `rothermel.reference.py` reads the same frozen fuel-table rows but independently evaluates every algebraic step stated above. It does not import ELMFIRE source or use a simulation output as a reference. Each predicted I_R and R_0 is therefore a deterministic known response. A bounded planar signed-distance front turns the scalar normal speed into a second independent prediction, $t(x) = t_0 + (x - x_0)/(R_0/60)$.

Each reference is calculated independently with

$$R_0 = \frac{I_R \xi}{\rho_b \epsilon Q_{ig}}, \quad Q_{ig} = 250 + 1116M_f,$$

using the generated common moisture inputs and the named standard-fuel record. The resulting

substitutions are

fuel model	R_{ref} (m min ⁻¹)	$I_{R,\text{ref}}$ (kW m ⁻²)
2	0.823796	675.184
9	0.306627	477.714
102	0.228507	186.041
165	0.325722	1595.730
188	0.275244	646.706

(for example, Fuel Model 2 gives $2.702741 \text{ ft min}^{-1}(0.3048) = 0.823796 \text{ m min}^{-1}$). For variant i , direct errors are $e_{R,i} = |R_{\text{max},i} - R_{\text{ref},i}| / \max(|R_{\text{ref},i}|, 10^{-8})$ and the analogous $e_{I,i}$. The TOA fit is $t = a + bx$, so $R_{\text{TOA}} = 60/b$ and $R^2 = 1 - \sum(t - \hat{t})^2 / \sum(t - \bar{t})^2$. The reported substitutions are $\max_i e_{R,i} \leq 0.02$, $\max_i e_{I,i} \leq 0.01$, $\max_i |R_{\text{TOA},i} - R_{\text{ref},i}| / R_{\text{ref},i} \leq 0.05$, and $\min_i R_i^2 \geq 0.995$.

4 Verification metrics and acceptance criteria

The direct spread raster tests the calculated speed with maximum relative error 0.02; the reaction-intensity raster tests the source term with maximum relative error 0.01. Their medians use colocated valid cells at the planar head, beginning one cell behind the initial east face, so direct verification does not require the front to traverse a downstream corridor. Linear regression of positive arrival time against interior x coordinate tests propagation independently: maximum ROS error is 0.05 and minimum R^2 is 0.995. TOA is required only for variants whose planned duration supports a multi-cell fit. All five variants and all four aggregate metrics are required; missing or malformed required evidence gives NOT EVALUABLE.

5 Simulation configuration and predicted outcome

The finite-strip PHI raster is a signed-distance field clipped to ELMFIRE’s supported $[-1, 1]$ interval, preserving both an ignited interior and a resolved front gradient. For every positive oracle ROS, **SIMULATION_DT** and **SIMULATION_DTMAX** use the whole-second value obtained by rounding the 5%-of-cell target downward, with a one-second minimum; every generated value remains below the configured CFL limit of 0.2. The stop time is rounded upward to an exact integer multiple of that step. This avoids a shortened final step that can trigger ELMFIRE’s stalled-front branch and corrupt the terminal timestamp. A zero-ROS variant retains a 5 s step and is evaluated from direct fields without requiring TOA propagation. The evaluator always requires uniquely named rasters tied to one final dump record. It accepts a stalled-final record only when subtracting the configured meteorology interval reconstructs the requested stop time within one configured timestep. A TOA-required variant must reach the **SIMULATION_TSTOP** in its generated namelist; a direct-only variant may terminate early because lack of propagation is part of the state being tested. The domain is 1400 m square with 10 m cells. A finite 50 m-wide ignition strip has a planar east face initially at $x = -400$ m and terminates well inside the north and south boundaries; the interior analysis corridor is therefore one-dimensional without presenting ELMFIRE with an edge-to-edge contained front. Moistures are 5, 7, and 9% for dead classes and 90% for both live classes. Wind and slope are zero. Run duration is selected from the predicted rate to provide approximately 300 m of travel. The interior results should be planar but need not have a monotonic ordering across fuel models because packing, SAV, load, depth, and extinction moisture interact.

Parameter	Configured value	Units	Role
Domain/grid	1400 m square; 10	m/cell	Planar interior propagation
Fuel sweep	FBFM2, FBFM9, GR2, TU5, and TL8	code/name	Independent variable
Moisture	dead 5/7/9; live 90/90	percent	Fixed damping state
Wind/slope	0 / 0	mph / degrees	Isolates base fuel response
Timing	generated whole-second step and exact-multiple stop	s	CFL-safe terminal evidence

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

Whole-domain input configuration — variants/fuel_102; WS=0 mph, WD=270 deg

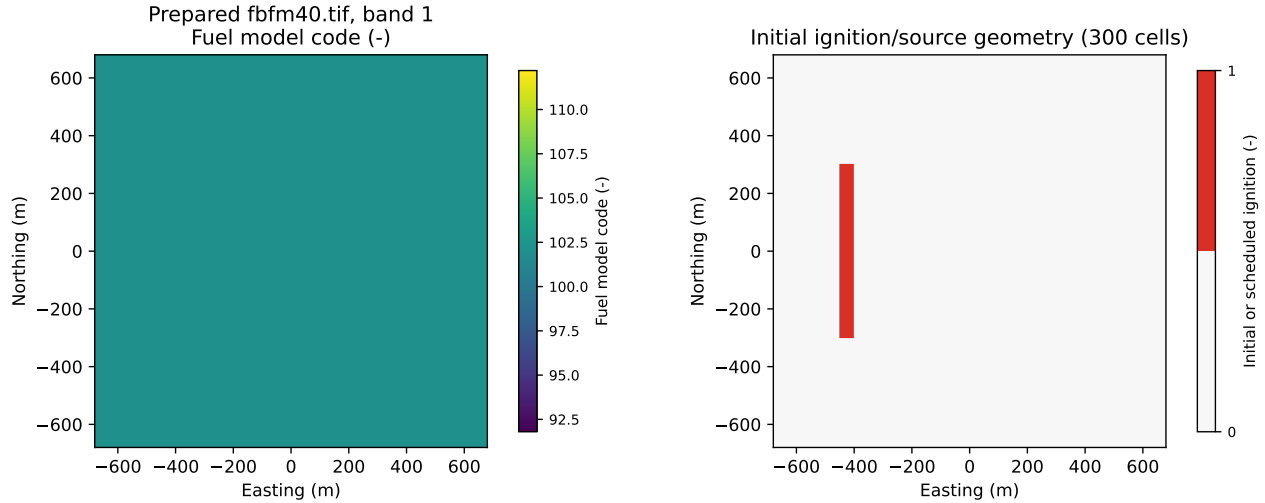


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The generated record contains 5 of 5 required variants. Its authoritative state is **PASS**. No plotted curve is substituted for missing simulation artifacts.

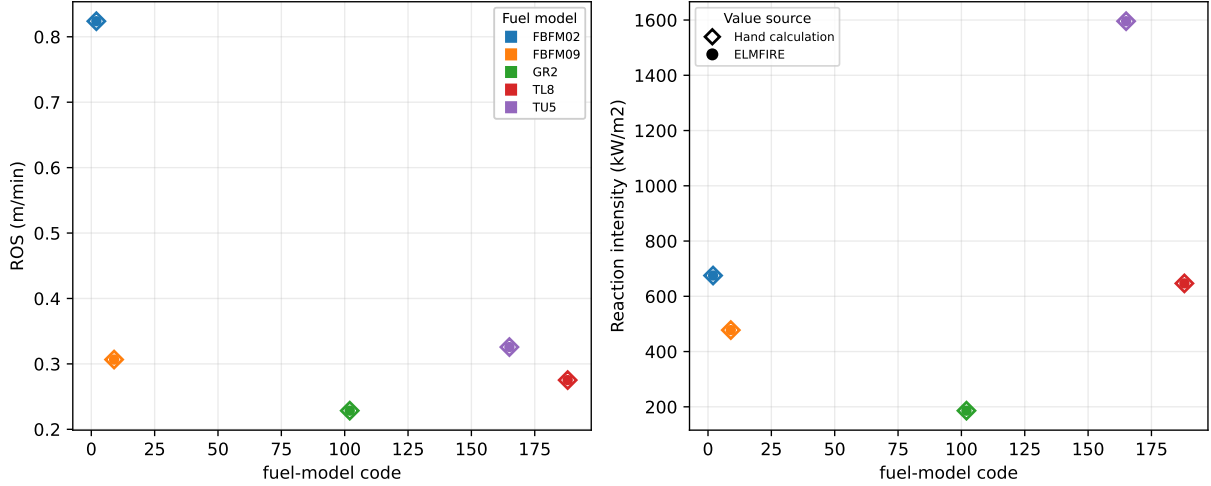


Figure 2: Hand-calculated and ELMFIRE rate of spread (ROS, m/min) and reaction intensity (kW/m²) for all five completed standard-fuel variants. Each fuel model uses one color in both panels; open diamonds denote the independent hand calculation and filled circles denote ELMFIRE.

7 Calculated metrics and verification decision

Metric	Limit	Calculated	Status
maximum direct-ROS relative error	≤ 0.02	3.86591e-07 fraction	PASS
maximum reaction-intensity relative error	≤ 0.01	2.93279e-07 fraction	PASS
maximum TOA-ROS relative error	≤ 0.05	0.0115779 fraction	PASS
minimum TOA regression R2	≥ 0.995	0.999961 dimensionless	PASS

The overall decision is **PASS**; a successful executable exit alone is not a pass.

8 Technical interpretation

An isolated intensity discrepancy implicates fuel parsing, class weighting, or moisture damping. Correct intensity with incorrect direct ROS implicates flux ratio, density, heat sink, adjustment, or output units. Correct direct ROS with a TOA discrepancy implicates level-set transport, raster orientation, or time units.

9 Scope and limitations

The case verifies five representative standard fuels at one moisture state. It does not validate Rothermel against experiments, exhaust all fuel models, or test wind, slope, curing, crown fire, or spatial heterogeneity; those mechanisms have separate cases.

Case 35: Midflame wind-speed Rothermel sweep

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

This case verifies ELMFIRE’s Rothermel wind response for FBFM2, FBFM9, GR2, TU5, and TL8 over 0, 1, 2, 4, 6, 8, and 12 mph midflame. It isolates the midflame wind factor while separately accounting for ELMFIRE’s public 20-ft wind input and unsheltered wind-adjustment factor (WAF). It can expose errors in wind units, WAF inversion, exponentiation, fuel-dependent coefficients, or wind limiting. The question is whether every variant agrees with the independent implementation-limited ROS within 2%, reaction intensity within 1%, and TOA-derived ROS within 5%.

2 Mathematical formulation and numerical implementation

Rothermel wind enhancement is

$$R = R_0(1 + \phi_w), \quad \phi_w = CU^B(\beta/\beta_{op})^{-E}, \quad (1)$$

with U in ft min^{-1} at midflame. Coefficients are $C = 7.47e^{-0.133\sigma^{0.55}}$, $B = 0.02526\sigma^{0.54}$, and $E = 0.715e^{-0.000359\sigma}$. The reference is Andrews (2018), RMRS-GTR-371. The inspected ELMFIRE implementation limits U to $0.9I_R$ in native units. ELMFIRE inputs 20-ft wind in mph, so preprocessing divides each target midflame speed by the independently computed unsheltered WAF.

3 Derivation of expected behavior

Below the limit, $(R/R_0 - 1)$ follows the fuel-specific power law in U ; once the limit is reached, the wind contribution is constant. The case-local oracle computes R_0 , I_R , WAF, ϕ_w , limit location, and final ROS from the paper equations and the documented ELMFIRE limiting policy. This separates paper-equation behavior from the implementation’s explicit cap.

The wind series uses the Rothermel wind factor

$$\phi_w = C U_{mf}^B, \quad R = 0.3048 R_0^{(\text{ft}/\text{min})}(1 + \phi_w),$$

with U_{mf} in ft min^{-1} and fuel-specific B, C . For a concrete generated substitution, Fuel Model 2 at $U_{mf} = 4 \text{ mph} = 352 \text{ ft min}^{-1}$ has $C = 0.0002125869$, $B = 1.8304427$, and

$$\phi_w = 0.0002125869(352)^{1.8304427} = 9.746235,$$

$$R_{\text{ref}} = 0.3048(2.702741)(1 + 9.746235) = 8.852700 \text{ m min}^{-1}, \quad I_{R,\text{ref}} = 675.184 \text{ kW m}^{-2}.$$

The same substitution is applied to every generated fuel/wind pair (0, 1, 2, 4, 6, 8, 12 mph); the 20-ft raster value is $U_{20} = U_{mf}/WAF$, e.g. $4/0.362674 = 11.02919 \text{ mph}$. For each variant, $e_R = |R_{\text{max}} - R_{\text{ref}}|/\max(|R_{\text{ref}}|, 10^{-8})$ and e_I is analogous. A TOA regression $t = a + bx$ gives $R_{\text{TOA}} = 60/b$ and $R^2 = 1 - SSE/SST$. The aggregated tests substitute $\max e_R \leq 0.02$, $\max e_I \leq 0.01$, $\max e_{\text{TOA}} \leq 0.05$, and $\min R^2 \geq 0.995$.

4 Verification metrics and acceptance criteria

Maximum relative errors are 0.02 for the final direct ROS, 0.01 for reaction intensity, and 0.05 for TOA-derived ROS. The minimum TOA regression R^2 is 0.995. Direct pointwise comparison is more diagnostic than monotonicity alone because it detects an incorrect exponent or premature plateau even when the curve still increases. All 35 fuel–wind combinations are required.

5 Simulation configuration and predicted outcome

The finite-strip PHI raster is a signed-distance field clipped to ELMFIRE’s supported $[-1, 1]$ interval, preserving both an ignited interior and a resolved front gradient. For every positive oracle ROS, `SIMULATION_DT` and `SIMULATION_DTMAX` use the whole-second value obtained by rounding the 5%-of-cell target downward, with a one-second minimum; every generated value remains below the configured CFL limit of 0.2. The stop time is rounded upward to an exact integer multiple of that step. This avoids a shortened final step that can trigger ELMFIRE’s stalled-front branch and corrupt the terminal timestamp. A zero-ROS variant retains a 5 s step and is evaluated from direct fields without requiring TOA propagation. The evaluator always requires uniquely named rasters tied to one final dump record. It accepts a stalled-final record only when subtracting the configured meteorology interval reconstructs the requested stop time within one configured timestep. A TOA-required variant must reach the `SIMULATION_TSTOP` in its generated namelist; a direct-only variant may terminate early because lack of propagation is part of the state being tested. Each homogeneous 1400 m square variant uses 10 m cells and a finite ignition strip whose east face is planar at $x = -400$ m within the interior analysis corridor. The strip ends inside the lateral boundaries so it remains an active fire rather than an edge-to-edge contained front. Slope is zero, dead moisture is 5/7/9%, and live moisture is 90%. The wind is from the west. Crown fire, spotting, buildings, fluctuations, acceleration, and Monte Carlo variation are disabled. Curves should increase as U^B and may flatten at fuel-dependent limits.

Parameter	Configured value	Units	Role
Domain/grid	1400 m square; 10	m/cell	Planar interior propagation
Fuel sweep	FBFM2, FBFM9, GR2, TU5, and TL8	code/name	Fuel-dependent response
Wind sweep	0, 1, 2, 4, 6, 8, and 12 mph midflame	mph	Independent midflame target
Moisture/slope	dead 5/7/9, live 90/90; 0	percent / degrees	Fixed damping and terrain
Timing	generated whole-second step and exact-multiple stop	s	CFL-safe terminal evidence

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

Whole-domain input configuration — variants/fuel_102_u_2p0; WS=5.51 mph, WD=270 deg

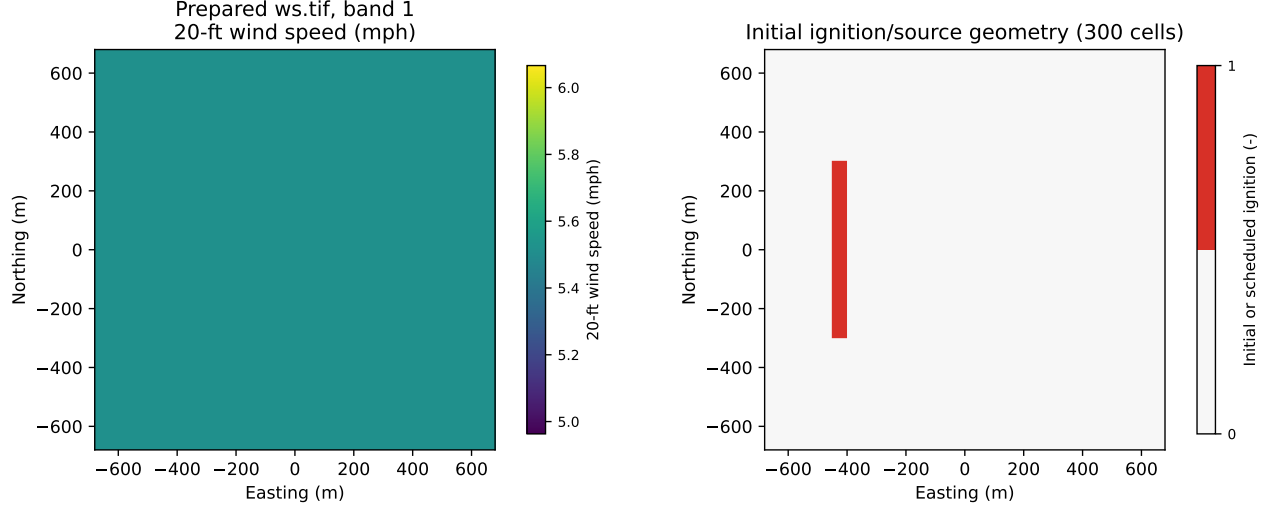


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The generated record contains 35 of 35 required variants and reports **PASS**.

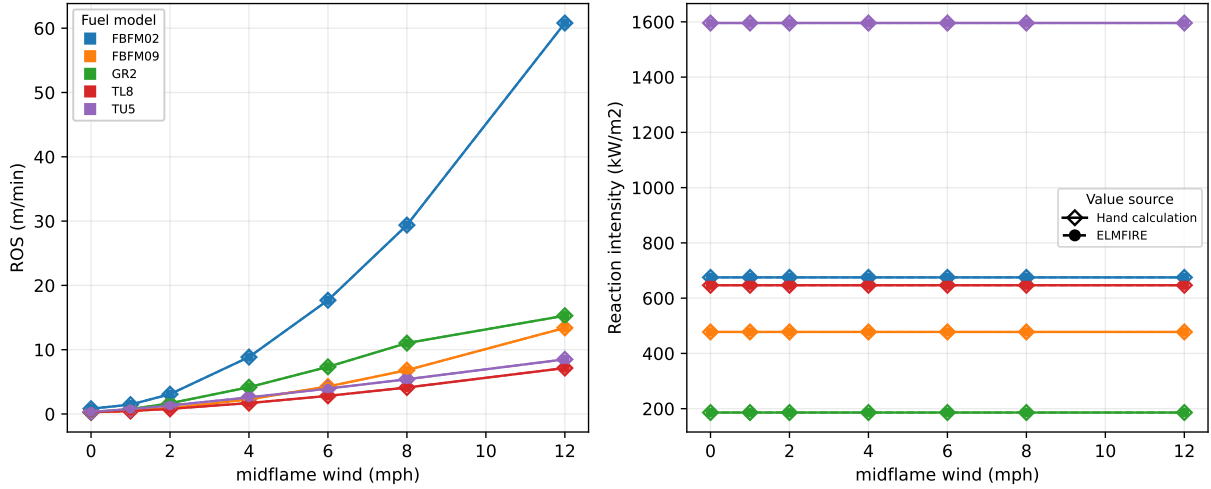


Figure 2: Hand calculations (open diamonds, solid lines) and ELMFIRE predictions (filled circles, dashed lines). Each fuel-model pair uses the same color. Target values are midflame wind; namelists contain the corresponding 20-ft wind.

7 Calculated metrics and verification decision

Metric	Limit	Calculated	Status
maximum direct-ROS relative error	≤ 0.02	1.17774e-06 fraction	PASS
maximum reaction-intensity relative error	≤ 0.01	2.93279e-07 fraction	PASS
maximum TOA-ROS relative error	≤ 0.05	0.028404 fraction	PASS
minimum TOA regression R2	≥ 0.995	0.999958 dimensionless	PASS

The authoritative overall decision is **PASS**.

8 Technical interpretation

A common scale error across fuels suggests mph, ft/min, or WAF conversion. Fuel-dependent curvature errors suggest B , C , E , or packing-ratio handling. A wrong plateau indicates a changed or incorrect wind-limit policy. Correct direct ROS but incorrect TOA points to transport rather than the Rothermel calculation.

9 Scope and limitations

The case verifies steady, aligned wind and the currently selected original wind limit. If a future ELMFIRE release intentionally adopts a corrected or optional limit, the scientific oracle must be reviewed rather than changing only the namelist. Canopy-dependent WAF is tested in CASE42.

Case 36: Slope-factor Rothermel sweep

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

This case verifies the scalar Rothermel slope response for FBFM2, FBFM9, GR2, TU5, and TL8 over 0, 10, 20, 30, 50, 75, and 100% rise over run. It tests slope-raster units, conversion to $\tan^2 \phi$, fuel-dependent packing response, and ELMFIRE's slope-factor cap without wind. The verification question is whether all 35 variants reproduce independently calculated intensity and slope-enhanced ROS within the stated tolerances.

2 Mathematical formulation and numerical implementation

The no-wind response is

$$R = R_0(1 + \phi_s), \quad \phi_s = 5.275 \beta^{-0.3} \tan^2 \phi. \quad (1)$$

Here reported slope percent is rise/run times 100 and the raster value is the angle $\phi = \tan^{-1}(s/100)$ in degrees. The formulation follows Andrews (2018), RMRS-GTR-371. The inspected ELMFIRE source additionally caps ϕ_s at the maximum wind factor derived from its wind limit; the oracle reproduces and reports that implementation policy.

3 Derivation of expected behavior

Before the cap, plotting ϕ_s against $\tan \phi$ on logarithmic axes has slope two. For each fuel and raster angle, the local oracle calculates R_0 , the uncapped factor, the cap, and final R . Because the initial level set is planar and aligned with the upslope direction, the output normal speed equals this scalar head-spread prediction.

The slope-only reference uses

$$\phi_s = 5.275 \beta^{-0.3} \tan^2 \theta, \quad R = R_0(1 + \phi_s).$$

As an explicit generated substitution, the requested 20 percent slope is stored as the integer raster angle $\theta = 11$ degrees ($100 \tan 11^\circ = 19.4380$ percent). For Fuel Model 2, $\beta = 0.00575$, $R_0 = 0.8237955 \text{ m min}^{-1}$, and

$$\phi_s = 5.275(0.00575)^{-0.3} \tan^2(11^\circ) = 0.936756, \quad R_{\text{ref}} = 0.8237955(1+0.936756) = 1.595491 \text{ m min}^{-1}.$$

Reaction intensity is unchanged by slope; here it is $675.184 \text{ kW m}^{-2}$. Every other generated fuel/slope pair uses the same equations with its stored integer angle and fuel packing ratio. Variant errors are $e_R = |R_{\text{max}} - R_{\text{ref}}| / \max(|R_{\text{ref}}|, 10^{-8})$ and analogously e_I . From $t = a + bx_{\text{surface}}$, $R_{\text{TOA}} = 60/b$ and $R^2 = 1 - SSE/SST$. The aggregate substitutions are $\max e_R \leq 0.02$, $\max e_I \leq 0.01$, $\max e_{\text{TOA}} \leq 0.05$, and $\min R^2 \geq 0.995$.

4 Verification metrics and acceptance criteria

Maximum relative errors are 0.02 for direct ROS, 0.01 for reaction intensity, and 0.05 for TOA-derived ROS; TOA regression requires $R^2 \geq 0.995$. The direct and oracle ROS values are slope-surface speeds. Accordingly, the TOA regression converts map easting to distance along the sloping surface using $x_s = x_0 + (x - x_0)/\cos \theta$ before fitting speed. Pointwise analytical errors are used instead of a trend-only test because degrees-versus-percent mistakes can preserve monotonicity. All variants and output fields must exist.

5 Simulation configuration and predicted outcome

The finite-strip PHI raster is a signed-distance field clipped to ELMFIRE’s supported $[-1, 1]$ interval, preserving both an ignited interior and a resolved front gradient. For every positive oracle ROS, `SIMULATION_DT` and `SIMULATION_DTMAX` use the whole-second value obtained by rounding the 5%-of-cell target downward, with a one-second minimum; every generated value remains below the configured CFL limit of 0.2. The stop time is rounded upward to an exact integer multiple of that step. This avoids a shortened final step that can trigger ELMFIRE’s stalled-front branch and corrupt the terminal timestamp. A zero-ROS variant retains a 5 s step and is evaluated from direct fields without requiring TOA propagation. The evaluator always requires uniquely named rasters tied to one final dump record. It accepts a stalled-final record only when subtracting the configured meteorology interval reconstructs the requested stop time within one configured timestep. A TOA-required variant must reach the `SIMULATION_TSTOP` in its generated namelist; a direct-only variant may terminate early because lack of propagation is part of the state being tested. Each 1400 m square domain uses 10 m cells and a finite ignition strip with a planar east face in the interior analysis corridor. The strip does not touch the lateral grid boundaries. Wind is zero, dead moisture is fixed at 5/7/9%, and live moisture is 90%. Aspect is chosen so upslope is aligned with propagation. Slope inputs are generated as explicit integer degrees after applying the positive-value equivalent of Fortran NINT; the expected response and slope-surface coordinate conversion use that same generated angle. ROS should rise quadratically in $\tan \phi$ before any source-defined plateau.

Parameter	Configured value	Units	Role
Domain/grid	1400 m square; 10	m/cell	Planar upslope propagation
Fuel sweep	FBFM2, FBFM9, GR2, TU5, and TL8	code/name	Fuel-dependent packing response
Slope sweep	0, 10, 20, 30, 50, 75, and 100%	percent rise/run	Independent terrain target
Wind/moisture	calm; dead 5/7/9, live 90/90	mph / percent	Isolates slope factor
Timing	generated whole-second step and exact-multiple stop	s	CFL-safe terminal evidence

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

Whole-domain input configuration — variants/fuel_102_s_030; WS=0 mph, WD=270 deg

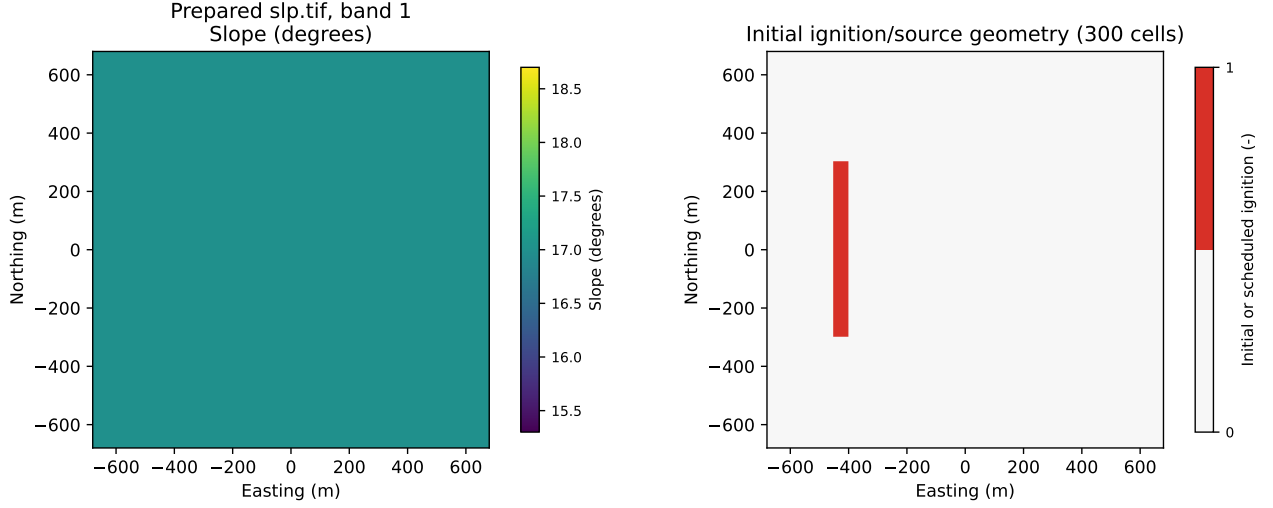


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The generated record contains 35 of 35 variants and has state **PASS**.

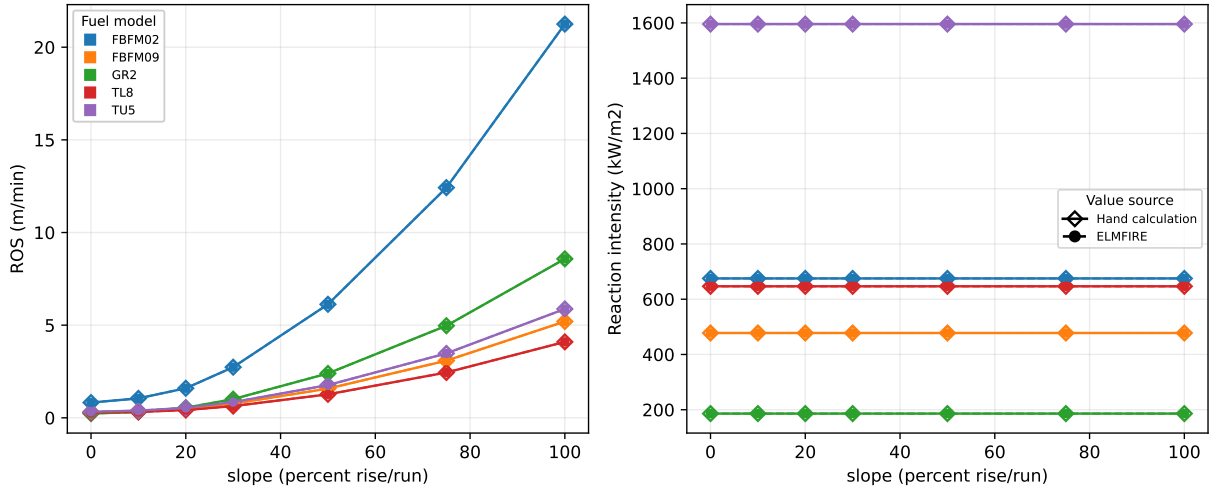


Figure 2: Hand calculations (open diamonds, solid lines) and ELMFIRE predictions (filled circles, dashed lines). Each fuel-model pair uses the same color.

7 Calculated metrics and verification decision

Metric	Limit	Calculated	Status
maximum direct-ROS relative error	≤ 0.02	5.13349e-07 fraction	PASS
maximum reaction-intensity relative error	≤ 0.01	2.93279e-07 fraction	PASS
maximum TOA-ROS relative error	≤ 0.05	0.0265981 fraction	PASS
minimum TOA regression R2	≥ 0.995	0.999961 dimensionless	PASS

The overall decision is **PASS**.

8 Technical interpretation

Large errors growing with slope indicate a units or tangent error. Fuel-dependent errors implicate packing ratio or table selection. A premature or absent plateau indicates changed cap semantics. An accurate direct field but biased TOA indicates propagation or coordinate handling.

9 Scope and limitations

This test covers uniform positive slopes aligned with the front. It does not verify vector competition between wind and slope, complex terrain, or experimental fire behavior; CASE40 covers the directional coupling.

Case 37: Dead-fuel moisture and extinction sweep

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

This case verifies dead-fuel moisture damping, heat of preignition, and extinction for the dead-dominated static fuels FBFM3 and TL6. Moisture is varied as a fraction of each model's configured dead extinction moisture. It can expose percent/fraction errors, incorrect polynomial evaluation, wrong extinction comparison, or omission of moisture from the heat sink. The verification question is whether reaction intensity and ROS follow the independent calculation through extinction for every required variant.

2 Mathematical formulation and numerical implementation

For $r_M = M_f/M_x$, Rothermel moisture damping is

$$\eta_M = \max\{0, \min[1, 1 - 2.59r_M + 5.11r_M^2 - 3.52r_M^3]\}. \quad (1)$$

Moisture also increases $Q_{ig} = 250 + 1116M_f$. Thus it reduces the reaction source and increases the heat sink. The equations follow Andrews (2018), RMRS-GTR-371. ELMFIRE receives percent rasters; the oracle uses fractions.

3 Derivation of expected behavior

For each fuel, the 1-h, 10-h, and 100-h moistures are equal, so their weighted dead moisture is the selected ratio times M_x . The damping polynomial and therefore dead reaction intensity reach zero at and above $r_M = 1$. Because these variants have no live load contribution, $I_R = R_0 = 0$ there. Below extinction, the independent fuel weighting and heat sink give the exact expected curve.

Dead-fuel moisture damping is evaluated from the generated ratio $r = M_d/M_x$:

$$\eta_M = \begin{cases} 1 - 2.59r + 5.11r^2 - 3.52r^3, & 0 \leq r \leq 1, \\ 0, & r > 1. \end{cases}$$

For Fuel Model 3 at $r = 0.50$, $\eta_M = 1 - 2.59(0.5) + 5.11(0.5)^2 - 3.52(0.5)^3 = 0.5425$. Substitution through the weighted reaction-intensity and Rothermel-rate equations gives $I_{R,\text{ref}} = 477.3889 \text{ kW m}^{-2}$ and $R_{\text{ref}} = 1.069455 \text{ m min}^{-1}$. At $r = 1.00$, the polynomial is zero, giving numerical references $I_R = 3.91 \times 10^{-13}$ and $R = 6.45 \times 10^{-16}$, both treated as zero; at $r = 1.10$ they are exactly zero by the piecewise definition. For nonzero references, $e_R = |R_{\text{max}} - R_{\text{ref}}| / \max(|R_{\text{ref}}|, 10^{-8})$ and e_I is analogous. The near-extinction intensity metric instead uses $\max |I_{R,\text{sim}} - I_{R,\text{ref}}| \leq 0.001 \text{ kW m}^{-2}$. Propagating variants additionally use $R_{\text{TOA}} = 60/b$ from $t = a + bx$ and $R^2 = 1 - SSE/SST$, with $\max e_{\text{TOA}} \leq 0.05$ and $\min R^2 \geq 0.995$; the direct limits are 0.02 for ROS and 0.01 for reaction intensity.

4 Verification metrics and acceptance criteria

Maximum relative errors are 0.02 for direct ROS and nonzero reaction intensity. Analytically extinguished intensity instead uses an absolute tolerance of 0.001 kW m^{-2} , so floating-point residuals are not divided by a near-zero reference. Variants predicted to spread at least 0.10 m min^{-1} additionally require TOA-derived ROS error at most 0.05 and regression $R^2 \geq 0.995$; slower variants cannot cross three 10m cells within the capped practical duration and are decided by direct ROS and intensity. Extinguished variants require zero ROS and reaction intensity within the stated absolute residual tolerance; TOA is not used to invent a velocity where no propagation occurs. All 16 variants are mandatory.

5 Simulation configuration and predicted outcome

The finite-strip PHI raster is a signed-distance field clipped to ELMFIRE’s supported $[-1, 1]$ interval, preserving both an ignited interior and a resolved front gradient. For every positive oracle ROS, **SIMULATION_DT** and **SIMULATION_DTMAX** use the whole-second value obtained by rounding the 5%-of-cell target downward, with a one-second minimum; every generated value remains below the configured CFL limit of 0.2. The stop time is rounded upward to an exact integer multiple of that step. This avoids a shortened final step that can trigger ELMFIRE’s stalled-front branch and corrupt the terminal timestamp. A zero-ROS variant retains a 5 s step and is evaluated from direct fields without requiring TOA propagation. The evaluator always requires uniquely named rasters tied to one final dump record. After subtracting the configured meteorology interval, an ordinary stalled-final record must reconstruct the stop time within one timestep. An early stalled-final record is accepted only for a variant whose independent oracle predicts zero ROS, and only when the reconstructed pre-jump time is finite, lies between zero and the stop time, and is aligned to the configured timestep grid. Making TOA optional does not authorize an early stall. The sweep ratios are 0.10, 0.25, 0.50, 0.70, 0.85, 0.95, 1.00, and 1.10. Domains use 10m cells and a finite ignition strip with a planar east face in the interior measurement corridor; the strip terminates inside the lateral boundaries. Wind and slope are zero, live moisture is fixed, and no coupled fire processes are enabled. Intensity and ROS should decrease toward zero and remain zero beyond extinction.

Parameter	Configured value	Units	Role
Domain/grid	1400 m square; 10	m/cell	Planar interior propagation
Fuel models	FBFM3 and TL6	code/name	Two extinction-moisture states
Dead-moisture ratio	0.10 through 1.10 times dead extinction mois- ture	fraction	Independent damping variable
Wind/slope	0 / 0	mph / de- grees	Isolates moisture damping
Timing	generated whole-second step and exact-multiple stop	s	Terminal-evidence compatibility

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

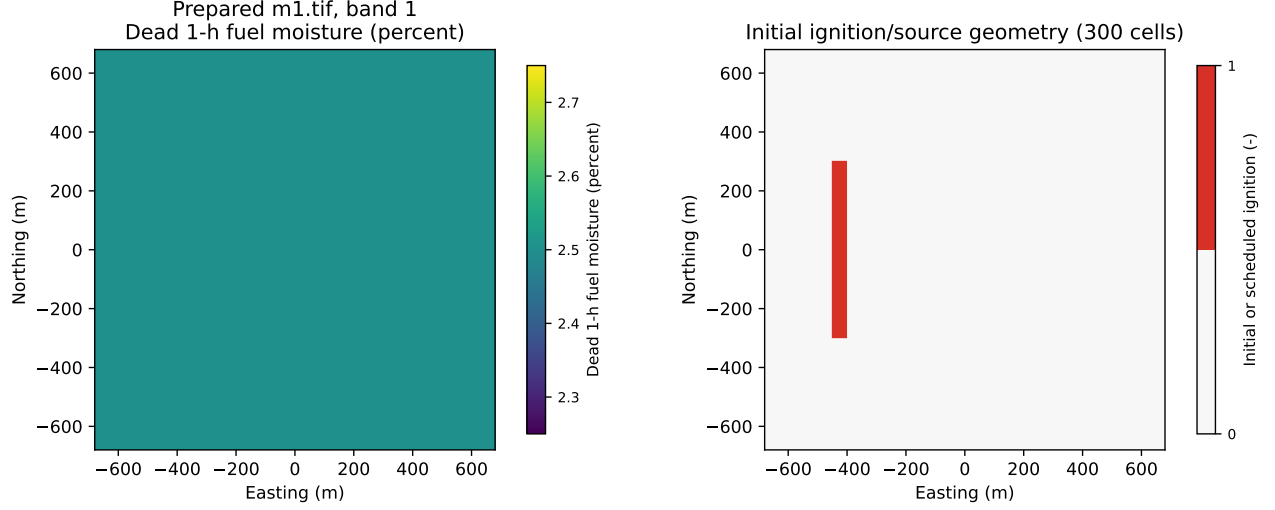


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The generated record contains 16 of 16 required variants and reports **PASS**. For a stalled extinction variant, the machine-readable result records the configured timestep, stop time, meteorology interval, recorded final time, reconstructed pre-jump time, and evidence-selection mode.

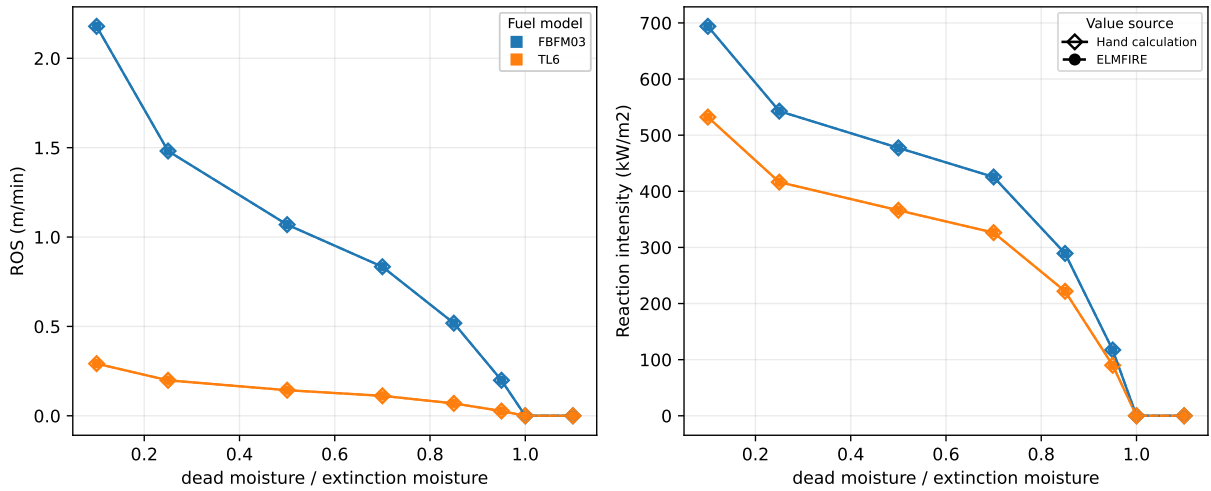


Figure 2: Hand calculations (open diamonds, solid lines) and ELMFIRE predictions (filled circles, dashed lines) for the dead-moisture response through extinction. Each fuel-model pair uses the same color.

7 Calculated metrics and verification decision

Metric	Limit	Calculated	Status
maximum direct-ROS relative error	≤ 0.02	3.66859e-06 fraction	PASS
maximum reaction-intensity relative error	≤ 0.01	3.31587e-06 fraction	PASS
maximum near-extinction reaction-intensity absolute error	≤ 0.001	0.000263903 kW/m ²	PASS
maximum TOA-ROS relative error	≤ 0.05	0.028024 fraction	PASS
minimum TOA regression R ²	≥ 0.995	0.999896 dimensionless	PASS

The authoritative decision is **PASS**.

8 Technical interpretation

The PASS decision is driven by the component statuses in the generated table. Direct observables use the upper envelope of the sparse diagnostic raster, excluding zero-filled and off-axis cells from the head-spread comparison. For direct ROS, the largest relative discrepancy is 3.66859e-06. Its independent value is 0.0266137 m/min; ELMFIRE gives 0.0266138 m/min. For reaction intensity, the largest relative discrepancy is 3.31587e-06. Its independent value is 89.9786 kW/m²; ELMFIRE gives 89.9789 kW/m². A horizontal shift of the extinction point indicates an M_x or units error. Correct intensity with wrong ROS suggests heat-sink or flux-ratio handling. Nonzero propagation or intensity at extinction indicates an unintended live contribution, adjustment, numerical residual, or stale output.

9 Scope and limitations

The case varies all three dead moisture classes together. It does not identify an isolated class-indexing defect or validate moisture observations. Static live moisture is addressed separately by CASE38.

Case 38: Static live-fuel moisture sweep

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

This case verifies live-fuel moisture damping and calculated live extinction while fuel loads remain fixed. The static shrub models SH2, SH6, and SH8 are intentionally used so changing live moisture cannot transfer herbaceous load into a dead class. It can expose live/dead index errors, percent conversion errors, incorrect live-extinction calculation, or unintended dynamic curing. The question is whether all static-fuel variants match independent I_R and ROS predictions over 30 to 200%.

2 Mathematical formulation and numerical implementation

The live term uses the same cubic damping function $\eta_M(r_M)$ as dead fuel, but $r_M = M_L/M_{x,L}$ and $M_{x,L}$ is calculated from dead/live weighting and dead moisture rather than read directly. Live moisture also appears in $Q_{ig} = 250 + 1116M_f$. The formulation follows Andrews (2018), RMRS-GTR-371. For a static fuel, W_{1h} , W_{LH} , and W_{LW} are invariant throughout the sweep.

3 Derivation of expected behavior

The oracle first calculates the live extinction moisture for each frozen fuel row and fixed dead moisture. It then evaluates live damping, reaction intensity, heat sink, and R_0 at each moisture. Since no loads move, changes arise only from moisture physics. This controlled result is the baseline needed to interpret the additional load-transfer mechanism in CASE39.

Live-fuel damping uses the same cubic with the generated live-moisture ratio $r_l = M_l/M_{x,l}$:

$$\eta_{M,l} = \max\{0, 1 - 2.59r_l + 5.11r_l^2 - 3.52r_l^3\}.$$

For generated SH2, $M_{x,l} = 0.7080597$. At $M_l = 0.70$, $r_l = 0.70/0.7080597 = 0.988617$, giving $\eta_{M,l} = 0.0326506$; substitution in the weighted Rothmel calculation gives $I_R = 354.3991 \text{ kW m}^{-2}$ and $R = 0.0800849 \text{ m min}^{-1}$. At $M_l = 0.90$, $r_l = 1.27108 > 1$, so the live contribution is zero and the remaining dead-fuel contribution gives $I_R = 304.8611 \text{ kW m}^{-2}$ and $R = 0.0579864 \text{ m min}^{-1}$. The calculation is repeated for each generated SH2, SH6, and SH8 moisture value. Direct metrics use $e_R = |R_{\max} - R_{\text{ref}}|/\max(|R_{\text{ref}}|, 10^{-8})$ and the analogous e_I , with $\max e_R \leq 0.02$ and $\max e_I \leq 0.01$. Where TOA is required, $t = a + bx$ gives $R_{\text{TOA}} = 60/b$, $R^2 = 1 - SSE/SST$, $\max e_{\text{TOA}} \leq 0.05$, and $\min R^2 \geq 0.995$.

4 Verification metrics and acceptance criteria

Maximum relative errors are 0.02 for direct ROS and 0.01 for reaction intensity. Variants predicted at or above 0.10 m min^{-1} also require TOA-derived ROS error at most 0.05 and minimum $R^2 = 0.995$;

slower variants use the direct field because the practical run-duration cap cannot provide a multi-cell TOA regression. Pointwise analytical comparison is selected because a merely monotonic curve would not detect an incorrect live-extinction threshold. All 21 variants are required and the preprocessor rejects a fuel row marked dynamic.

5 Simulation configuration and predicted outcome

The finite-strip PHI raster is a signed-distance field clipped to ELMFIRE’s supported $[-1, 1]$ interval, preserving both an ignited interior and a resolved front gradient. For every positive oracle ROS, **SIMULATION_DT** and **SIMULATION_DTMAX** use the whole-second value obtained by rounding the 5%-of-cell target downward, with a one-second minimum; every generated value remains below the configured CFL limit of 0.2. The stop time is rounded upward to an exact integer multiple of that step. This avoids a shortened final step that can trigger ELMFIRE’s stalled-front branch and corrupt the terminal timestamp. A zero-ROS variant retains a 5 s step and is evaluated from direct fields without requiring TOA propagation. The evaluator always requires uniquely named rasters tied to one final dump record. It accepts a stalled-final record only when subtracting the configured meteorology interval reconstructs the requested stop time within one configured timestep. A TOA-required variant must reach the **SIMULATION_TSTOP** in its generated namelist; a direct-only variant may terminate early because lack of propagation is part of the state being tested. These shrub models carry live woody but no live herbaceous load. Live-woody moisture is set to 60, 70, 90, 120, 150, 180, or 200%; 60% is the supported ELMFIRE lower bound; dead classes remain at 8%. The homogeneous domain uses a finite ignition strip whose east face is planar in the interior measurement corridor and whose ends remain inside the lateral boundaries. Wind and slope are zero and all coupled processes are disabled. Fuel loads remain invariant; reaction intensity and ROS should generally decline as live moisture rises, with behavior governed by each fuel’s calculated live extinction.

Parameter	Configured value	Units	Role
Domain/grid	1400 m square; 10	m/cell	Planar interior propagation
Fuel models	SH2, SH6, and SH8	code/name	Static live-fuel responses
Live moisture	30 to 200%	percent	Independent damping variable
Wind/slope	0 / 0	mph / degrees	Isolates live damping
Timing	generated whole-second step and exact-multiple stop	s	Terminal-evidence compatibility

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

Whole-domain input configuration — variants/fuel_146_m_090; WS=0 mph, WD=270 deg

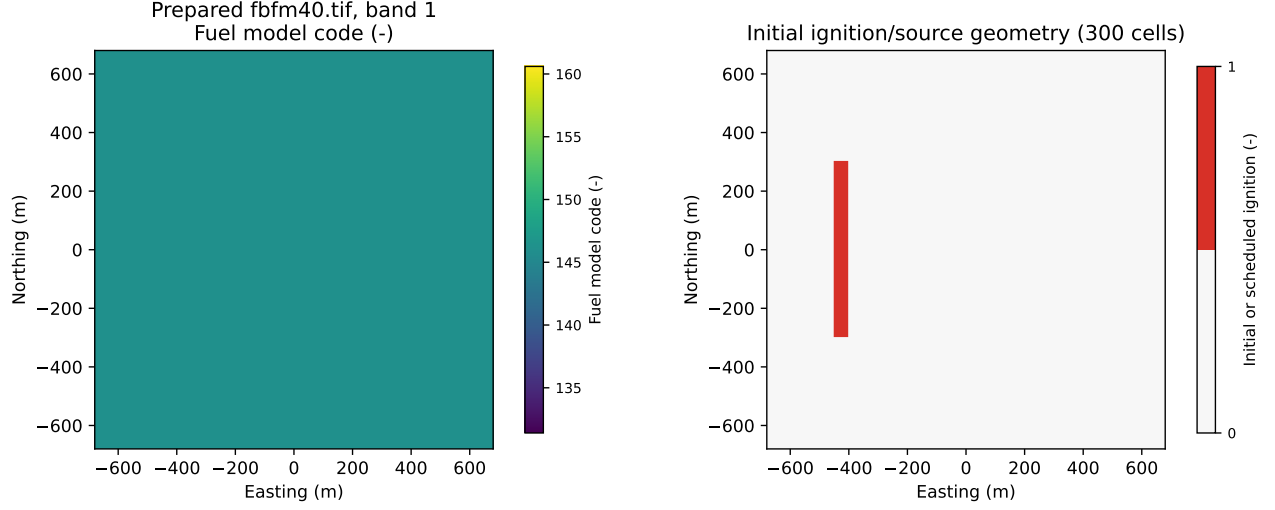


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The generated record contains 21 of 21 required variants and reports **PASS**.

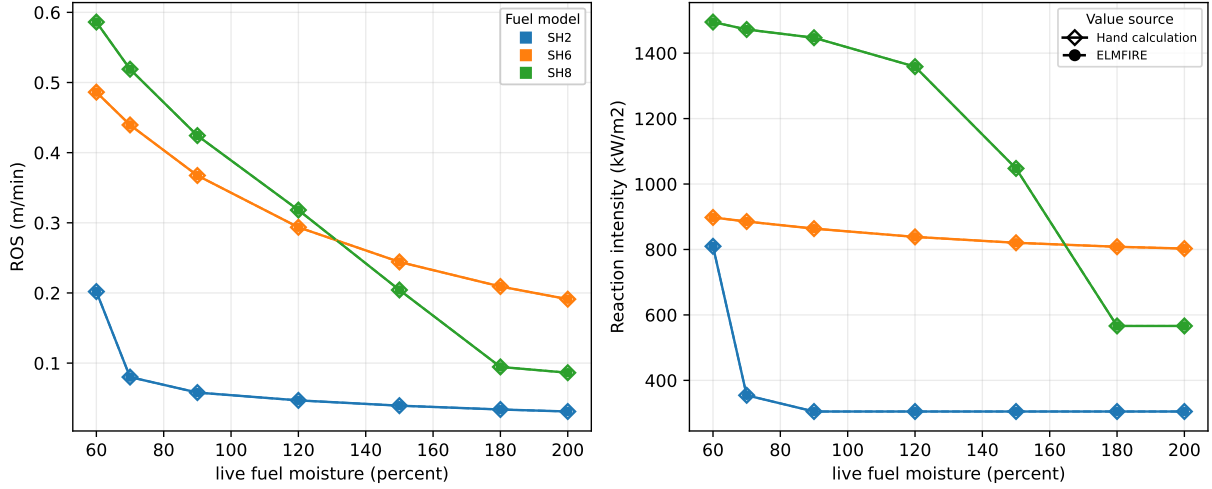


Figure 2: Hand calculations (open diamonds, solid lines) and ELMFIRE predictions (filled circles, dashed lines) for the static live-moisture response. Each fuel-model pair uses the same color.

7 Calculated metrics and verification decision

Metric	Limit	Calculated	Status
maximum direct-ROS relative error	≤ 0.02	6.50329e-06 fraction	PASS
maximum reaction-intensity relative error	≤ 0.01	6.30732e-06 fraction	PASS
maximum TOA-ROS relative error	≤ 0.05	0.0170672 fraction	PASS
minimum TOA regression R2	≥ 0.995	0.999953 dimensionless	PASS

The authoritative decision is **PASS**.

8 Technical interpretation

A shifted transition implicates calculated live extinction or moisture units. Unexpected sharp changes suggest that a supposedly static row was treated dynamically. Agreement in intensity but not ROS narrows the defect to heat sink or spread conversion.

9 Scope and limitations

This case covers three static shrub fuels at one dead-moisture state. It does not verify dynamic curing, spatial moisture mosaics, temporal interpolation, or physical observations. Dynamic herbaceous transfer is isolated in CASE39.

Case 39: Dynamic herbaceous curing sweep

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

This case verifies dynamic herbaceous curing in GR2 and GS3. Live-herb moisture is the controlled input; ELMFIRE must transfer the corresponding fraction of herbaceous load to the dead 1-h class and then evaluate the full Rothermel model. It can expose reversed curing, incorrect 30/120% bounds, failure to conserve load, wrong class SAV weighting, or confusion between curing and live moisture damping. The verification question is whether all required curing states reproduce independently calculated intensity and ROS within 1% and 2%, respectively.

2 Mathematical formulation and numerical implementation

For live-herb moisture M_{LH} expressed as a fraction, the cured fraction is

$$T = \max[0, \min(1, 1.33 - 1.11M_{LH})]. \quad (1)$$

ELMFIRE implements the equivalent integer-percent interpolation between 30 and 120%. If $W_{LH,0}$ is original live-herb load, $W'_{1h} = W_{1h} + TW_{LH,0}$ and $W'_{LH} = (1 - T)W_{LH,0}$. The transferred material adopts dead 1-h moisture while remaining live material uses live damping and heat of preignition. The formulation follows Andrews (2018), RMRS-GTR-371.

3 Derivation of expected behavior

At 30% moisture all live-herb load is cured; at 75% half is cured; at 120% and above none is transferred. Above 120%, curing is constant but live moisture damping continues to change, so ROS need not form a plateau. The case-local oracle conserves total load, recomputes effective SAV and every dependent Rothermel coefficient, applies moisture damping, and applies ELMFIRE's selected wind limit at the target 6 mph midflame wind.

For dynamic grass fuels, the cured fraction and load transfer are

$$f_c = \min\left(1, \max\left[0, \frac{120 - M_{lh}}{90}\right]\right), \quad w_{lh}^* = (1 - f_c)w_{lh}, \quad w_{1h}^* = w_{1h} + f_c w_{lh}.$$

For the generated GR2 variant at $M_{lh} = 75$ percent, $f_c = (120 - 75)/90 = 0.5$. With original $w_{lh} = 0.04591$ and $w_{1h} = 0.00459 \text{ lb ft}^{-2}$,

$$w_{lh}^* = 0.5(0.04591) = 0.022955, \quad w_{1h}^* = 0.00459 + 0.5(0.04591) = 0.027545.$$

The weighted fuel calculation gives $R_0 = 1.109830 \text{ ft min}^{-1}$, $I_R = 206.9424 \text{ kW m}^{-2}$, and $\phi_w = 31.095160$; therefore

$$R_{\text{ref}} = 0.3048(1.109830)(1 + 31.095160) = 10.857030 \text{ m min}^{-1}.$$

Every GR2 and GR3 moisture variant repeats these equations. Direct metrics use the maximum over variants of $e_R = |R_{\max} - R_{\text{ref}}| / \max(|R_{\text{ref}}|, 10^{-8})$ and the analogous e_I , with limits 0.02 and 0.01. The TOA fit $t = a + bx$ supplies $R_{\text{TOA}} = 60/b$ and $R^2 = 1 - SSE/SST$, tested by $\max e_{\text{TOA}} \leq 0.05$ and $\min R^2 \geq 0.995$.

4 Verification metrics and acceptance criteria

Maximum relative errors are 0.02 for direct ROS and 0.01 for reaction intensity. Variants predicted at or above 0.10 m min^{-1} also require TOA-derived ROS error at most 0.05 and minimum $R^2 = 0.995$; the most moisture-limited variants use direct fields because they cannot cross a multi-cell corridor within the practical duration cap. Direct effective loads are not exposed by the normal executable, so their conservation is an oracle invariant and the implementation is observed through intensity and ROS. All 20 variants are required; no qualitative trend alone can pass the case.

5 Simulation configuration and predicted outcome

The finite-strip PHI raster is a signed-distance field clipped to ELMFIRE’s supported $[-1, 1]$ interval, preserving both an ignited interior and a resolved front gradient. For every positive oracle ROS, `SIMULATION_DT` and `SIMULATION_DTMAX` use the whole-second value obtained by rounding the 5%-of-cell target downward, with a one-second minimum; every generated value remains below the configured CFL limit of 0.2. The stop time is rounded upward to an exact integer multiple of that step. This avoids a shortened final step that can trigger ELMFIRE’s stalled-front branch and corrupt the terminal timestamp. A zero-ROS variant retains a 5 s step and is evaluated from direct fields without requiring TOA propagation. The evaluator always requires uniquely named rasters tied to one final dump record. It accepts a stalled-final record only when subtracting the configured meteorology interval reconstructs the requested stop time within one configured timestep. A TOA-required variant must reach the `SIMULATION.TSTOP` in its generated namelist; a direct-only variant may terminate early because lack of propagation is part of the state being tested. Live-herb moisture is 30, 45, 60, 75, 90, 100, 110, 120, 135, or 200%. Dead classes are fixed at 5%, live woody moisture at 90%, slope at zero, and target midflame wind at 6 mph; the 20-ft input is back-calculated from each fuel’s WAF. Each homogeneous domain uses a finite ignition strip with a planar east face inside the analysis corridor and with lateral ends clear of the grid boundary. Crown fire, spotting, buildings, and acceleration are excluded. Strong nonlinear response is expected in the curing interval.

Parameter	Configured value	Units	Role
Domain/grid	1400 m square; 10	m/cell	Planar interior propagation
Dynamic fuels	GR2 and GS3	code/name	Enables herbaceous load transfer
Live-herb moisture	30 to 200% live-herb moisture	percent	Independent curing variable
Wind/slope	fixed wind; 0 slope	—	Holds directional forcing fixed
Timing	generated whole-second step and exact-multiple stop	s	Terminal-evidence compatibility

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells

selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

Whole-domain input configuration — variants/fuel_123_lh_030; WS=14.7 mph, WD=270 deg

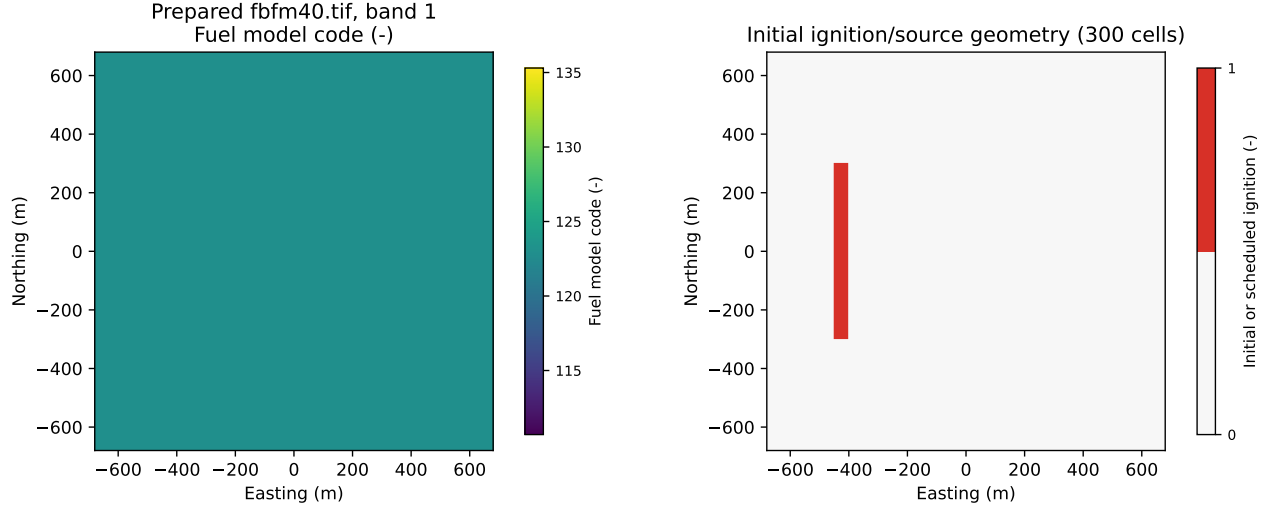


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The generated record contains 20 of 20 variants and reports **PASS**.

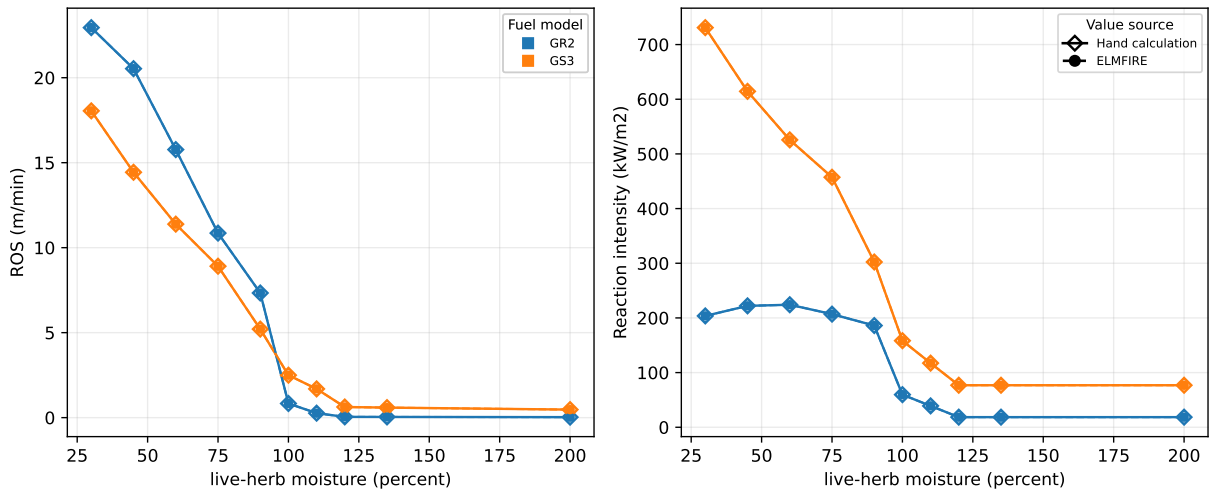


Figure 2: Hand calculations (open diamonds, solid lines) and ELMFIRE predictions (filled circles, dashed lines) for the dynamic-curing response. Each fuel-model pair uses the same color.

7 Calculated metrics and verification decision

Metric	Limit	Calculated	Status
maximum direct-ROS relative error	≤ 0.02	1.51489e-06 fraction	PASS
maximum reaction-intensity relative error	≤ 0.01	2.3555e-07 fraction	PASS
maximum TOA-ROS relative error	≤ 0.05	0.0226402 fraction	PASS
minimum TOA regression R2	≥ 0.995	0.999981 dimensionless	PASS

The authoritative decision is **PASS**.

8 Technical interpretation

A reversed curve indicates a reversed transfer fraction. Discontinuities away from integer moisture selection suggest class indexing or interpolation defects. Correct behavior above 120% but disagreement below it isolates curing rather than live damping. Correct intensity and wrong ROS implicate heat sink or wind coupling.

9 Scope and limitations

This executable-level test observes curing indirectly; a direct internal-load assertion would require a compiled component diagnostic. It covers two dynamic fuel models, constant weather, and no temporal curing. CASE38 separately isolates live moisture without load transfer.

Case 40: Wind-slope vector and direction sweep

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

This coupling case verifies how ELMFIRE combines Rothermel wind and slope factors into a direction of maximum spread (DMS), head speed, and fire ellipse. Wind direction rotates through 0, 45, 90, 135, and 180 degrees relative to upslope while magnitudes remain fixed. It can expose degree/radian errors, wind-from/wind-to reversal, aspect reversal, swapped raster axes, scalar addition in place of vector addition, or incorrect ellipse construction. The question is whether all variants meet the DMS, head-ROS, length/width, and intensity limits.

2 Mathematical formulation and numerical implementation

Let upslope azimuth be zero and wind-to azimuth be α . The dimensionless forcing vector is

$$\phi = \phi_s(0, 1) + \phi_w(\sin \alpha, \cos \alpha), \quad (1)$$

so $\theta_{DMS} = \text{atan2}(\phi_x, \phi_y)$ and $R_H = R_0(1 + |\phi|)$. ELMFIRE derives an effective wind from $|\phi|$ and uses its Rothermel length-to-width relation, capped at eight. Because these quantities are defined on the sloping surface, the preprocessor makes the initial circle and the postprocessor measures distances in slope-surface coordinates rather than mixing them with map-plane northing. The scalar factors follow Andrews (2018), RMRS-GTR-371; vectoring and ellipse propagation exercise ELMFIRE's level-set implementation.

3 Derivation of expected behavior

Aligned wind and slope add; opposed forcing subtracts in vector space and the stronger component sets direction. At intermediate angles, the expected DMS lies between the two forcing directions, not generally at the arithmetic mean of angles. The case-local oracle independently evaluates ϕ_s , ϕ_w , their vector magnitude and azimuth, head ROS, effective wind, and length/width.

Let upslope define zero azimuth and let α be the generated wind-to-upslope angle. With $\phi_s = 6.436561$, $\phi_w = 9.746235$, and $R_0 = 0.9178563 \text{ m min}^{-1}$, the vector reference is

$$\phi_x = \phi_s + \phi_w \cos \alpha, \quad \phi_y = \phi_w \sin \alpha, \quad \phi = \sqrt{\phi_x^2 + \phi_y^2}, \quad \theta = \text{atan2}(\phi_y, \phi_x), \quad R_h = R_0(1 + \phi).$$

For example, at $\alpha = 90$ degrees, $\phi = \sqrt{6.436561^2 + 9.746235^2} = 11.679829$, $\theta = \tan^{-1}(9.746235/6.436561) = 56.5586$ degrees, and $R_h = 0.9178563(1 + 11.679829) = 11.638261 \text{ m min}^{-1}$. The effective vector wind is $4.41571 \text{ mph} = 1.97394 \text{ m s}^{-1}$; substitution in $L/W = 0.936e^{0.2566U} + 0.461e^{-0.1548U} - 0.397$ gives $L/W = 1.495859$. Across $\alpha = \{0, 45, 90, 135, 180\}$ degrees, the reference $(\theta, R_h, L/W)$ rows are

(0, 15.7713, 1.63745), (27.3422, 14.6898, 1.60074), (56.5586, 11.6383, 1.49586), (93.7779, 7.25716, 1.33838), (180, 3.95)

Direction error is circular, $e_\theta = \min(|\theta_m - \theta_r|, 360 - |\theta_m - \theta_r|)$. The other metrics are relative errors in R_h , L/W , and $I_R = 686.5297 \text{ kW m}^{-2}$; their maximum substitutions are $e_\theta \leq 5$ degrees, $e_R \leq 0.05$, $e_{LW} \leq 0.10$, and $e_I \leq 0.01$.

4 Verification metrics and acceptance criteria

Maximum circular DMS error is 5° , maximum head-ROS relative error is 0.05, maximum length/width relative error is 0.10, and maximum intensity relative error is 0.01. DMS is inferred from the farthest one percent of the burned footprint, subject to at least 100 burned cells and at least five farthest-front samples. Head ROS is a TOA regression in a narrow predicted-axis corridor with at least ten samples. Length/width uses full-footprint slope-surface extents projected along and across the DMS. Reaction intensity is the median over valid burned cells and requires at least 90 percent coverage of that footprint. These complementary metrics distinguish direction, speed, shape, and output completeness. Every variant is required.

5 Simulation configuration and predicted outcome

Fuel model 2 is used on a 1400 m square, 10 m grid with a 30 m-radius signed-distance ignition clipped to $[-1, 1]$. Slope is requested as 50% and generated as the integer 27-degree value used by ELMFIRE (50.95% effective rise/run), with upslope north; target midflame wind is 4 mph. Dead moistures are 5/7/9%, live-herb moisture is 60%, and live-woody moisture is 90% in both every generated namelist and the independent oracle. Wind-from azimuth is generated so the wind-to vector has the required relative angle. Each variant uses the largest conservative whole-second timestep no greater than the 5%-of-cell target that divides the fixed 1800 s duration exactly. This retains CFL margin and prevents a shortened final step from triggering ELMFIRE's stalled-front timestamp defect. Crown fire, spotting, WUI, stochastic wind, and acceleration are disabled. The footprint should rotate smoothly and change head extension as the vectors move from aligned to opposed. The evaluator accepts only unique TOA and reaction-intensity rasters identified by a final dump record. A stalled-final record is accepted only when subtracting the configured meteorology interval reconstructs the 1800 s stop time within one configured timestep.

Parameter	Configured value	Units	Role
Domain/grid	1400 m square; 10	m/cell	Whole-footprint direction and shape
Fuel/slope	FM2; 50	code / percent	Fixed slope forcing
Midflame wind	4	mph	Fixed wind-factor magnitude
Relative angles	0, 45, 90, 135, and 180 degrees	degrees	Vector-coupling sweep
Time control	generated whole-second divisor; stop 1800	s	Exact timestep multiple

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

Whole-domain input configuration — variants/angle_090; WS=11 mph, WD=270 deg

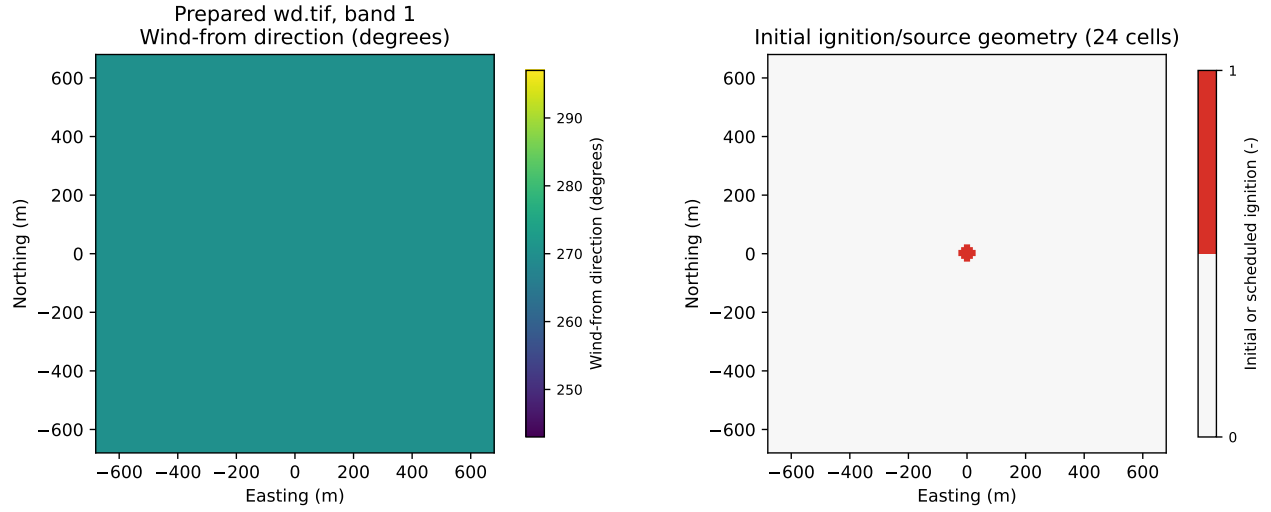


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The generated record contains 5 of 5 variants and reports **PASS**. The generated diagnostic is: $-$.

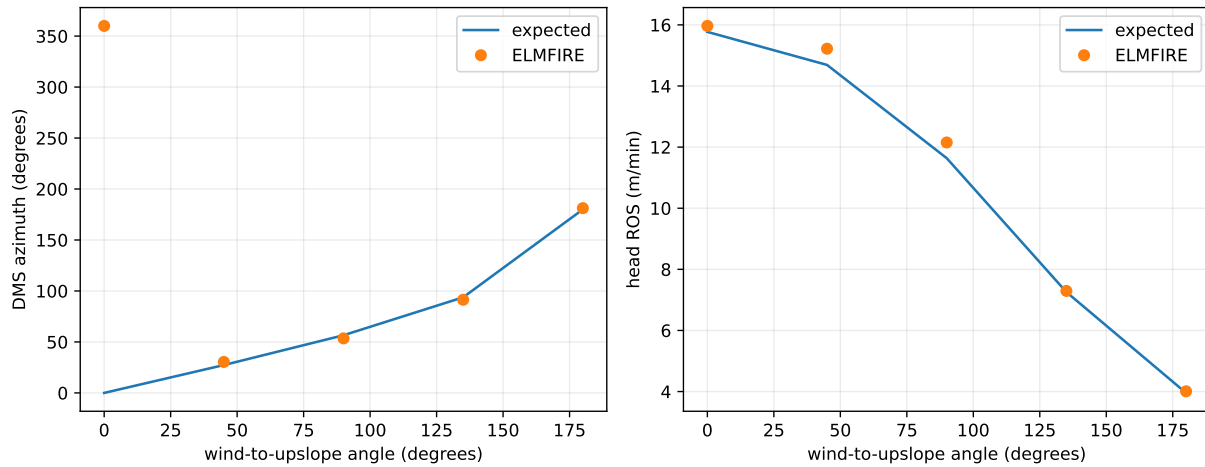


Figure 2: Expected and measured DMS azimuth and head ROS.

7 Calculated metrics and verification decision

Metric	Limit	Calculated	Status
maximum DMS direction error	≤ 5.0	3.05989 degrees	PASS
maximum head-ROS relative error	≤ 0.05	0.0441104 fraction	PASS
maximum length/width relative error	≤ 0.1	0.0961658 fraction	PASS
maximum reaction-intensity relative error	≤ 0.01	2.01156e-07 fraction	PASS

The authoritative decision is **PASS**.

8 Technical interpretation

A 180-degree error indicates a wind-from or aspect reversal. A 90-degree error suggests swapped axes or azimuth convention. Correct direction but excessive head speed suggests scalar addition. Correct DMS and head speed with wrong length/width isolates effective-wind or ellipse construction.

9 Scope and limitations

This test uses one fuel, one slope magnitude, one wind magnitude, uniform terrain, and a single grid. It verifies deterministic code behavior rather than observed fire shape. Complex terrain and resolution convergence remain separate concerns.

Case 41: Custom fuel-parameter Rothermel sweep

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

This case verifies ELMFIRE’s public custom fuel-table interface and the Rothermel response to one-at-a-time changes in dead 1-h load, 1-h SAV, fuel-bed depth, dead extinction moisture, and heat content. It can expose column misalignment, ignored custom values, stale precomputed coefficients, or unit conversion errors. The question is whether every generated code-250 row produces independently calculated reaction intensity and ROS within 1% and 2%.

2 Mathematical formulation and numerical implementation

The custom row supplies class loads w_0 , surface-area-to-volume ratios σ , depth δ , dead extinction M_x , and heat content h . These feed packing ratio $\beta = \sum w_0/(\delta\rho_p)$, optimum packing, reaction velocity, propagating flux, damping, heat sink, and

$$R_0 = I_R\xi/(\rho_b\epsilon Q_{ig}). \quad (1)$$

The formulation follows Andrews (2018), RMRS-GTR-371. Particle density and mineral constants are fixed by the inspected ELMFIRE interface and therefore are not claimed as user-sweepable variables.

3 Derivation of expected behavior

The baseline row is copied and exactly one public column is changed in each variant. A case-local fuel table is written beside that variant and read independently by the oracle. The full equation is recalculated rather than imposing a monotonic trend: load, SAV, and depth have competing effects through packing and reaction velocity, so a simple ordering would be scientifically unjustified.

Each custom-fuel reference is recalculated from $R_0 = I_R\xi/(\rho_b\epsilon Q_{ig})$, changing only the indicated fuel-table column. The substitutions produced from the generated table are

variant	R_{ref} (m min ⁻¹)	$I_{R,\text{ref}}$ (kW m ⁻²)
baseline	0.411242	1045.499
$w_{1h} = 0.05$	0.253350	688.121
$w_{1h} = 0.15$	0.551674	1379.611
$SAV = 1000$	0.241692	960.426
$SAV = 3000$	0.662626	1074.561
$d = 0.5$	0.225975	910.468
$d = 2.0$	0.694647	991.942
$M_x = 15\%$	0.371790	945.198
$M_x = 35\%$	0.438500	1114.797
$h = 6000$	0.308432	784.125
$h = 10000$	0.514053	1306.874

For example, replacing the baseline 1-h load 0.10 by 0.05 lb ft⁻² changes $R_0 = 1.349221$ to 0.831202 ft min⁻¹; multiplication by 0.3048 gives 0.253350 m min⁻¹. Each variant uses $e_R = |R_{\max} - R_{\text{ref}}| / \max(|R_{\text{ref}}|, 10^{-8})$ and the analogous e_I . The TOA fit $t = a + bx$ gives $R_{\text{TOA}} = 60/b$ and $R^2 = 1 - SSE/SST$. The aggregate substitutions are $\max e_R \leq 0.02$, $\max e_I \leq 0.01$, $\max e_{\text{TOA}} \leq 0.05$, and $\min R^2 \geq 0.995$.

4 Verification metrics and acceptance criteria

Maximum relative errors are 0.02 for direct ROS and 0.01 for reaction intensity. Variants predicted at or above 0.10 m min⁻¹ additionally require TOA-derived ROS error at most 0.05 and minimum $R^2 = 0.995$; slower variants are assessed from direct ROS and intensity. Pointwise equality to a complete recalculation detects both ignored columns and wrong nonlinear propagation. The baseline and ten low/high variants are all required.

5 Simulation configuration and predicted outcome

The finite-strip PHI raster is a signed-distance field clipped to ELMFIRE’s supported $[-1, 1]$ interval, preserving both an ignited interior and a resolved front gradient. For every positive oracle ROS, **SIMULATION_DT** and **SIMULATION_DTMAX** use the whole-second value obtained by rounding the 5%-of-cell target downward, with a one-second minimum; every generated value remains below the configured CFL limit of 0.2. The stop time is rounded upward to an exact integer multiple of that step. This avoids a shortened final step that can trigger ELMFIRE’s stalled-front branch and corrupt the terminal timestamp. A zero-ROS variant retains a 5 s step and is evaluated from direct fields without requiring TOA propagation. The evaluator always requires uniquely named rasters tied to one final dump record. It accepts a stalled-final record only when subtracting the configured meteorology interval reconstructs the requested stop time within one configured timestep. A TOA-required variant must reach the **SIMULATION_TSTOP** in its generated namelist; a direct-only variant may terminate early because lack of propagation is part of the state being tested. All variants use fuel code 250, 5/7/9% dead moisture, 90% live moisture, zero wind and slope, and 10 m cells. A finite ignition strip supplies a planar east face in the interior analysis corridor while remaining clear of the lateral grid boundaries. The baseline row uses 0.10/0.05/0.02 dead loads, 0.08 live-woody load, 1-h SAV 2000 ft⁻¹, depth 1 ft, dead extinction 25%, and heat content 8000 Btu lb⁻¹. Each output should match its own oracle; no universal increasing trend is predicted.

Parameter	Configured value	Units	Role
Domain/grid	1400 m square; 10	m/cell	Planar interior propagation
Custom fuel code	250	code	Selects variant-local table row
Changes	load, SAV, depth, extinction, heat	native table units	One-at-a-time interface exercise
Wind/slope	0 / 0	mph / degrees	Isolates custom fuel response
Timing	generated whole-second step and exact-multiple stop	s	Terminal-evidence compatibility

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells

selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

Whole-domain input configuration — variants/load_high; WS=0 mph, WD=270 deg

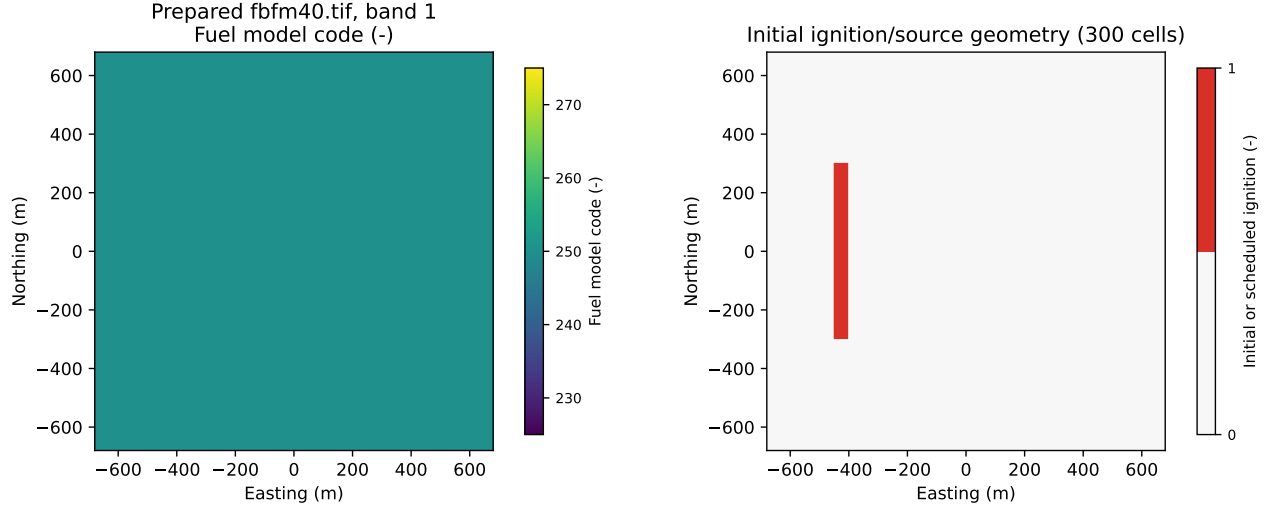


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The generated record contains 11 of 11 variants and reports **PASS**.

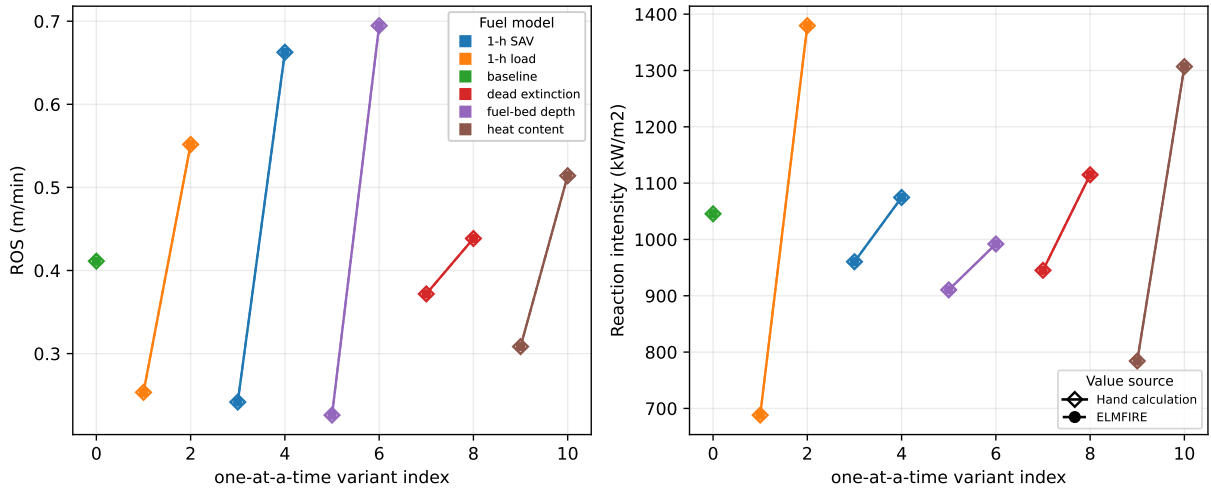


Figure 2: Hand calculations (open diamonds, solid lines) and ELMFIRE predictions (filled circles, dashed lines) for the custom fuel-table variants. Each comparison pair uses the same color.

7 Calculated metrics and verification decision

Metric	Limit	Calculated	Status
maximum direct-ROS relative error	≤ 0.02	4.62672e-07 fraction	PASS
maximum reaction-intensity relative error	≤ 0.01	1.10118e-07 fraction	PASS
maximum TOA-ROS relative error	≤ 0.05	0.0151881 fraction	PASS
minimum TOA regression R2	≥ 0.995	0.999968 dimensionless	PASS

The authoritative decision is **PASS**.

8 Technical interpretation

If only one low/high pair fails, its fuel-table column or dependent coefficient is suspect. A common intensity error suggests table parsing or units; correct intensity with wrong ROS points to heat sink, density, or flux ratio. Identical results across a changed pair indicate the column was ignored.

9 Scope and limitations

This case tests only parameters exposed by the current custom table and one synthetic baseline. It does not test particle density, mineral contents, spatial mixtures, dynamic curing, or physical realism of the synthetic fuel.

Case 42: Wind-adjustment-factor coupling sweep

ELMFIRE Verification and Validation Suite

September 2, 2026

1 Verification purpose

This case verifies the upstream coupling from 20-ft wind, canopy cover, canopy height, and fuel-bed depth through ELMFIRE’s wind-adjustment factor (WAF) into Rothermel ROS. It distinguishes an input-height conversion error from a defect in the Rothermel wind factor. It can expose percent/fraction canopy errors, height-times-ten errors, sheltered/unsheltered branch errors, or missing WAF application. The question is whether all canopy variants reproduce the independently predicted final ROS within 3% while reaction intensity remains within 1%.

2 Mathematical formulation and numerical implementation

ELMFIRE calculates $U_{mf} = WAF U_{20}$. Without canopy it uses a logarithmic fuel-bed-height expression. With canopy it evaluates a sheltered factor from canopy cover and height, limits it by the unsheltered factor, and blends the two when $0.3333CC < 0.05$. The resulting midflame speed enters $\phi_w = CU_{mf}^B(\beta/\beta_{op})^{-E}$. The case oracle restates these equations directly and does not read ELMFIRE output to estimate WAF.

3 Derivation of expected behavior

For fixed 10 mph and fuel model 2, reaction intensity is invariant because canopy affects wind adjustment but not the configured surface fuel or moisture. ROS differences are completely predicted by the independently calculated WAF and Rothermel wind factor. The zero-canopy variant selects the unsheltered formula; canopy combinations exercise transition and sheltered branches.

The unsheltered adjustment for fuel-bed depth d (ft) is

$$WAF_u = \frac{1.36}{\ln[(20 + 0.36d)/(0.13d)]} [\ln(1.36/0.13) - 1].$$

For canopy cover C , height H (ft), and $s = 0.3333C$, the sheltered value is

$$WAF_s = \frac{1}{\ln[(20 + 0.36H)/(0.13H)]} \frac{0.555}{\sqrt{sH}}.$$

The generated blend uses $WAF = \min(WAF_s, WAF_u)$ when $s \geq 0.05$; otherwise $f_u = 1 - 20s$ and $WAF = f_u WAF_u + (1 - f_u) \min(WAF_s, WAF_u)$. For the zero-canopy Fuel Model 2 control, $WAF_u = 0.362674$, so the 10 mph 20-ft wind gives $U_{mf} = 3.626740$ mph, $\phi_w = 8.146356$, and $R = 0.8237955(1 + 8.146356) = 7.534727 \text{ m min}^{-1}$. For $C = 0.10$, $H = 15/0.3048 = 49.2126$ ft, $s = 0.03333$, $WAF_s = 0.244254$, and $f_u = 0.3334$; hence $WAF = 0.3334(0.362674) + 0.6666(0.244254) = 0.283735$, $U_{mf} = 2.837354$ mph, $\phi_w = 5.197957$, and $R = 0.8237955(1 + 5.197957) = 5.105849 \text{ m min}^{-1}$. All

variants retain $I_R = 675.184 \text{ kW m}^{-2}$. Variant relative errors use $|q_{\max} - q_{\text{ref}}| / \max(|q_{\text{ref}}|, 10^{-8})$. From $t = a + bx$, $R_{\text{TOA}} = 60/b$ and $R^2 = 1 - SSE/SST$. The aggregate substitutions are $\max e_R \leq 0.03$, $\max e_I \leq 0.01$, $\max e_{\text{TOA}} \leq 0.06$, and $\min R^2 \geq 0.995$.

4 Verification metrics and acceptance criteria

Maximum direct-ROS relative error is 0.03, reaction-intensity error is 0.01, and TOA-derived ROS error is 0.06; TOA regression requires $R^2 \geq 0.995$. Constant reaction intensity is retained as a control that detects unintended crown or canopy-fuel coupling. All ten variants are required and missing fields give NOT EVALUABLE.

5 Simulation configuration and predicted outcome

The finite-strip PHI raster is a signed-distance field clipped to ELMFIRE’s supported $[-1, 1]$ interval, preserving both an ignited interior and a resolved front gradient. For every variant, **SIMULATION_DT** and **SIMULATION_DTMAX** use a conservative whole-second value no greater than the 5 s nominal step or the configured 0.2 CFL limit. The stop time is rounded upward to an exact integer multiple of that step. This avoids both discrete-front resonance and a shortened final step that can trigger ELMFIRE’s stalled-front timestamp defect. A zero-ROS variant retains a 5 s step and is evaluated from direct fields without requiring TOA propagation. The evaluator requires uniquely named rasters tied to one final dump record and accepts a stalled-final record only when subtracting the configured meteorology interval reconstructs the requested stop time within one configured timestep. A TOA-required variant must reach the **SIMULATION_TSTOP** in its generated namelist; a direct-only variant may terminate early because lack of propagation is part of the state being tested. Canopy cover is 0, 10, 30, or 60%; canopy heights are 5, 15, and 30 m where applicable. The raster stores cover in percent and height times ten, matching explicit namelist flags. Fuel model 2, dead moisture 5/7/9%, live moisture 90%, and zero slope are fixed. A finite west-to-east ignition strip has a planar east face in the interior analysis corridor and ends inside the lateral boundaries. Crown fire is disabled even though canopy inputs are present. ROS should vary only through WAF while I_R remains fixed.

Parameter	Configured value	Units	Role
Domain/grid	1400 m square; 10	m/cell	Planar interior propagation
Fuel / 20-ft wind	FM2 / 10 mph	code / mph	Fixed surface forcing
Canopy cover	0, 10, 30, 60	percent	WAF branch and blend sweep
Canopy height	0, 5, 15, 30	m	Sheltered-profile sweep
Timing	generated whole-second step and exact-multiple stop	s	Terminal-evidence compatibility

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

Whole-domain input configuration — variants/cc_030_ch_015; WS=10 mph, WD=270 deg

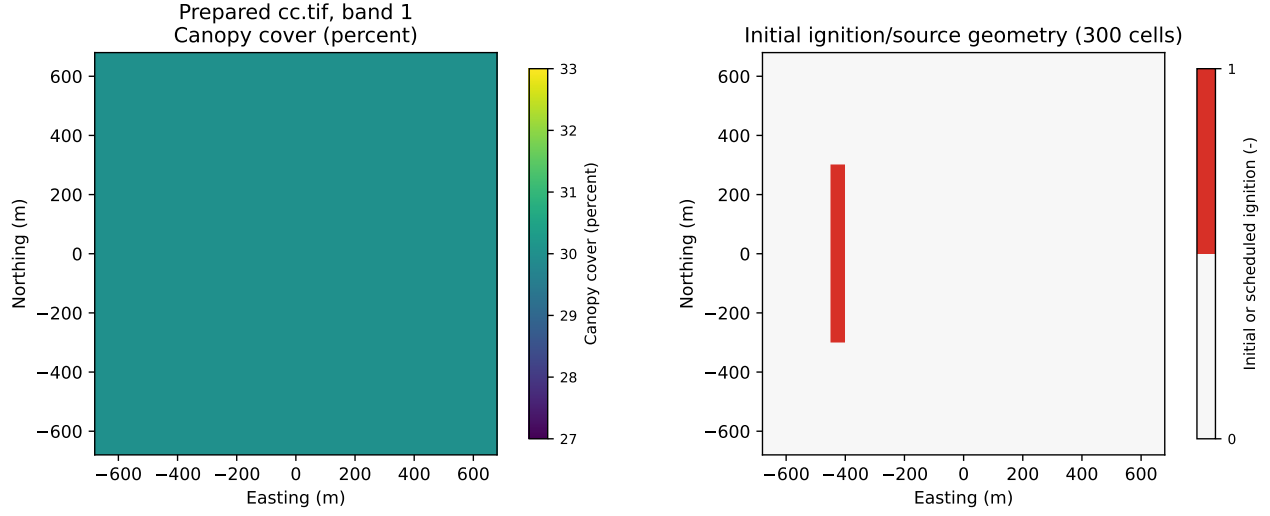


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The generated record contains 10 of 10 variants and reports **PASS**.

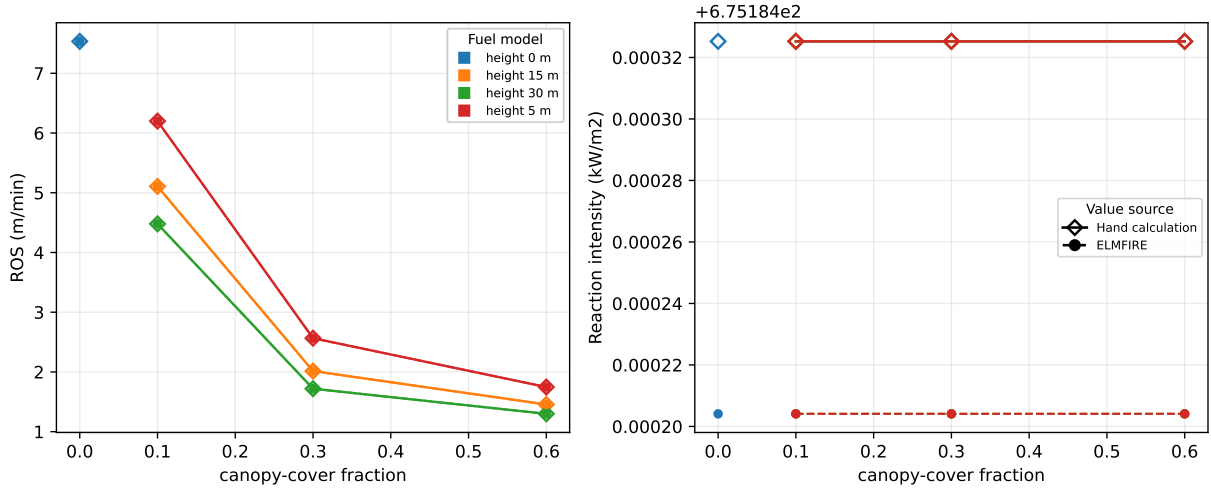


Figure 2: Expected and simulated canopy/WAF response at fixed 20-ft wind.

7 Calculated metrics and verification decision

Metric	Limit	Calculated	Status
maximum direct-ROS relative error	≤ 0.03	6.68301e-07 fraction	PASS
maximum reaction-intensity relative error	≤ 0.01	1.79416e-07 fraction	PASS
maximum TOA-ROS relative error	≤ 0.06	0.0245548 fraction	PASS
minimum TOA regression R2	≥ 0.995	0.999972 dimensionless	PASS

The authoritative decision is **PASS**.

8 Technical interpretation

A uniform wind-scale error suggests incorrect 20-ft units. Errors confined to canopy variants implicate sheltered WAF or canopy scaling. Varying reaction intensity indicates an unintended canopy or crown-fire pathway. Correct direct ROS but biased TOA implicates level-set propagation.

9 Scope and limitations

This is an implementation-specific coupling test, not a validation of canopy wind profiles. It uses one fuel, one 20-ft wind, uniform canopy fields, and the current ELMFIRE WAF algorithm. A deliberate algorithm change requires oracle and report review.